Sorting into (In)formality: Job Offers and Worker Preferences*

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Abstract

When do formal job offers move workers out of informality? I study informal workers' willingness to accept formal employment and the role of wages and working conditions in their acceptance decisions. I use labor inspections in Brazil that require employers to register informal workers, creating opportunities to formalize within existing worker–employer matches. Within six months of an inspection, new formal hires correspond to 78% of detected informal men and 68% of detected informal women. Take-up varies substantially across workplaces: women are more likely than men to formalize in female-dominated industries, while the opposite holds in male-dominated industries. To better understand take-up rates, I develop and estimate a choice model in which workers decide whether to accept a formal contract at their current employer or pursue an informal alternative. Firm-level estimates show that men and women respond similarly to wages but differ in their valuations of non-wage attributes, particularly flexibility and parental leave. A complementary survey experiment elicits workers' decisions under a guaranteed formal offer. Most respondents accept, and preserving take-home pay raises acceptance from 61 to 79% while substantially narrowing the gender gap in acceptance. The contrast between high take-up under inspections and low same-employer formalization rates in the broader labor market suggests that persistent informality largely reflects limited formal opportunities rather than workers' unwillingness to formalize. When formal offers arise, wage and non-wage contract terms and the value of the informal alternative shape acceptance decisions.

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1. Introduction

Informal jobs are often viewed as an inferior form of employment. Informal workers lack protections such as pension contributions, severance payments, and unemployment insurance. Yet roughly six in ten workers worldwide hold such jobs (ILO, 2018).[1] The standard explanation is that workers lack access to formal jobs: firms avoid regulatory costs by operating informally or employing workers *off the books* (Ulyssea, 2018, 2020), while search and matching frictions can prevent workers from reaching employers willing to offer formal contracts. This explanation, however, assumes that informal workers would accept a formal job if offered one, with little evidence from actual acceptance decisions.

Workers may also *voluntarily* sort into informal jobs. A formal contract provides benefits, but it can lower take-home pay and restrict working arrangements that workers value, including autonomy and flexibility in working hours (Beylis et al., 2026).[2] Whether workers would accept a formal job, and on what terms, therefore matters for understanding why workers remain informal and whether policies that expand formal job offers will move workers out of informality.

In this paper, I study informal workers' willingness to accept formal employment and the role of wages and non-wage job attributes in their decisions. Disentangling workers' preferences from firm or market frictions is challenging because formalization rates reflect the decisions of both workers and firms, as well as the labor market's ability to match them. Firms choose the wages and non-wage attributes they offer alongside a formal or informal contract; workers choose among these employment packages. A worker who remains informal may never have received a formal offer or may prefer the informal job to the offers available. Observing a transition into formal employment does not resolve this problem: the transition occurs only when an offer arrives and the worker accepts it. Neither employment status nor transitions between formal and informal jobs are sufficient to distinguish between these explanations.

Separating acceptance from access requires observing whether informal workers would accept a formal contract if their current employer offered one. Low take-up would suggest that workers prefer their informal alternatives to the terms offered; high take-up would suggest that many remain informal because they lack formal offers. I approximate this

[1] In Latin America and the Caribbean, informality has remained near 55% for nearly two decades despite efforts to encourage formalization (Maloney et al., 2026).

[2] Models incorporating formal and informal employment emphasize compensating differentials (Meghir, Narita & Robin, 2015; Haanwinckel & Soares, 2021). Recent evidence documents workers' valuations of informal job attributes, formal benefits, and the insurance provided by both types of employment (Felix et al., 2026; Beylis et al., 2026; Finamor, 2025).

setting through two complementary approaches: labor inspections in Brazil that require employers to register informal workers, and a survey experiment that elicits acceptance under a guaranteed formal offer. In both settings, workers can formalize without having to find a new employer.

Brazil combines widespread informality with a rigid formal labor market. Labor costs are high (Botero et al., 2004) and, before the 2017 labor reform, formal contracts offered little scope for flexible working hours (Srebot, 2026). Informal jobs may therefore provide working arrangements that workers value but cannot obtain in formal employment. Moreover, Brazil offers a setting in which many informal workers work for registered, formal firms,[3] where labor inspections monitor compliance with the labor code. When inspectors detect unregistered workers, they require the employer to register them.[4] Paying a fine does not replace this obligation, and dismissing a worker does not remove liabilities for past informal employment. Further inspections and legal claims can impose additional costs. These incentives make registration the expected employer response to enforcement. Workers, in turn, can accept the formal contract or leave for another job, potentially informal.

I develop and estimate a binary-choice model of this decision. A worker chooses between the formal job at the inspected establishment and an informal alternative in the same labor market. I estimate separate wage and amenity coefficients for men and women and allow the residual value of the informal alternative to differ by gender. The model relates differences in take-up across establishments to differences in the formal wage premium and the attributes of the formal job.

To measure take-up of formal jobs, I use novel inspection data that record labor-code violations, penalties imposed on firms, and the number of unregistered workers detected, separately by gender. I link these records to RAIS, Brazil's matched employer–employee data for formal employment from 2003 to 2021, to observe formal hiring before and after each inspection. I compare inspected establishments with never-inspected establishments matched on size, entry cohort, industry, state, and average worker earnings. Because the inspection records identify how many informal workers were detected, I can measure take-up as the inspection-induced inflow of new formal hires relative to the number of workers subject to the registration requirement. Information on working conditions comes from RAIS and collective bargaining agreements (CBAs) in *Sistema Mediador*. These

[3] At least 40% of informal employment in Brazil is located in formal firms (Ulyssea, 2018). The corresponding shares are 56% in Mexico, excluding the self-employed, and 32% in Peru (Samaniego de la Parra, 2016; Cisneros-Acevedo, 2021); see Ulyssea (2020).

[4] I use "informal" and "unregistered" interchangeably to refer to employees without a formal labor contract, as indicated by the absence of an employer-signed labor card (*carteira de trabalho*).

agreements specify negotiated provisions that apply to formal employees, allowing me to relate take-up to the conditions workers obtain upon registration.

One limitation of the inspection approach is that I observe registrations but not the offers made to individual workers. Interpreting the estimates as take-up requires employers to offer formal contracts in response to inspections. To complement this revealed-preference evidence, I survey Brazilian workers who have held an informal job. Respondents first consider a vignette in which their employer offers to register their existing job. I elicit acceptance when employee contributions lower take-home pay and when gross pay rises enough to offset those contributions. Because the offer is guaranteed in the vignette, the response measures stated willingness to accept its terms.

The survey also includes a discrete-choice experiment with 15 tasks that independently vary wages and non-wage job attributes. I use these choices to estimate how much workers value each attribute relative to wages. This approach builds on work eliciting preferences over job attributes (Wiswall & Zafar, 2018; Maestas et al., 2023) and formal and informal employment (Felix et al., 2026; Beylis et al., 2026). Comparing these estimates with the inspection-based valuations provides a separate test of the gender differences in preferences inferred from observed formalization decisions.

I first document the employment response to inspections. Relative to matched never-inspected establishments, formal employment at inspected establishments increases by 28% in the inspection year and remains 20–25% higher five years later. Most of the increase comes from new formal hires, consistent with employers registering workers previously employed off the books.[5] These hires are concentrated in full-time jobs near the minimum wage, in line with evidence that informal employment is concentrated among lower-paid, less-skilled workers (Haanwinckel & Soares, 2021). The concentration near the minimum wage is particularly pronounced among women: jobs paying one to one-and-a-half minimum wages account for 66% of the female employment response, compared with 50% of the male response.

Most workers formalize according to the inspection-based take-up measure. Within six months, new formal hires correspond to 78% of detected informal men and 68% of detected informal women. This 10-percentage-point gender gap in take-up narrows when the comparison is restricted to inspections that detect both men and women. Moreover, women have higher take-up in some workplaces: in female-dominated industries, their formalization rate exceeds men's by 8 percentage points. In male-dominated industries, men's rate is 18 percentage points higher. The aggregate gap thus conceals substantial

[5]These records can capture both the registration of existing informal workers and recruitment from other employers.

differences across the workplaces in which men and women are employed.

To better understand take-up of formal employment, I estimate the discrete-choice model from establishment-level take-up rates, recovering men's and women's valuations of wages and non-wage job attributes. The estimates show that men and women respond similarly to the formal wage premium. Averaging the model-implied elasticities across establishments, a 10% increase in the formal wage raises take-up by 11.4% for men and 9.6% for women.[6] Differences in wage sensitivity therefore explain little of the gender gap in take-up.

Gender differences are larger for non-wage attributes. Women value collective-agreement provisions on leave and on flexibility and part-time work at about 19% and 18% of the formal wage, compared with 4% and −8% for men. These differences connect evidence on gender-specific valuations of working conditions (Mas & Pallais, 2017; Maestas et al., 2023; Corradini, Lagos & Sharma, 2025) to workers' decisions to accept formal employment. Women's take-up is also higher where more managers are women and lower where jobs are more manual, while the opposite patterns hold for men. Together, wages and non-wage attributes allow the model to reproduce women's higher take-up in female-dominated industries and men's higher take-up in male-dominated industries.

Counterfactual exercises show how these differences in valuations interact with the attributes of formal jobs at inspected establishments. In male-dominated industries, assigning women men's amenity valuations raises their predicted formalization from 54.1% to 58.2%, narrowing the gender gap by about 4 percentage points. In female-dominated industries, assigning men women's valuations lowers their predicted formalization from 61.3% to 53.7%, widening the gap by 7.6 percentage points. Increasing provisions on maternity and childcare, leave, harassment and discrimination, and flexibility and part-time work to the sample mean at establishments below it raises women's predicted formalization by about 3 percentage points, while leaving men's unchanged. Higher wages raise take-up for both genders, while changes in non-wage attributes can also narrow the gender formalization gap.

The vignette responses indicate that most workers would accept a formal contract even at a monetary cost. When employee social-security contributions lower take-home pay, 61% of respondents accept; when gross pay adjusts to offset those contributions, 79% accept.[7] Protecting take-home pay has a larger effect on women: their acceptance rises from 56.7% to 77.3%, compared with an increase from 65.6% to 80.6% among men. The gender gap

[6] The estimated wage coefficients are $\hat{\delta}_m = 3.17$ and $\hat{\delta}_w = 3.08$.

[7] I count *definitely yes* and *probably yes* as acceptance, and *not sure* as non-acceptance.

consequently falls from 8.9 to 3.3 percentage points.[8]

Beyond these job attributes, the experiment also measures how much respondents value a signed formal contract (*carteira assinada*) relative to wages. I compare this valuation with the loss in take-home pay in the inspection-style vignette to predict whether respondents would accept formalization of their existing job. Predicted acceptance exceeds stated acceptance in the vignette, particularly among women. To account for this discrepancy, the model requires a higher residual value of informal employment for women than for men. Women can therefore value formal job attributes more highly and still be less willing to leave their informal jobs.

I examine two possible explanations for this pattern. First, an Oaxaca–Blinder decomposition attributes 3.1 percentage points of the 8.9-point acceptance gap to observed characteristics, of which 2.4 points come from household and caregiving constraints. Second, I measure preferences for gig-style work—schedule flexibility, on-call pay, and working off-site—relative to formal-sector protections. Workers with a one-standard-deviation higher value of this index are 2.3 percentage points less likely to accept the formal offer, although the estimate is imprecise. Together, these results suggest that household constraints and preferences for job attributes associated with informal employment help explain why some workers decline formalization and why women are less likely than men to accept it.

The final part of the analysis asks how often workers have a chance to formalize outside inspections. I use the *Pesquisa Nacional por Amostra de Domicílios Contínua* (PNAD-C), Brazil's household labor-force survey, to measure six-month transitions from informal to formal employment with the same employer. These transitions are substantially less frequent than formalization after inspections. The comparison suggests that low formalization in the broader labor market cannot be explained by low acceptance alone.

I quantify the availability of formal offers by combining these transition rates with inspection-based take-up for each gender, state, industry, and year. The calculation assumes that workers would accept comparable offers at the same rate inside and outside the inspection setting. Under this assumption, the implied probability of receiving a formal offer within six months is 38.3% for men and 29.4% for women. Equalizing women's offer probability to men's, while holding take-up fixed, would raise women's formalization by 30%. Although offers are not directly observed, this calculation indicates that limited access to them can account for a substantial part of persistent informality and the gender gap in formalization.

[8]When take-home pay falls, 31.8% of women and 22.5% of men answer *not sure*. Refusal rates are nearly identical, and counting unsure respondents as accepting eliminates the gender gap.

Related literature and contributions. The paper contributes to the literature on informality by providing evidence on workers' acceptance of formal employment when employers are required to offer it. Existing work studies firms' compliance decisions (Ulyssea, 2018), how firms' hiring and wage-setting decisions interact with workers' search and employment choices to determine formal and informal employment in equilibrium (Meghir, Narita & Robin, 2015; Haanwinckel & Soares, 2021; Bobba et al., 2021), and workers' valuations of these alternatives (Felix et al., 2026; Beylis et al., 2026; Campos-Vazquez et al., 2025; Rajan, 2025; Finamor, 2025). Studies of enforcement estimate how inspections affect compliance, employment, and firm outcomes (Almeida & Carneiro, 2012; Samaniego de la Parra & Fernández Bujanda, 2024; do Prado, Santos & Van Doornik, 2025; Brotherhood et al., 2024; Colonna, Pérez Pérez & Rodrian, 2025). I use inspections to estimate a different object: take-up of formal employment. This allows me to move from the employment response to enforcement to a measure of worker acceptance, under the assumption that employers offer formal contracts to detected informal workers. A survey with guaranteed offers provides independent evidence on willingness to formalize.

The paper also contributes to the literature that uses workers' sorting across employers to infer how they value firms and job attributes (Sorkin, 2018; Taber & Vejlin, 2020; Lamadon, Mogstad & Setzler, 2022; Corradini, Lagos & Sharma, 2025). A related theoretical framework is provided by Card et al. (2018), who model workers' choices among employers with differentiated non-wage attributes and show how heterogeneous workplace preferences can give firms wage-setting power. I extend this approach to sorting between formal and informal employment through workers' decisions to accept a formal contract at their current employer. The advantage of this setting is that accepting the formal job does not require finding a new employer, allowing me to study valuations through changes in contract status. I compare these valuations with those elicited in a separate choice experiment, linking experimental measures of gender differences in preferences to observed formalization decisions.

Finally, I connect research on gender differences in sorting across employers (Goldin, 2014; Card, Cardoso & Kline, 2016; Morchio & Moser, 2026) to workers' choices between formal and informal employment. Existing work highlights gender differences in amenity valuations (Mas & Pallais, 2017; Wiswall & Zafar, 2018; Maestas et al., 2023), employer attachment and access to desirable jobs (Sharma, 2023), and responses to wage information (Jalal, 2026). In developing countries, these preferences and constraints also shape sorting into informality, particularly after childbirth (Berniell et al., 2021, 2023). Model counterfactuals show that greater flexibility in formal jobs can reduce this shift (Andrade et al., 2026). I bring these strands together by estimating how gender differences in amenity

valuations shape acceptance of formal contracts, distinguishing willingness to accept the jobs offered from access to formal employment.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and data. Section 3 estimates the effect of inspections on formal employment, and Section 4 constructs take-up rates and documents their heterogeneity. Section 5 develops the choice model, Section 6 presents the estimates, and Section 7 examines how wages, preferences, and job attributes shape take-up. Section 8 presents the survey evidence. Section 9 shows what broader labor-market transitions imply about workers' access to formal job offers. Section 10 concludes.

2. Setting and Data

2.1 Formal and Informal Employment in Brazil

In Brazil, salaried employment is formal when the employer registers the worker's employment relationship in the *carteira de trabalho*, a work booklet that records the worker's formal employment history. Registration places the worker under the *Consolidação das Leis do Trabalho* (CLT) and provides access to statutory benefits and protections, such as a thirteenth-month salary and paid annual vacation, while imposing payroll obligations. Salaried workers without a registered employment relationship are considered informal. Table 1 summarizes the dimensions of formal employment most relevant to workers' decisions; Appendix A provides the statutory details.

Registration changes both current compensation and the non-wage attributes of the job. At a fixed gross wage, employee social-security contributions reduce take-home pay, with statutory rates ranging from 7.5 to 14% depending on earnings.[9] In exchange, registration provides statutory benefits that include social insurance, severance and unemployment protection, paid leave, and maternity benefits. It also subjects the employment relationship to regulations governing the minimum wage, hours, paid vacations, and working conditions. Beyond these statutory provisions, formal jobs differ in the amenities provided by employers and negotiated through collective bargaining agreements. The formalization decision therefore involves trade-offs between current earnings, future protections, and non-wage job attributes that workers may value differently.

In this paper, I focus on unregistered salaried employment. About one-quarter of

[9] Before 2020, contribution rates of 8, 9, or 11% applied by earnings bracket; since 2020, contributions follow marginal rates of 7.5, 9, 12, and 14% on successive brackets. Appendix A provides details.

Brazilian private-sector employees lack a formal labor contract. Importantly, informal employment is not confined to informal firms. While informal firms are those not registered with the tax authorities and therefore lacking a tax identification number (*Cadastro Nacional da Pessoa Jurídica*, CNPJ), formal firms are registered with the tax authorities but may still employ workers without formal labor contracts. In Brazil, at least 40% of informal employment occurs within formal firms (Ulyssea, 2018). Labor inspections directly target this margin of informality.

2.2 Labor Inspections and Formalization

Labor inspections enforce establishments' compliance with labor law, covering obligations that range from wages and hours to workplace safety; inspectors also verify whether employment relationships are formally registered. The Ministry of Labor conducts more than 300,000 inspections annually through regional offices in all 27 federative units.[10] For each inspection, the administrative records identify the establishment, inspection date, violations detected, and number of workers covered. When inspectors find informal employment, they record the number of unregistered workers separately for men and women. Approximately one-quarter of inspections detect at least one informal worker. Conditional on detection, the average inspection finds 4.9 informal workers; the median detects two, and 38% detect exactly one.

When inspectors detect unregistered workers, the establishment is legally required to register them as formal employees. Registering only a subset does not resolve the violations associated with the remaining workers, and the requirement applies equally to men and women. Employers report formal employment relationships through *eSocial*, the federal government's electronic system for recording employment contracts, payroll information, and labor and social security obligations. Inspectors can use this system to verify whether workers have been registered. Establishments also remain subject to subsequent inspections: nearly half of those found employing informal workers are inspected again, and approximately 70% of these establishments have no informal workers detected at the second inspection (Appendix D).[11]

Labor inspections create strong incentives for establishments to offer formal employment to workers previously employed informally. Paying an administrative fine does not substitute for registration. Under Article 47 of the CLT, the fine for employing an

[10] Inspections originate from planned enforcement operations, worker complaints, and routine enforcement activities (Almeida & Carneiro, 2012; Coslovsky, Pires & Bignami, 2017).

[11] Among those in which informal workers are detected again, the informal share of the workforce falls by 6.4 percentage points, from 31.9 to 25.5%.

unregistered worker was fixed in nominal terms before the 2017 labor reform, so its value fell from 1.7 monthly minimum wages per worker in 2003 to 0.4 in 2017. The reform raised it to 3.1 minimum wages per worker in 2018, or 0.8 for microenterprises and small enterprises, with higher penalties for repeated violations.[12] These penalties are separate from outstanding labor liabilities, including unpaid deposits into workers' mandatory severance savings accounts (*Fundo de Garantia do Tempo de Serviço*, FGTS), social-security contributions, thirteenth-month salary, and vacation benefits.

Following an inspection, the establishment has three options. First, it can formalize and retain the worker, settling outstanding obligations and assuming the ongoing costs of formal employment, including employer contributions and statutory benefits. Second, it can dismiss the worker, but dismissal does not eliminate the original fine or liabilities from past informal employment. Dismissal without cause also entails notice pay, accrued benefits, and a severance penalty generally equal to 40% of the FGTS balance, in addition to losing the existing employment match. Third, it can continue employing the worker informally, exposing itself to additional fines, accumulating labor liabilities, and potential legal claims. Persistent noncompliance may also trigger referral to the Public Labor Prosecutor's Office or exclusion from the simplified tax regime. Appendix Figure B.1 summarizes these options, and Appendix Table B.2 details the associated costs.

Crucially, these incentives do not depend on whether the worker wishes to formalize. Even if a worker prefers informality, an agreement to remain unregistered does not protect the establishment from penalties or future labor claims. An establishment wishing to retain the worker therefore has an incentive to offer a formal contract: if accepted, it preserves the productive match while avoiding immediate dismissal costs and further noncompliance; if declined, the employment relationship may end, subject to applicable termination obligations.

These institutional features make formalization the expected response for establishments seeking to preserve existing employment relationships. Conditional on a formal continuation offer, workers face a choice between accepting formal employment with their current employer and leaving to pursue alternative employment, potentially an informal job. Because accepting formalization does not require finding a new employer, inspections provide a setting in which to study workers' willingness to formalize within an existing match. Although establishments may still choose dismissal and individual offers are not observed, the substantial costs of noncompliance and separation support interpreting

[12]The statutory fine was R$402.57 per worker before Lei 13.467/2017 and R$3,000 thereafter, or R$800 for microenterprises and small enterprises. Appendix Table B.1 reports the fine relative to the monthly minimum wage in selected years.

subsequent registrations as a measure of revealed take-up when formal employment becomes available.

I measure this response using the increase in formal hires following an inspection.

2.3 Data

I combine administrative, survey, and contractual data. The administrative records identify the establishments where informal workers are detected and allow me to follow subsequent formal employment. Household survey data provide a benchmark for informal wages, while collective agreements characterize the non-wage attributes of formal jobs. I complement these sources with a discrete-choice survey that directly measures workers' valuations of formal employment and job amenities. Appendix C provides additional details on each source and the construction of the analysis variables.

Formal employment records. I use the *Relação Anual de Informações Sociais* (RAIS), Brazil's matched employer–employee administrative data, from 2003 to 2021 (Ministério do Trabalho e Emprego, 2021). Establishments are required to report all registered employment relationships to the Ministry of Labor each year. For each job spell, RAIS identifies the worker and establishment and records earnings, occupation, hours, tenure, and worker characteristics including gender, age, and education, as well as establishment location and industry. I use the worker identifiers to follow workers across registered jobs and aggregate the records to construct establishment–year measures of formal employment, hiring, earnings, and workforce composition. Most importantly for the analysis, RAIS allows me to measure formal registrations at inspected establishments following an inspection.[13]

From the RAIS records, I construct three additional measures. The first is an establishment-level measure of revealed firm value based on worker-to-worker mobility, following the PageRank approach of Sorkin (2018). I also construct occupation-level task measures using O*NET and industry-level female employment shares.[14] Together, these measures characterize revealed firm value, job content, and the gender composition of workers' labor markets.

[13]Because the inspection records identify the number of informal workers detected but cannot be linked to individual RAIS records, I interpret subsequent registrations at the establishment as formalization of the workers detected during the inspection.

[14]O*NET provides detailed information on the tasks, skills, and work activities associated with U.S. occupations. I map O*NET occupations to Brazilian occupational codes using occupation titles and descriptions and merge the resulting measures with workers in RAIS. See https://www.onetcenter.org/database.html.

Labor inspections. I link RAIS to administrative records covering labor inspections conducted by the Ministry of Labor between 1995 and 2023. For each enforcement action, the records identify the inspected establishment, the inspection date, the violations detected, and the number of workers covered by the inspection. When inspectors detect informal employment, the data report the number of unregistered workers separately for men and women. The records also identify repeat inspections, allowing me to track subsequent enforcement and compliance. I define the enforcement event as the establishment's first inspection and restrict the sample to establishments found employing at least one informal worker at that inspection. The matched analysis focuses on inspections conducted between 2005 and 2018. Combining the gender-specific numbers detected at the inspection with subsequent registrations in RAIS yields the formalization measures studied in Section 4. Appendix D describes inspection coverage, repeat inspections, compliance, and the construction of the estimation samples.

The informal employment benchmark. I use PNAD-C (IBGE, 2023), Brazil's nationally representative quarterly household labor-force survey, to measure wages and job amenities in informal employment. I restrict the sample to private-sector salaried workers aged 18–65 between 2012 and 2022. Using these data, I construct local informal wage benchmarks and measures of non-wage job attributes in informal employment—the part-time share, female representation in management, and the manual and interpersonal task content of jobs—for the discrete-choice model. These measures characterize the informal employment alternatives against which workers evaluate formalization at their current establishment.

Collective bargaining agreements. I use collective bargaining agreements registered in *Sistema Mediador*, the Brazilian Ministry of Labor's official registry, from 2009 to 2021 to measure negotiated job amenities. The data include the negotiating parties, geographic coverage, effective dates, and full text of each agreement. The Ministry of Labor classifies negotiated clauses into 137 categories, which I aggregate into broader groups capturing family-related benefits, working-time arrangements, workplace protections, and other employment conditions (Appendix Table C.1). Appendix B.3 describes the collective bargaining system and the procedure used to match agreements to establishments.

Survey experiment. Finally, I conducted a survey with 1,501 Brazilian workers through Netquest between June and August 2026. The survey includes 15 binary-choice tasks that vary wages and job attributes to estimate willingness to pay for formal employment and specific amenities. It also includes a vignette that mirrors the inspection setting, asking

respondents whether they would remain with their current employer if their informal job were converted into registered employment. Of the 1,501 respondents, 1,298 pass the attention screens, and 942 of these report having ever held an informal job. Appendix K provides the survey questionnaire, sampling procedures, experimental design, descriptive statistics, and details on the construction of survey outcomes.

3. Estimating the Impact of Labor Inspections on Formal Employment

3.1 Empirical Strategy

I compare inspected establishments with matched never-inspected establishments around the inspection year using pre-inspection size, entry cohort, industry, state, and average monthly earnings.[15]

For establishment i in year t, let T_i denote its inspection year and define

$$R_{it}^k = \mathbf{1}\{t = T_i + k\}.$$

For matched controls, T_i is the inspection year of the corresponding inspected establishment. I estimate

$$y_{it} = \alpha_i + \delta_t + \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k R_{it}^k + \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \theta_k \left(R_{it}^k \times Inspected_i \right) + \varepsilon_{it}, \tag{1}$$

where α_i and δ_t denote establishment and calendar-year fixed effects, respectively, and $Inspected_i$ equals one for inspected establishments and zero for matched controls. The coefficients of interest are θ_k.[16] The key identifying assumption is that outcomes at inspected and matched control establishments would have followed parallel trends in the absence of an inspection. Under this assumption, and given the normalization $\theta_{-1} = 0$,

[15] Using later-inspected establishments as controls conditions on remaining active in the formal market until inspection and may therefore confound event-time dynamics with differential selection. Matched never-inspected establishments provide an explicit counterfactual without requiring subsequent inspection. Appendix F reports an alternative specification using ever-inspected establishments; the results are qualitatively similar.

[16] The omitted period is $k = -1$, endpoints are binned at $|k| \geq 5$, the estimation window is $k \in [-7, 7]$, and standard errors are clustered at the firm level.

the coefficients θ_k for $k \geq 0$ capture the effect of the inspection on the outcome k years after enforcement, relative to the year before the inspection. Although this identifying assumption cannot be tested directly, I evaluate whether parallel trends hold in the years leading up to the inspection by examining the coefficients θ_k for $k < 0$, restricting the analysis to matched pairs observed at event time -3.

3.2 Matched Event-Study Sample

I construct the event-study sample from private-sector establishments whose first inspection, conducted between 2005 and 2018, detected at least one informal worker. Establishments must appear in RAIS in both the inspection year and the preceding year. To avoid contamination within firms, I exclude from the comparison pool any establishment belonging to a firm with an inspected site.

For each inspected establishment, I select one control establishment that was never inspected during the sample period. Controls are matched to inspected establishments based on pre-inspection employment size, year of first appearance in RAIS, industry, state, and average monthly earnings. I assign each control the inspection year of its matched treated establishment, so that both are observed over the same event-time window. The resulting matched sample contains 172,974 treated–control pairs and 3,005,668 establishment–year observations.

The main event-study sample further requires both establishments in a pair to be observed at event time -3, leaving 141,887 pairs. The structural analysis additionally requires wage, amenity, and local labor-market measures and includes 17,908 inspected establishments. Appendix Table D.3 traces these restrictions.

Table 2 compares inspected and never-inspected establishments before and after matching. Before matching, inspected establishments are substantially larger and differ along several other dimensions. Matching sharply reduces these differences, including in employment, wages, contracted hours, education, and the pre-inspection gender wage gap.[17] Appendix Table D.2 reports balance separately by inspection cohort.

3.3 Formal-Sector Activity and Employment

Establishment survival in the formal labor market. Figure 1 examines whether inspected establishments remain active in the formal labor market, as measured by their continued

[17] Given the large sample, I assess balance based on the economic magnitude of the remaining differences rather than their statistical significance.

appearance in RAIS with positive employment. The estimates show no evidence of differential pre-trends. After the inspection, inspected establishments are 5.6 percentage points more likely than their matched controls to remain active in the formal market. The gap rises to 7.1 percentage points by year four and remains at 6.5 percentage points in year five.[18] One potential explanation is that inspected establishments remain under closer scrutiny: an initial inspection increases the likelihood of subsequent inspections, as documented in Appendix D. This continued exposure to enforcement strengthens incentives to comply with labor regulations and maintain formally registered employment, rather than exiting the formal market or returning to informality.

This pattern is not specific to the matched sample. Appendix Figure E.1 plots the share of establishments observed in the administrative records by event time for the matched inspected establishments used here and for the universe of inspected establishments subject to the same pre-inspection presence requirement. The two series track each other closely, so the matching step does not select establishments with unusual survival in the formal sector. Appendix Figure E.6 reports the same activity outcome separately for all inspection cohorts and for establishments first inspected in 2012 or later. The post-inspection effects are somewhat smaller for the more recent cohorts but remain positive and persistent. As an additional robustness check, Appendix Figure E.12 shows that changes in establishment location remain uncommon and, if anything, decline slightly after inspection, suggesting that the activity results are not driven by establishments relocating. Appendix E reports the remaining outcomes under the same cohort split.

Formal employment. Inspected establishments are also more likely to expand formal employment, with large and persistent increases. Figure 2 shows that total annual formal employment increases by approximately 28% in the inspection year relative to matched controls. The effect remains economically large through year five. Although the pre-inspection coefficients are not uniformly zero, they are small relative to the discrete increase at the time of the inspection. I therefore interpret the evidence as showing a sharp and persistent expansion in formal employment following enforcement.[19]

New formal hires. I define a new formal hire as a worker registered at the inspected establishment after the inspection who had no active formal contract at any employer in the calendar month immediately preceding the inspection. The increase in new formal

[18]The inspection-year coefficient is mechanically zero because both establishments in each matched pair must appear in RAIS with positive employment during the inspection year.

[19]Appendix Figure E.7 reports the results separately for all inspection cohorts and for establishments first inspected in 2012 or later. The estimated patterns are broadly comparable across the two samples.

hires is concentrated around the inspection. Figure 3 reports these hires as a fraction of pre-inspection employment over progressively wider post-inspection windows. Across the four windows, inspected establishments register additional workers equivalent to 19–31% of their pre-inspection workforce relative to matched controls. Figure 4 reports the corresponding effects in counts. Inspected establishments register approximately 1.4 additional workers within one month of the inspection, with the cumulative response reaching approximately 2.9 workers by the end of the inspection year. Under both measures, the estimated effects in subsequent years are statistically indistinguishable from zero.

The concentration of new registrations around the inspection is consistent with establishments formalizing workers detected as informal rather than permanently increasing their hiring rates. Because the inspection records and RAIS cannot be linked at the worker level, this interpretation requires that the excess registrations correspond to workers detected during the inspection. The event-time patterns provide additional support for this interpretation: the windowed measures show no evidence of differential pre-trends.

Three additional exercises, reported in Appendix F, support this interpretation. First, a treated-only event study using all establishments ever found with informal workers reproduces the sharp inspection-year increase in formal registrations, indicating that the result is not driven by the matched sample. Second, new formal hires also increase at uninspected establishments belonging to the same firm, although by less than at the inspected establishment. This spillover suggests that part of the adjustment occurs at the firm level and that estimates based solely on the inspected establishment may not capture the full response. Third, inspections that identify violations unrelated to informal employment also generate a modest increase in formal hiring. Contact with enforcement authorities may therefore affect registration behavior even in the absence of a recorded informality violation. The substantially larger response when inspectors detect informal workers, however, is consistent with a distinct response to the requirement that those workers be formally registered.

3.4 Hiring Responses by Gender

Does enforcement formalize women's and men's jobs at the same rate? Formalization and the gender gap in formalization are the outcomes at the center of the paper, and the gender-specific hiring responses estimated here are the ingredients from which the take-up rates in Section 4 are built. I therefore estimate the response separately for inspections that detect informal women and informal men. Figure 5 reports additional formal hires within six months of the inspection, expressed as a fraction of pre-inspection employment

in the top row and in counts in the bottom row. Over this window, inspections detecting informal women generate an estimated 1.50 new formal female hires, compared with 1.99 new formal male hires when informal men are detected. Because these estimates are based on different sets of establishments, I normalize the effects by pre-inspection employment to make them comparable. The resulting increases are 0.210 for women and 0.233 for men, a difference of 2.3 percentage points.[20] Appendix Figures E.2 and E.3 report estimates for alternative windows, expressed as employment shares and counts, respectively. Section 4 examines heterogeneity in these responses across establishments and industries.

3.5 Distributional Effects

Which jobs do inspections add to the formal payroll? The increase in registered employment could in principle arise anywhere in the earnings, hours, and occupation distributions, and where it is concentrated determines which workers face the formalization trade-offs at the center of the paper. To answer this question, I partition workers by earnings, contracted hours, and occupation. I estimate

$$\begin{aligned}\frac{n_{ibt}}{\bar{L}_i} = &\sum_{b=1}^{B}\sum_{\substack{k=-5\\k\neq-1}}^{5}\beta_{bk}\left[R_{it}^{k}\times\mathbf{1}\{\text{bin}=b\}\right]\\ &+\sum_{b=1}^{B}\sum_{\substack{k=-5\\k\neq-1}}^{5}\theta_{bk}\left[Inspected_i\times R_{it}^{k}\times\mathbf{1}\{\text{bin}=b\}\right]+\mu_{ib}+\rho_{bt}+u_{ibt},\end{aligned} \tag{2}$$

where n_{ibt} is the number of workers in bin b at establishment i in year t, $\bar{L}_i$ is formal employment in the year before the inspection, and μ_{ib} and ρ_{bt} denote establishment-by-bin and bin-by-calendar-year fixed effects, respectively. As above, $R_{it}^{k} = \mathbf{1}\{t = T_i + k\}$, with matched controls assigned the inspection year of their inspected counterpart. The specification jointly estimates a separate event study for each bin, with $k = -1$ omitted. The coefficients θ_{bk} capture the differential change in employment in bin b at event time k for inspected establishments relative to matched controls. Figure 6 reports the inspection-year coefficients θ_{b0}, and Figure 7 reports the earnings and hours estimates separately by gender.

The increase in registered employment is concentrated at the bottom of the formal earnings distribution. The number of workers earning between one and one-and-a-half

[20] Newly formalized jobs are short-lived for both genders: after one year, 33% of newly formalized men and 37% of newly formalized women remain at the inspected establishment, compared with about half of incumbents, a pattern consistent with separation rates declining with tenure (Appendix F.4).

minimum wages, expressed as a share of pre-inspection employment, increases by 28.7 percentage points in the inspection year. The corresponding increases are 4.8 percentage points below one minimum wage and 10.0 percentage points between one-and-a-half and two minimum wages. The estimates decline sharply above two minimum wages, indicating that the expansion of the formal workforce is concentrated near the bottom of the earnings distribution.[21]

This concentration is present for both genders but is more pronounced among women. The one to one-and-a-half minimum wage bin accounts for 66% of the female response and 50% of the male response. The two lowest earnings bins together account for 78% of the female response and 58% of the male response. Conversely, earnings above two minimum wages account for 8% of the female response and 18% of the male response.

The increase in registered employment is concentrated in full-time jobs. The number of workers contracted for at least forty hours per week, expressed as a share of pre-inspection employment, increases by 46.5 percentage points in the inspection year and accounts for 91% of the response across hours bins. The concentration in full-time employment is weaker among women: jobs with fewer than thirty weekly hours account for 11% of the female response, compared with 5% of the male response. This difference is consistent with part-time work being more prevalent among informal women than among informal men. Among workers assigned to an occupational group, low-skill white-collar jobs account for 49% of the increase, blue-collar jobs for 40%, and high-skill white-collar jobs for the remaining 11%.

Inspections also broaden the range of occupations held by registered workers. The number of distinct six-digit occupations increases by 0.55 in the inspection year, or 12% of the pre-inspection mean, and remains high in subsequent years (Appendix Figure E.13). This corresponds to roughly one occupation not previously registered for every five new formal hires. Together with the concentration in low-skill white-collar and blue-collar jobs, this suggests that formalization brings some previously unregistered positions onto the payroll, rather than simply adding workers to existing formal occupations.

Taken together, the estimates show that inspections primarily add formally registered workers in full-time, lower-skill jobs near the minimum wage. I also examine whether these induced formalization patterns affect the measured gender pay gap: the establishment-level gap widens where detected informal workers are disproportionately female, reflecting

[21] Using the same event-study design with log average monthly earnings as the outcome, I estimate a decline of approximately 1–2% following the inspection. This modest response is consistent with related evidence: Samaniego de la Parra & Fernández Bujanda (2024) find no detectable short-run wage effect following randomized inspections in Mexico, while do Prado, Santos & Van Doornik (2025) estimate an approximately 1% decline in average formal wages in the inspection year in Brazil.

the concentration of women's new registrations near the wage floor—a composition effect rather than a change in individual wages (Appendix E.3).

4. Worker Take-Up of Formal Contracts

4.1 Measurement

I define the formalization rate as the number of formal hires an inspection generates per informal worker it detects, and I compute it separately by gender using the estimated inspection-year hiring responses. For gender d, let H_{it}^d be new formal hires of gender d at establishment i in year t; estimating equation (1) on H_{it}^d gives $\hat{\theta}_0^d$, the inspection-year jump relative to matched controls. Let $\bar{N}_d$ be the mean number of informal workers of gender d recorded at treated establishments at event time zero. The formalization rate for gender d is then

$$\hat{\lambda}_d = \frac{\hat{\theta}_0^d}{\bar{N}_d}. \tag{3}$$

I report $\hat{\lambda}_d$ as a percentage of detected informal workers, although it is formally a ratio rather than a probability and can therefore exceed one. Interpreting it as a take-up rate requires that, after differencing relative to matched controls, the additional hires within the window correspond to workers detected during the inspection rather than other workers hired in response to the inspection. The difference-in-differences numerator removes common seasonal hiring and employment growth, while narrow post-inspection windows limit contamination from unrelated hires.[22]

To assess sensitivity to the timing of registration, I define new formal hires over four windows: admission to the inspected establishment within one, three, or six months of the inspection, or during the remaining months of the inspection year. Each measure includes only workers with no active formal contract anywhere in Brazil in the month preceding the inspection, thereby excluding workers moving from other formal jobs.[23]

[22]This interpretation also assumes that the number of informal workers recorded by inspectors accurately measures the workers subject to formalization.

[23]I estimate the gender gap, $\lambda_w - \lambda_m$, by stacking the gender-specific outcomes after scaling each by its corresponding number of detected informal workers.

4.2 Take-Up of Formal Contracts Is High but Incomplete

Table 3 reports estimates of equation (3) across the four new-hire definitions and shows that formalization is rapid but incomplete. Columns (1)–(4) estimate gender-specific rates using inspections that detected at least one informal worker of the relevant gender. In Panel A, take-up within one month is 45.6% for men and 43.1% for women; within six months, it reaches 77.6 and 67.8%, respectively. The gender gap therefore widens from 2.5 to 9.8 percentage points. Columns (5)–(8) restrict the sample to establishments in which inspectors detected both informal women and men. Within these establishments, take-up is similar across genders, with a six-month gap of only 1.6 percentage points. Panel B restricts to inspections from 2012 onward—the cohort used in the structural estimation, for which the informal wage benchmark and collective-agreement data are available (Section 5.4)—and shows lower take-up but the same broad pattern.[24]

4.3 The Gender Gap Across Establishments and Industries

The aggregate gender gap arises across rather than within establishments. Table 3 shows that women's six-month take-up is 9.8 percentage points lower than men's across all inspections, but only 1.6 points lower among establishments where inspectors detect informal workers of both genders. The latter difference is statistically indistinguishable from zero, including in the post-2012 sample. Lower take-up among women could reflect fewer formal offers rather than lower willingness to accept them; repeat inspections provide a partial test, since establishments found with no informal workers at a second inspection are those that brought their workforce into compliance. Even among these complying establishments, the gender gap persists overall but disappears where inspectors initially detected both informal women and men, providing no evidence of selective formalization of men within complying workplaces (Appendix G.4).

Table 4 shows that the gap reverses across industries. Columns (1)–(4) report estimates for female-dominated industries. In Panel A, women's six-month take-up exceeds men's by 7.9 percentage points. Columns (5)–(8) show that women's take-up is instead 18.4 points lower in male-dominated industries. Panel B documents the same reversal after 2012, with

[24]The shortfall from full registration—take-up remains below one through the end of the inspection year—is unlikely to be mechanical: establishment closure would lower the ratio, yet inspected establishments are more likely than their matched controls to remain active, and undercounting of informal workers would raise it. Turnover can lower it when detected workers obtain formal jobs elsewhere; the estimates thus measure whether the observed informal match becomes formal at the same establishment, not whether the worker obtains any formal job. Appendix G.1 examines these alternatives.

gaps of 10.9 and −13.7 percentage points, respectively. Establishment-level differences are similar: women's six-month take-up is 21.8 points lower where the pre-inspection gender pay gap is large, compared with 2.7 points lower where it is small.[25]

The gender gap in take-up is therefore concentrated across establishments and reverses with industry gender composition. This pattern points to establishment-level job attributes that women and men may value differently; I examine this mechanism in Section 5.

4.4 Heterogeneity in Take-Up

The descriptive evidence points to substantial heterogeneity in formalization across establishments, industries, and local labor markets. First, the aggregate gender gap depends on the composition of enforcement. Appendix Figure G.1 shows that the share of detected informal workers who are women increases noticeably after 2014, as inspections become more concentrated in female-dominated industries. This shift matters because the gender gap reverses across industries: women have higher take-up than men in female-dominated industries, while men have higher take-up in male-dominated industries (Table 4). Reweighting enforcement toward female-dominated industries therefore narrows the aggregate gender gap. This helps explain why the gap is smaller in the post-2012 sample even though the within-industry pattern remains similar.[26]

Second, take-up varies systematically across local labor markets. Formalization rates are higher in the South and Southeast and lower in the North and Northeast (Appendix Figure G.2), a gradient that mirrors regional informality rates: take-up is lowest precisely where informal employment is most prevalent. This pattern is consistent with the informal outside option shaping formalization—where informal work is widespread, remaining informal is easier and the value of a formal contract at the inspected establishment is correspondingly lower.

Taken together, these patterns sharpen the main descriptive result. Formalization after inspections is high but incomplete, and women have lower take-up on average. Yet the gender gap disappears when men and women are compared within the same establishment and reverses across female- and male-dominated industries. Moreover, its aggregate magnitude changes as enforcement shifts across industries, while overall take-up is lower where informal employment is more prevalent. The workers who remain informal after

[25] Appendix Table G.1 reports estimates by the establishment's pre-inspection gender pay gap.

[26] This change in enforcement composition motivates including inspection-year fixed effects throughout the analysis and restricting the structural estimation to the post-2012 period. Pooling inspections across years without accounting for changes in targeting would conflate changes in take-up with changes in the establishments and workers exposed to enforcement.

enforcement are therefore systematically selected across jobs and local labor markets. These facts motivate the model below, which allows the value of formalization to depend on both formal job attributes and the informal outside option.

5. A Model of the Formalization Decision

The evidence in the previous section shows that take-up of formal contracts is high but incomplete and that gender differences emerge primarily across establishments, industries, and regions. These patterns suggest that formalization depends on the characteristics of the job being offered and the informal alternative workers would forgo. I formalize this idea with a binary-choice model in which workers compare the wages and non-wage attributes of the offered formal job with the value of remaining informal. The random-utility formulation follows the differentiated-workplace approach of Card et al. (2018), in which workers value both pay and non-wage job attributes. Here, I model acceptance of a formal contract within an existing match, taking the offered wage and amenities as given rather than solving for employers' wage-setting decisions.

5.1 Environment

Establishment j operates in industry–state market $s(j) = (z(j), r(j)) \in \mathcal{S}$.[27] When an inspection detects N_{dj} informal workers of gender $d \in \{m, w\}$, the establishment must offer formal registration. As Figure 8 illustrates, each worker then either continues the match as a registered employee or takes an informal alternative. The formal job pays wage w_{dj} and provides amenities $\mathbf{a}_j$. The informal alternative pays the market wage $\pi_{ds(j)}$, provides amenities $\bar{\mathbf{a}}_{s(j)}$, and has residual value γ_d—the net value of informality beyond wages and measured amenities, such as government transfers, search opportunities, and other differences the data do not capture.

The informal branch represents moving to informal work elsewhere in the same labor market. Remaining informal at the inspected establishment is particularly costly for the employer after detection, given registration obligations and the risk of further penalties. Subsequent inspections detect informal workers in only 30% of the matched establishments inspected at least twice (Appendix Table D.1), although these records do not identify whether they are the same workers. Because the data do not reveal establishment-specific

[27] I use "establishment" and "firm" interchangeably throughout this section.

informal wages or workers' subsequent informal jobs, I model the outside option as informal employment elsewhere in the same market. I use "acceptance" to denote subsequent formal employment at the inspected establishment and "rejection" to denote its absence; these labels describe observed outcomes.

5.2 The Formalization Choice

The utility of worker i of gender d from formal employment at inspected establishment j and from informal employment in labor market $s(j)$ is

$$\begin{aligned} u_{ij}^F &= \delta_d w_{dj} + \mathbf{a}_j' \boldsymbol{\beta}_d + \varepsilon_{ij}^F, \\ u_{ij}^I &= \delta_d \pi_{ds(j)} + \bar{\mathbf{a}}_{s(j)}' \boldsymbol{\beta}_d + \gamma_d + \varepsilon_{ij}^I, \end{aligned} \tag{4}$$

where w_{dj} is the formal wage at establishment j, $\pi_{ds(j)}$ is the wage available in the informal market, and $\mathbf{a}_j$ and $\bar{\mathbf{a}}_{s(j)}$ are the corresponding amenity bundles. Both wages are measured in logs, with the log operator suppressed. The parameter $\delta_d > 0$ captures the effect of log wages on utility; it is not an elasticity of formalization. Because the model is estimated separately for men and women, δ_d and the amenity coefficients $\boldsymbol{\beta}_d$ are identified relative to each gender's scale of unobserved utility and cannot be directly compared across genders without an additional assumption on that scale. I therefore compare amenity valuations in wage units, using $\boldsymbol{\beta}_d/\delta_d$, which removes this scale dependence.[28] The parameter γ_d is the residual value of the informal alternative introduced above.

Assuming that ε_{ij}^F and ε_{ij}^I are independently distributed Type-I Extreme Value, the probability that the worker appears as formally employed at the inspected establishment is

$$\lambda_{dj} = \Pr\left(u_{ij}^F \geq u_{ij}^I\right) = \frac{\exp(V_{dj}^F)}{\exp(V_{dj}^F) + \exp(V_{dj}^I)}, \tag{5}$$

where V_{dj}^F and V_{dj}^I denote the deterministic components of formal and informal utility. Define the log-odds $\tilde{\lambda}_{dj} \equiv \ln[\lambda_{dj}/(1-\lambda_{dj})]$, the formal wage premium $\Delta w_{dj} \equiv w_{dj} - \pi_{ds(j)}$, and the net amenity gain $\Delta \mathbf{a}_j \equiv \mathbf{a}_j - \bar{\mathbf{a}}_{s(j)}$. Equation (5) then implies

$$\tilde{\lambda}_{dj} = \underbrace{\delta_d \Delta w_{dj}}_{\text{wage premium}} + \underbrace{\Delta \mathbf{a}_j' \boldsymbol{\beta}_d}_{\text{amenity gain}} - \underbrace{\gamma_d}_{\text{informal value}}. \tag{6}$$

[28] With wages measured in logs, the relative monetary value in equation (10) expresses an amenity's value as a proportional wage equivalent.

The logit assumption makes formalization log-odds linear in the wage and amenity gains from registration relative to the informal option. This permits estimation from establishment-level formalization rates even though individual choices are not observed.

The wage premium. I construct the gender-specific wage premium by comparing the formal wage offered at establishment j, measured in RAIS, with the wage a comparable worker earns in informal employment in the same industry–state market, measured in PNAD-C. The formal component, ψ_{dj}, is the establishment effect from a two-way fixed-effects wage regression (AKM; Abowd, Kramarz & Margolis, 1999). For the informal benchmark, I estimate an analogous worker fixed-effects regression in PNAD-C, replacing the establishment effect with an employment-status-by-industry-by-state effect.[29] The omitted category is formal employment in restaurants and food services, providing a common normalization across the two estimates. Letting $\pi_{ds(j)}$ denote the informal wage component for gender d in the industry–state market of establishment j, the formal wage premium is

$$\Delta w_{dj} = \psi_{dj} - \pi_{ds(j)}. \tag{7}$$

Appendix H describes the estimation samples, normalization, connected-set diagnostics, and alternative wage measures.

The amenity bundle. The estimation includes four establishment characteristics measured in RAIS—the part-time share, female representation in management, and the nonroutine-manual and nonroutine-interpersonal task content of the occupational mix—as well as the fifteen collective-agreement clause groups defined in Appendix Table C.1. For each RAIS amenity, I construct an informal-market counterpart from PNAD-C, which records each worker's occupation, contractual hours, and gender alongside formality status; the part-time share, the female share among workers in management occupations, and the occupation-based task measures are therefore computed for informal employees in each industry–state market (Appendix C). The net amenity gain is

$$\Delta a_{\ell j} = a_{\ell j} - \bar{a}_{\ell s(j)}, \qquad \bar{a}_{\ell s(j)} = \mathbb{E}\big[a_\ell \mid s(j), \text{informal}\big], \tag{8}$$

[29] Because PNAD-C does not identify employers, the wage component is defined at the employment-status-by-industry-by-state level. Employment status indicates whether a private-sector employee has a signed *carteira de trabalho*; self-employed workers, employers, and unpaid family workers are excluded. With worker fixed effects, the component is identified from employees who change registration status while observed in the rotating panel.

where $a_{\ell j}$ is the amenity at establishment j and $\bar{a}_{\ell s(j)}$ is its mean among informal employees of both genders in the same industry–state market, pooled over 2012–2022. Collective-agreement provisions have no informal counterpart and enter in levels.

The informal benchmark is constant within an industry–state market. In the main specification, which includes industry-by-state fixed effects (Section 5.4.2), it is therefore absorbed. Identification of $\boldsymbol{\beta}_d$ comes from differences in formal amenities $a_{\ell j}$ across establishments within the same industry–state market, holding fixed the informal alternative they face.

Because these measures do not capture every dimension of job quality, the wage premium may partly reflect compensating differentials for unobserved amenities, as documented in Brazilian matched employer–employee data by Morchio & Moser (2026). I assess this concern using a revealed-preference measure of establishment value constructed from worker mobility following Sorkin (2018), estimated separately for men and women. Its correlation with the corresponding gender-specific wage premium is modest—0.16 for men and 0.15 for women, falling to 0.11 and 0.10 after removing the fixed effects in the estimating equation. Appendix H reports this comparison. As a further robustness check, Appendix Table J.6 controls for the component of revealed establishment value that is orthogonal to the wage premium and re-estimates δ_d. This allows the wage coefficient to be estimated conditional on revealed non-wage job quality.

5.3 Willingness to Pay and the Gender Gap

The model maps wage and amenity valuations into formalization responses. Differentiating equation (5) yields

$$\frac{\partial \lambda_{dj}}{\partial \Delta w_{dj}} = \delta_d \lambda_{dj}(1-\lambda_{dj}), \qquad \frac{\partial \lambda_{dj}}{\partial a_{\ell j}} = \beta_{d\ell} \lambda_{dj}(1-\lambda_{dj}). \tag{9}$$

The wage elasticity of formalization is therefore $\delta_d(1-\lambda_{dj})$. Because the wage and amenity derivatives share the same logistic term, their ratio does not depend on the establishment's formalization rate. I express the value of amenity ℓ through its Relative Monetary Value (RMV), the proportional reduction in the formal wage that leaves workers indifferent:

$$\mathrm{RMV}_{d\ell} \equiv 1 - \exp\left(-\frac{\beta_{d\ell}}{\delta_d}\right) \approx \frac{\beta_{d\ell}}{\delta_d}. \tag{10}$$

The RMV is positive for an amenity and negative for a disamenity.[30]

Comparing men and women. The logit identifies utility parameters relative to the scale of unobserved taste shocks. With Type-I Extreme Value shocks of scale σ_d, the estimated coefficients are δ_d/σ_d and $\boldsymbol{\beta}_d/\sigma_d$. Because I estimate the model separately by gender, differences in coefficients may reflect either preferences or unobserved dispersion. I therefore compare amenity valuations in wage units, as in equation (10), where the scale cancels. I also compare model-implied wage elasticities and effects on formalization probabilities.

With the model estimated separately by gender, the predicted gender gap in formalization log-odds decomposes into wage, amenity, and informal-alternative components:

$$\begin{aligned}\tilde{\lambda}_{wj} - \tilde{\lambda}_{mj} = \underbrace{\delta_w \Delta w_{wj} - \delta_m \Delta w_{mj}}_{\text{wage channel}} \\ + \underbrace{\Delta \mathbf{a}_j'(\boldsymbol{\beta}_w - \boldsymbol{\beta}_m)}_{\text{amenity channel}} - \underbrace{(\gamma_w - \gamma_m)}_{\text{informal alternative}} .\end{aligned} \tag{11}$$

Each term is expressed in log-odds and uses the estimated, scale-normalized coefficients. The wage channel combines differences in wage premia with differences in their estimated effects on formalization. The amenity channel captures how the same amenity bundle affects men's and women's formalization differently. The final term captures the difference in the normalized residual contribution of the informal alternative. These components describe differences in predicted choice behavior, rather than differences in underlying preferences alone. I compare valuations across genders using RMVs, which remove the scale dependence.

5.4 Identification and Estimation

5.4.1 Unit of Observation and Outcome

The unit is an establishment–gender pair entering at most once in its inspection year, so the sample is a cross-section.[31] The outcome is the log-odds of the establishment-level take-up

[30] Because the RMV is a nonlinear function of estimated coefficients, I construct confidence intervals using a cluster bootstrap.

[31] Restricting to inspections from 2012 onward is forced by the inputs: collective agreements are digitized from 2009 and the PNAD-C informal benchmark begins in 2012.

rate:

$$\hat{\lambda}_{dj} \;=\; \frac{h_{dj}(0) - h_{dj}(-1)}{N_{dj}}, \tag{12}$$

where $h_{dj}(e)$ counts new formal hires of gender d in event year e who held no active formal contract in the month preceding the inspection, and N_{dj} is the number of informal workers of gender d detected. I use the single difference to avoid introducing idiosyncratic hiring at the matched control into the establishment-level outcome. Section 3 shows that the inspection-year increase is attributable to the inspection rather than seasonality. The baseline specification uses new hires within six months of the inspection; Appendix J.5 shows that the estimates are similar using a three-month window or a window extending through the end of the inspection year.

Because $\hat{\lambda}_{dj}$ is a ratio of estimated new formal hires to informal workers detected, it need not lie within the unit interval: it equals zero when the inspection generates no new hires and exceeds one when estimated hiring is greater than the number of workers detected. Nonetheless, 71.4% of rates in the men's sample and 71.1% in the women's sample lie between zero and one, inclusive; the remaining 28.5% and 28.8%, respectively, exceed one, and none are negative. To define the log-odds outcome, I winsorize rates at 0.001 and 0.999, leaving values inside the unit interval unchanged.

5.4.2 Accounting for the Informal Outside Option

I estimate equation (6) separately by gender using two definitions of the informal outside option. The first enters the constructed wage premium directly:

$$\tilde{\lambda}_{dj} = \delta_d \Delta w_{dj} + \Delta \mathbf{a}_j' \boldsymbol{\beta}_d + \mathbf{c}_j' \boldsymbol{\beta}_d^c + \mu_{dz(j)} + \mu_{dr(j)} + \mu_{dt(j)} + \eta_{dj}, \tag{13}$$

where $\Delta \mathbf{a}_j$ contains the four formal-minus-informal amenity differences, $\mathbf{c}_j$ contains the fifteen CBA clause-group indicators, total clause intensity, and CBA coverage, and $\mu_{dz(j)}$, $\mu_{dr(j)}$, and $\mu_{dt(j)}$ are gender-specific industry, state, and inspection-year fixed effects. This specification maps directly into the model but requires estimating the informal wage component $\pi_{ds(j)}$.

The main specification instead enters the formal establishment wage component, ψ_{dj}, and absorbs the informal wage component with industry-by-state fixed effects:

$$\tilde{\lambda}_{dj} = \delta_d \psi_{dj} + \Delta \mathbf{a}_j' \boldsymbol{\beta}_d + \mathbf{c}_j' \boldsymbol{\beta}_d^c + \mu_{z(j)\times r(j)} + \mu_{t(j)} + \eta_{dj}. \tag{14}$$

Because $\pi_{ds(j)}$ and $\bar{\mathbf{a}}_{s(j)}$ are constant within an industry–state market, $\mu_{z(j)\times r(j)}$ absorbs

both, without introducing their sampling error into the regressors. This specification also retains establishments for which the informal wage component is unavailable, increasing the estimation sample by approximately 60%. The trade-off is that the residual value of informality, γ_d, is absorbed by the fixed effects and cannot be recovered from the intercept. The wage and amenity coefficients remain identified from within-market variation. I therefore use equation (14) as the main specification and report equation (13) as a complementary estimate.

5.4.3 Measurement Error and the Instrument

The primary concern with estimating equation (14) by OLS is measurement error in the establishment wage premium. The premium ψ_{dj} is estimated from workers of gender d employed at establishment j, and establishments in the inspection sample are typically small. The resulting estimation error attenuates $\hat{\delta}_d$ toward zero. Because women are a minority of workers at most establishments, the female premium is generally estimated from fewer observations, potentially producing greater attenuation in $\hat{\delta}_w$ than in $\hat{\delta}_m$. A lower estimated wage coefficient for women could therefore arise mechanically from differential measurement error. Because RMVs divide each amenity coefficient by the wage coefficient, this attenuation would also inflate women's estimated valuations relative to men's.

To address this concern, I instrument each gender's wage premium with the other gender's premium at the same establishment. The two premia reflect a common establishment wage policy but are estimated from nonoverlapping sets of workers. The other gender's premium is therefore correlated with the true own-gender premium without sharing its mechanical estimation error.

This instrument corrects attenuation from measurement error; it does not provide an exogenous source of variation in establishment pay. In particular, it does not address correlation between the establishment's true wage policy and unobserved job attributes that also affect formalization. The specification limits this concern through industry-by-state and inspection-year fixed effects and the observed amenity measures, but cannot rule out unobserved establishment attributes correlated with wages. The survey experiment provides complementary evidence because wages and job attributes are randomly assigned. Section 8.2 compares the two designs using valuations expressed in wage units, which, unlike coefficient levels, do not depend on the dispersion of unobserved utility in each design or for each gender.[32]

[32]The estimated wage coefficients are 3.27 for men and 2.99 for women in the survey, compared with 3.17

5.4.4 Identifying Assumptions

Assumption 5.1 (Informal wage conditions)**.** *Within each market $s = (z, r)$, workers face similar informal wage opportunities, so that the market-level informal wage π_{ds} provides a valid measure of the informal wage available to workers at establishments in that market.*

This assumption does not require informal wages to be identical across establishments. Rather, it requires the relevant informal labor market to be sufficiently local to the industry–state cell: workers employed at establishments in the same market must face similar informal wage opportunities. Under this condition, π_{ds} captures the relevant informal outside option, and

$$\Delta w_{dj} = \psi_{dj} - \pi_{ds(j)}$$

measures cross-establishment variation in the wage gain from formal employment.

The identifying restriction is therefore that any residual variation in informal wage opportunities within an industry–state market is not systematically related to establishments' formal pay premia. A violation would arise, for example, if workers at high-ψ establishments also have systematically better informal wage opportunities than other workers in the same market. In that case, π_{ds} understates their informal outside option and Δw_{dj} overstates the formal wage premium, attenuating $\hat{\delta}_d$ toward zero.

A second requirement concerns the interpretation of Δw_{dj} as the wage gain relevant to the worker's choice. The formal wage decomposes into a worker component and an establishment component, while the informal wage analogously contains a worker component and a market component. Thus, $\psi_{dj} - \pi_{ds(j)}$ isolates the employer–market component of the formal–informal wage gap. It coincides with the wage premium entering the worker's choice when worker-specific earnings capacity is comparable across sectors, so that the worker component differences out of the formal–informal comparison. This restriction is substantive because the formal and informal wage components are estimated separately using RAIS and PNAD-C.

Departures from this restriction have a transparent implication. If workers with greater earnings capacity command systematically higher relative returns in formal employment, the constructed Δw_{dj} understates the formal wage gain for these workers. This measurement error again attenuates the estimated wage sensitivity. Both identifying restrictions therefore work in the same direction: violations that generate systematic within-market informal wage differences or sector-specific returns to worker skill tend to bias $\hat{\delta}_d$ toward zero.

and 3.08 in the administrative data, so the female-to-male ratios are 0.91 and 0.97, respectively. These ratios equal the ratio of underlying wage preferences only if the dispersion of unobserved utility is the same for men and women within each design (Section 5).

Assumption 5.2 (Informal amenity conditions)**.** *Within each market s, variation in the informal working conditions available to workers is not systematically related to the formal amenities $\mathbf{a}_j$ offered by establishments in that market.*

In the main specification, the market-level benchmark $\bar{\mathbf{a}}_s$ is absorbed by the industry-by-state fixed effects, so its level plays no role in identifying $\boldsymbol{\beta}_d$: the amenity coefficients are identified from differences in formal amenities across establishments in the same market. What Assumption 5.2 requires is that these differences are not mirrored by differences in the informal alternative. It fails if, within a market, workers at establishments offering more of an amenity formally also have better access to it informally. In that case, the variation in $\mathbf{a}_j$ overstates the variation in the formal–informal gap and the corresponding coefficient is attenuated. The level of $\bar{\mathbf{a}}_s$, measured in PNAD-C and pooled across genders, matters only for the complementary specification in equation (13), where it enters the regressor.

Assumption 5.3 (Conditional exogeneity of job attributes)**.** *Conditional on the formal wage premium and the fixed effects, the formal–informal amenity differences $\Delta\mathbf{a}_j$ and collective-agreement provisions $\mathbf{c}_j$ are uncorrelated with the unobserved determinants of formalization:*

$$\mathbb{E}\left[\eta_{dj} \mid \Delta\mathbf{a}_j, \mathbf{c}_j, \psi_{dj}, \mu_{z(j)\times r(j)}, \mu_{t(j)}\right] = 0.$$

Assumption 5.3 is the key identifying restriction for interpreting $\boldsymbol{\beta}_d$ and $\boldsymbol{\beta}_d^c$ as workers' responses to non-wage job attributes. Conditional on the establishment wage premium and the included fixed effects, establishments offering different amenities or collective-agreement provisions must not differ systematically in unobserved determinants of workers' propensity to formalize.

The main threat is endogenous sorting across establishments. Workers who place greater value on particular job attributes may select into establishments that already provide them. Similarly, establishments with more extensive collective-agreement provisions may differ along unobserved dimensions that independently affect formalization. For example, if establishments offering maternity provisions disproportionately employ women with a stronger latent preference for formal employment, the estimated coefficient on maternity benefits would combine workers' valuation of the provision with this underlying selection. More generally, if establishments with more extensive collective agreements have stronger unions, better management, or other unobserved characteristics that raise take-up, the corresponding CBA coefficients would also capture these differences.

The construction of the regressors addresses reverse causality from the inspection. I measure formal amenities and collective-agreement provisions using establishment characteristics and CBAs observed during the three years preceding the inspection. I

average establishment characteristics over this period and record a clause group as present if the agreement contains it in any of these years. Measuring attributes over the pre-inspection period captures persistent differences in establishments' working conditions and provides a strong predictor of the attributes available when the inspection occurs. Because these measures are determined before the inspection, they cannot adjust in response to enforcement or to workers' subsequent formalization decisions. The specification further absorbs industry-by-state and inspection-year fixed effects and conditions on the establishment wage premium, so identification comes from differences in pre-existing job attributes across establishments within the same labor market.

Predetermination does not eliminate sorting on pre-existing job attributes or omitted establishment characteristics. I assess this concern by conditioning on the component of the Sorkin (2018) revealed-preference measure of establishment value that is orthogonal to the establishment wage premium. This residual captures the non-wage attractiveness of an establishment revealed by worker flows. If the amenity and CBA coefficients primarily reflected sorting toward otherwise desirable establishments, conditioning on this measure should materially change the estimates. Appendix Table J.6 shows that the resulting valuations are stable.

Finally, I complement the administrative estimates with the survey experiment, in which the job attributes included in the experimental choice tasks are randomly assigned across alternatives. Random assignment eliminates sorting between worker preferences and those experimentally varied attributes, providing an independent estimate of workers' willingness to pay for them. This comparison applies only to attributes that are directly varied in the experiment; the identification of the remaining CBA provisions continues to rely on Assumption 5.3.

Thus, the administrative variation identifies $\boldsymbol{\beta}_d$ and $\boldsymbol{\beta}_d^c$ from differences in predetermined job attributes across establishments within labor markets. The pre-inspection construction rules out adjustment to the enforcement event, while the fixed effects and revealed-preference firm-value control address observed and unobserved differences in establishment attractiveness. The remaining identifying restriction is that, conditional on these controls, workers do not sort across establishments on unobserved preferences in a way that is systematically related to the pre-existing amenities and CBA provisions whose valuations I estimate.

5.4.5 Estimation and Inference

Estimation proceeds in three steps. First, I construct the establishment-level formalization rate $\hat{\lambda}_{dj}$ from equation (12), winsorize it at 0.001 and 0.999, and transform it to log-odds. Second, I estimate the main specification in equation (14) separately by gender using two-stage least squares, instrumenting each gender's establishment wage premium with the other gender's premium at the same establishment, as described in Section 5.4.3. I estimate equation (13), which enters the constructed formal–informal wage premium directly, as a complementary specification.

Third, I use the estimated wage, amenity, and CBA coefficients to construct the RMVs defined in equation (10). These express the valuation of each non-wage job attribute relative to wages and provide a common scale for comparing valuations across attributes and genders.

I conduct inference using an establishment-level cluster bootstrap. In each bootstrap replication, I resample establishments and re-estimate the full specification, including the first stage and the resulting RMVs. Male and female observations from the same establishment are resampled jointly, preserving their dependence when constructing cross-gender comparisons. Confidence intervals for the RMVs and their gender differences are obtained from the bootstrap distribution.

The main objects of interest are the model-implied wage elasticities of formalization and the RMVs of non-wage job attributes. The residual value of the informal alternative, γ_d, is absorbed by the fixed effects and is not separately identified in the administrative data; Section 8.3 recovers it using the survey experiment.

6. Estimating Who Formalizes, and What They Value

6.1 Wage Responses

Formalization responds strongly to the establishment pay premium for both men and women. The preferred IV estimates in columns (1) and (3) of Table 5 yield precisely estimated wage coefficients of $\hat{\delta}_m = 3.17$ and $\hat{\delta}_w = 3.08$. As discussed in Section 5, these scale-normalized coefficients do not separately identify wage preferences and the dispersion of unobserved utility. I therefore compare model-implied wage elasticities, which describe behavioral responsiveness rather than underlying preferences. Evaluating $\hat{\delta}_d(1 - \hat{\lambda}_{dj})$ at each establishment's predicted formalization rate and averaging across establishments

yields elasticities of 1.14 for men and 0.96 for women. These imply that a 10% increase in the formal wage would raise formalization by approximately 11.4% for men and 9.6% for women, using a local approximation.[33] The estimates thus indicate substantial wage responsiveness for both genders.

The OLS estimates in columns (2) and (4) are smaller, at 1.67 for men and 1.42 for women, consistent with attenuation from measurement error in the estimated establishment premia. This attenuation may be greater for women because their premia are estimated from fewer workers on average.[34]

The alternative specification in columns (5) and (6), which explicitly incorporates the industry–state informal wage benchmark, yields wage coefficients of 2.63 for men and 2.84 for women. At the reported mean predicted take-up rates, these imply average wage elasticities of 0.94 for men and 0.88 for women: a 10% increase in formal wages would raise formalization by approximately 9.4% and 8.8%, respectively. The implied elasticities remain similar across genders and somewhat smaller than in the preferred specification.

6.2 The Value of Non-Wage Job Attributes

I next examine how men and women respond to the non-wage attributes of formal employment. To make the magnitude of these responses economically interpretable, Table 6 reports the RMVs defined in equation (10). The RMV expresses the value of each non-wage attribute relative to wages and can therefore be interpreted as the share of the formal wage a worker would be willing to forgo for that attribute. The table reports the RMVs for every non-wage regressor in the preferred IV specification, together with the underlying coefficients.

The strongest responses are concentrated among women. Among collective-agreement provisions, women's largest valuation is for provisions governing the negotiation process,[35] with an RMV of 0.44, equivalent to 44% of the formal wage, compared with 0.12 for men.

[33] Because the elasticity is linear in $\hat{\lambda}_{dj}$, its average equals $\hat{\delta}_d(1-\bar{\lambda}_d)$, where $\bar{\lambda}_d$ is average predicted take-up: 64.1% for men and 69.0% for women in the estimation sample (Table 7).

[34] The comparison between OLS and IV is made within gender and is therefore unaffected by the scale normalization. However, the OLS and IV samples differ because the instrument requires an estimated premium for the other gender at the same establishment. I therefore treat OLS as a benchmark rather than attributing the entire OLS–IV difference to measurement error.

[35] The negotiation category captures provisions governing how the collective agreement is negotiated, implemented, enforced, and renewed. These include rules and timelines for renegotiation; mechanisms for resolving disputes between the employer and union; rules governing the application and enforcement of the agreement; penalties for non-compliance; and procedures for renewal or termination. For example, agreements may require negotiations over the subsequent agreement to begin 60 days before expiration, establish joint employer–union committees to resolve disputes, or impose monetary penalties for violations of negotiated provisions.

Leaves and flexibility and part-time provisions have RMVs of 0.19 and 0.18 for women, compared with 0.04 and −0.08 for men. The underlying female coefficients are statistically significant for all three categories, although the women–men difference is significant only for flexibility and part-time provisions.

Formal–informal amenity differences reveal additional gender contrasts. A higher formal-sector part-time share has an RMV of 0.21 for women and approximately zero for men. Female management has an RMV of 0.09 for women and −0.29 for men, with a gender difference significant at the 1% level. Manual task intensity shows the opposite pattern: its RMV is −0.33 for women and 0.06 for men, also a significant gender difference. Interpersonal task intensity is negatively valued by both genders, with no statistically significant difference.

Overall, the estimates show that non-wage attributes contribute to formalization decisions and that their valuation differs by gender. These results provide revealed-preference evidence that non-wage job attributes contribute to gender differences in sorting into formal employment. Section 8.2 provides complementary evidence from experimentally varied job attributes.

6.3 Model-Implied Formalization Rates

I next ask whether the estimated model reproduces the reversal in the gender gap in formalization across male- and female-dominated workplaces documented in the inspection data in Section 4. For each establishment and gender, I use the preferred specification to compute the fitted log-odds of formalization and transform them into predicted probabilities. I then average these probabilities across establishments within each workplace type. Table 7 compares these model-implied rates with the corresponding observed six-month formalization rates.

In male-dominated workplaces, the model implies formalization rates of 73.3% for men and 54.1% for women, a women–men gap of −19.2 percentage points. In female-dominated workplaces, the ordering reverses: the corresponding rates are 61.3% for men and 70.9% for women, a gap of +9.6 points.

The six-month women–men gap in the inspection data is −13.7 percentage points in male-dominated workplaces and +10.9 points in female-dominated workplaces. The model therefore closely reproduces the observed gap in female-dominated workplaces and predicts a larger gender gap in male-dominated workplaces.[36]

[36]The counterfactuals of Section 7 are evaluated on this same one-third classification, so the model-implied rates in columns (1) and (3) are the baseline against which every exercise there is measured. As a robustness

The comparison is more informative about gender differences than about formalization levels. The model produces establishment-level probabilities, whereas the observed measure is the ratio of post-inspection formal hires to detected informal workers and is not bounded above by one. The two measures therefore need not coincide in levels.

7. Worker Preferences and Formalization Take-Up

In this section, I use the estimated model to compute counterfactuals. The exercises quantify how much of the variation in take-up, and of the gender gap in take-up, is accounted for by differences in workers' preferences over wages and job attributes, and how take-up would respond to policies that change the terms of formal employment. All exercises condition on an existing formalization opportunity; how often such opportunities arise is the subject of Section 9.

Each exercise changes a set of preference parameters or job attributes in the fitted log-odds of the estimating equation and maps the shifted index back into a probability. I average these probabilities across establishments within gender, separately for female- and male-dominated industries, and report the gender gap as the women's minus the men's average. Panel A of Table 8 reports the baseline, which by construction reproduces the model-implied rates in Table 7. The exercises are partial equilibrium; Appendix I states each one formally, with inference throughout from 1,000 establishment-level bootstrap replications.

Equalizing preferences. Panel B of Table 8 asks how much of the gender gap in take-up reflects differences in estimated responses to wages and job attributes. Holding each job fixed, I give the minority group the coefficients of the majority gender: women in male-dominated industries receive men's coefficients, and men in female-dominated industries receive women's. I transfer the coefficients on the wage premium, on establishment amenities, and on collective-agreement provisions separately, and then jointly; the job attributes themselves are unchanged.

Transferring establishment-amenity coefficients changes the gender gap substantially more than transferring the wage coefficient. In male-dominated industries, assigning women men's amenity coefficients raises women's predicted formalization from 54.1 to 58.2% and narrows the gender gap from −19.2 to −15.2 percentage points; in female-

check, raising the threshold from one third to one half yields model-implied gaps of −14.9 and +9.9 percentage points, leaving the reversal unchanged.

dominated industries, assigning men women's amenity coefficients lowers men's predicted formalization from 61.3 to 53.7% and widens the gap from 9.6 to 17.2 percentage points. Both changes are statistically significant, with the part-time share the largest single contributor. Transferring the wage coefficient or collective-agreement coefficients leaves the gap essentially unchanged, and transferring all coefficients jointly produces changes close to those from establishment amenities alone.

Equalizing wage premia. Gender differences in take-up could reflect the formal wage premia of the jobs men and women are offered rather than how workers value them. Panel C holds preferences and non-wage attributes fixed and assigns the minority group the reference group's premium. Differences in premia account for little of the predicted gap: equalizing them narrows women's advantage in female-dominated industries from 9.6 to 7.8 percentage points and slightly widens their disadvantage in male-dominated industries, from 19.2 to 19.8 percentage points.

Policy counterfactuals: wages versus job attributes. Panel D of Table 8 considers a 10% increase in formal wages, allowing minimum-wage-induced gains to pass through to informal wages at rate ρ; I consider $\rho = 0$ and a preferred calibration of $\rho = 0.66$ (Derenoncourt et al., 2025). With no pass-through, predicted take-up rises by 4.1 percentage points for women in male-dominated industries and by 4.7 points for men in female-dominated industries. The gains fall to roughly one-third of their size under the preferred pass-through rate and are smaller still when the increase is restricted to establishments in the bottom quartile of the pay distribution. In every specification the gender gap moves by less than one percentage point, reflecting men's and women's similar wage-premium coefficients: higher formal wages raise take-up for both genders but do little to the gap.

Panel E instead raises the provision of non-wage attributes, lifting each provision in the amenity bundle to its sample mean at establishments initially below it. The first bundle combines work–family and workplace-condition provisions—maternity and childcare, leaves, harassment and discrimination protections, and flexibility and part-time arrangements; the second consists of union-negotiation provisions, the category with the largest estimated RMV for women in the administrative data. The work–family floor primarily raises women's take-up: women's predicted formalization increases by 2.8–2.9 percentage points across workplace types while men's is unchanged, so the gap narrows from −19.2 to −16.4 percentage points in male-dominated industries and widens from 9.6 to 12.5 percentage points in female-dominated ones; both changes are statistically significant. Negotiation provisions raise take-up for both genders but more for women,

moving the gaps to −16.2 and 12.6 percentage points, although these changes are less precisely estimated.

What the worker margin explains. Conditional on a formal job opportunity, the wage premium raises take-up for both genders but accounts for little of the gender gap; valuations of non-wage attributes account for most of the gap the model can attribute to preferences, through their interaction with the attributes of the jobs men and women face. The remaining source, the residual value of the informal alternative γ_d, is absorbed by the fixed effects of the estimating equation and cannot be recovered from the administrative data. Section 8 therefore turns to the survey experiment to examine what lies behind incomplete take-up.

8. Survey Evidence: Take-Up Under a Guaranteed Offer

The inspection setting provides evidence on actual formalization behavior and on how take-up varies with wages and job attributes. Its interpretation as worker take-up, however, rests on employers offering formal contracts in response to inspections. Employers face strong legal and financial incentives to do so (Section 2.2), but I do not observe these offers directly. The administrative data also cannot recover the residual value of the informal alternative, which the fixed effects of the estimating equation absorb (Section 7). I address both limitations with a survey experiment that elicits workers' decisions under a guaranteed formal offer. The analysis focuses on the 942 respondents who have previously held an informal job, the subsample closest to the population detected by inspectors. Appendix K describes the sampling, the experimental design, and the estimation.

8.1 Stated Take-Up Under a Guaranteed Offer

I use a vignette that mirrors the inspection setting. The vignette is embedded in the online survey: respondents read a short written scenario in which a labor inspector arrives unannounced at their current employer, who then offers to register them formally by signing their *carteira*. They report on a five-point scale whether they would accept the offer and remain in the job. Because the offer is guaranteed and the worker–employer match is held fixed, stated acceptance measures take-up conditional on a formal offer with known terms. Each respondent answers two versions of the question. In the first, gross salary,

hours, and duties remain unchanged, so formal registration reduces take-home pay by the employee's social-security contribution, which respondents are told amounts to around 7 to 11% of the gross salary. In the second, gross salary increases enough to preserve take-home pay. I classify respondents as accepting if they answer *definitely yes* or *probably yes*; those who are unsure are classified as not accepting.

Most respondents accept. When take-home pay falls, 60.9% accept the formal contract; when it is protected, acceptance rises to 78.9% (Appendix Table K.4). These results support the finding from the inspection setting that most workers formalize when offered the opportunity, and they show that acceptance depends strongly on the immediate monetary cost of formalization.[37]

Stated acceptance is high even among workers actively looking for another job, suggesting that willingness to formalize extends to those less attached to their current employer (Appendix Table K.6). When take-home pay falls, 77.6% of active job seekers accept, compared with 55.0% of workers not interested in changing jobs ($p = 0.001$). When pay is protected, acceptance among active job seekers rises to 84.2%, with similar rates for men and women (83.3% and 85.3%). The only significant gender gap under protected pay occurs among workers not interested in changing jobs, where 70.8% of women accept compared with 87.5% of men. High acceptance among workers seeking to leave their current employer suggests that attachment to that employer is not the primary explanation for stated take-up.

The cost of formalization also accounts for the gender gap in stated take-up. When take-home pay falls, 56.7% of women and 65.6% of men accept, a gap of 8.9 percentage points ($p = 0.005$).[38] When take-home pay is protected, acceptance increases to 77.3% for women and 80.6% for men, and the remaining 3.3-point gap is not statistically significant ($p = 0.221$). Comparing responses across the two scenarios helps explain part of the gender gap. The shares rejecting formalization under both scenarios are nearly identical, 13.4% of women and 13.8% of men. The difference comes instead from respondents who accept only when take-home pay is preserved: 30.0% of women, compared with 20.5% of men. Women are therefore not systematically more likely to reject formal employment; their decisions respond more strongly to the price at which formalization is offered, the reduction in take-home pay from the employee's social-security contribution. Once registration no

[37]Respondents who have experienced a labor inspection answer the vignette in much the same way. Among the 342 main-wave respondents (of all 1,501, before sample restrictions) who report that an inspector visited one of their workplaces, 66.4% accept when take-home pay falls and 85.4% when it is protected. Of the 57 who were informal at that inspection and were offered a *carteira*, 78.9% accepted it: 81.5% of men and 76.7% of women (Appendix Table K.5).

[38]The survey gap is consistent with the lower formalization rate of women in male-dominated workplaces in the administrative data (13.7 percentage points among inspections from 2012 onward).

longer reduces take-home pay, women accept at nearly the same rate as men.[39]

8.2 Valuations of Wages and Job Attributes

The vignette fixes the terms of the offer. To recover how workers value those terms, I use a discrete-choice experiment in which each respondent makes fifteen binary choices between jobs that differ in the wage and in one non-wage attribute, with both randomly assigned. The experiment is embedded in the same online survey. Each task displays two jobs side by side: Job A pays the respondent's current salary, or reservation wage if not employed, and omits the attribute, while Job B offers the attribute at a salary lower by a randomly drawn 5 to 40%. Task order and the left–right position of the two jobs are also randomized. Random assignment addresses the main limitation of the administrative estimates, the possible correlation between establishment pay and unobserved job attributes (Section 5.4.3).

I first examine how workers respond to wages. The estimated wage coefficients are $\hat{\delta}_m = 3.27$ for men and $\hat{\delta}_w = 2.99$ for women, close to the administrative estimates of 3.17 and 3.08, respectively. Because these coefficients are identified relative to the dispersion of unobserved utility, their magnitudes are not directly comparable across genders or designs (Section 5). Under the common utility scale imposed by the pooled survey logit, the gender difference is not statistically significant. I therefore focus on valuations of non-wage attributes expressed relative to wages.

Women value every experimental attribute more than men, although many differences are imprecisely estimated (Appendix Table K.3). The largest differences are for parental leave (0.15), home office (0.14), advance notice (0.12), and unemployment insurance (0.10). Consistent with this pattern, the administrative estimates show higher relative valuations among women for leave, maternity and childcare provisions, and flexibility and part-time arrangements.

To assess whether these preferences are reflected in actual formalization decisions, Figure 9 compares gender differences in wage-equivalent valuations across 13 attributes measured in both datasets. The Pearson correlation is 0.42 and increases to 0.65 when restricting the comparison to the eight attributes with a one-to-one correspondence.[40] Thus, attributes that women value relatively more in the experiment also tend to be

[39]The gap when take-home pay falls also comes entirely from respondents who are unsure: 31.8% of women answer *not sure*, compared with 22.5% of men, while the shares answering *probably* or *definitely no* are nearly identical (11.5 and 11.8%). Counting unsure respondents as accepting, acceptance is 88.3% when take-home pay falls and 95.6% when it is protected, with no significant gender difference in either scenario (88.5% for women and 88.2% for men when pay falls; 96.2 and 95.1% when it is protected).

[40]The corresponding rank correlations are 0.45 and 0.81.

valued relatively more in observed formalization decisions, particularly when the measures are directly comparable. This correspondence provides further evidence that gender differences in the valuation of non-wage attributes shape sorting into formal employment.

8.3 The Outside Option

The choice experiment also prices the formal contract itself: one choice task varies whether the job comes with a signed *carteira*. Women value it at least as much as men, with a relative monetary value of 0.40, against 0.35 for men. I use this valuation to predict whether respondents would accept the vignette's offer to formalize their existing job and compare the prediction with their stated acceptance.

Let β_{1d} denote the utility value of the *carteira*, δ_d the wage coefficient, and γ_d the residual value of remaining informal. Acceptance in the vignette satisfies

$$\Pr(\text{accept} \mid \text{net pay falls by } p) = \Lambda^{-1}\left(\beta_{1d} + \delta_d \ln(1-p) - \gamma_d\right). \tag{15}$$

Here p is the reduction in take-home pay in the vignette. Respondents are told that the employee's social-security contribution would lower take-home pay by around 7 to 11% of the gross salary, and I set $p = 0.09$, the midpoint of this range. Because the *carteira* task in the choice experiment offers the signed contract against randomly assigned wage reductions of 5 to 40%, this value lies within the support of the experiment. The vignette identifies the left-hand side, while the choice experiment identifies β_{1d} and δ_d. Evaluating the right-hand side at $p = 0.09$ with $\gamma_d = 0$ gives the acceptance rate predicted by the choice experiment alone. I therefore recover γ_d as the residual value of the informal alternative required to reconcile the vignette and the choice experiment.

Predicted acceptance exceeds stated acceptance, particularly among women. Based on the choice experiment, predicted acceptance at the vignette price is 78.0% for women and 74.6% for men, compared with stated acceptance of 56.7 and 65.6%, respectively.[41] The implied residual value of remaining informal is therefore larger for women ($\gamma_w = 1.00$) than for men ($\gamma_m = 0.43$). As with the relative monetary values of job attributes, I express these residuals as a share of wages to make them comparable across genders. The resulting RMVs are 0.28 for women and 0.12 for men. Thus, women place more than twice the relative monetary value on remaining informal, despite their higher valuations of formal job attributes. Their lower stated acceptance is therefore consistent with a more valuable informal alternative rather than a lower valuation of formal employment.

[41] A test that observed and predicted acceptance coincide rejects at $p = 0.004$.

I interpret γ_d as an outside-option residual rather than a primitive preference for informality. It captures dimensions of the informal alternative that are not priced in the choice experiment, potentially including control over hours and tasks, expected job duration, eligibility for transfers, and household circumstances that make a formal work arrangement costly. The survey does not separately identify these components, so γ_d locates the source of the discrepancy between the vignette and the choice experiment without establishing which component generates it.

8.4 Household Constraints and Informal Job Attributes

Two additional results speak to what the residual captures. The first asks how much of the 8.9-point gender gap in stated acceptance is associated with observable differences between men and women, including household and caregiving constraints. I estimate an Oaxaca–Blinder decomposition (Oaxaca, 1973; Blinder, 1973) of a linear probability model with separate coefficient vectors by gender. Covariates are grouped into household and caregiving constraints, children, knowledge of statutory entitlements and trust in employers, prior formal-sector experience, demographics, and state and two-digit industry.[42]

Table 9 reports the three-fold decomposition. Differences in observed characteristics account for 3.1 percentage points, or 35% of the raw gender gap. Household and caregiving constraints account for most of this component: differences in these characteristics contribute 2.4 percentage points to lower take-up among women. Children, knowledge and trust, prior formal-sector experience, demographics, and sorting across states and industries contribute little through differences in observed characteristics. The remaining gap is associated with differences in the coefficients relating these characteristics to take-up and with the interaction between differences in coefficients and differences in characteristics, which account for 3.4 and 2.4 percentage points, respectively. Their detailed contributions are imprecisely estimated and offset one another, so I do not interpret individual coefficient differences as evidence of specific mechanisms.

The second result examines whether workers who value the flexibility associated with alternative work arrangements are less willing to formalize. Such arrangements may offer greater control over schedules, work location, and earnings opportunities, increasing the

[42]The knowledge and trust group combines a twelve-item score measuring knowledge of the entitlements associated with a *carteira*, indicators for knowing that formal employment provides dismissal payments and parental leave, and whether the respondent trusts employers to provide these benefits. The last component captures expected employer compliance rather than information. State and industry indicators are normalized to sum to zero within each categorical variable, making their detailed contributions invariant to the omitted category. Structurally missing categories are retained using explicit indicators.

value of remaining informal. I construct an index comparing price-adjusted valuations of schedule flexibility, on-call pay, and working off-site with valuations of formal-sector protections, excluding the *carteira* task because it directly captures the vignette outcome. A one-standard-deviation increase in this index is associated with 2.3 percentage points lower stated acceptance (s.e. 1.6), or 1.9 points with demographic controls (s.e. 1.6), relative to a baseline of 60.9% (Appendix Table K.7). Figure 10 shows a negative relationship for both genders. Although imprecise, these estimates are consistent with flexible work arrangements increasing the value of the informal alternative.

9. Access to Formal Job Offers and Informality

The preceding sections show that most informal workers would accept formal employment, although acceptance depends on wages, working conditions, and the value of the informal alternative. I now ask how often workers receive opportunities to formalize outside inspections. I combine informal-to-formal transitions in the broader labor market with inspection-based take-up to distinguish access to formal offers from willingness to accept them. Because offers are not directly observed, this calculation relies on workers accepting comparable offers at the same rate inside and outside the inspection setting.

9.1 Separating Acceptance from Access

An informal worker becomes formal when an offer arrives and the worker accepts it. The probability of formalization can therefore be written as

$$\Pr(\text{Formal}_{dc}) = \Pr(\text{Formal}_{dc} \mid \text{Offer}_{dc}) \times \Pr(\text{Offer}_{dc}), \tag{16}$$

where d indexes gender and c an industry × state × year labor-market cell. The first term on the right-hand side measures acceptance conditional on an offer, the focus of Sections 4–8. The second measures access to formal offers.[43]

Household surveys measure transitions into formal employment but do not distinguish the arrival of an offer from its acceptance. Inspections provide a measure of acceptance

[43] Access depends on employers' incentives to register workers, including compliance costs and enforcement (Ulyssea, 2018; Almeida & Carneiro, 2012), and on search and matching frictions that restrict access to formal jobs (Meghir, Narita & Robin, 2015; Bosch & Esteban-Pretel, 2015; Bobba et al., 2021; Haanwinckel & Soares, 2021).

when employers are required to register workers already employed informally. Combining the two sources therefore allows me to infer how frequently comparable formal offers arise outside inspections.

I use the PNAD-C to measure the six-month probability that an informal worker becomes formal with the same employer, p_{dc}. Restricting transitions to the same employer makes them more comparable to formalization following inspections. I relate these transitions to inspection-based take-up over the same horizon, λ_{dc}:

$$p_{dc} = q_d \lambda_{dc} + \varepsilon_{dc}. \tag{17}$$

I estimate this equation separately by gender across industry × state × year cells, weighting by the number of informal workers underlying p_{dc}.

Interpreting $\hat{q}_d$ as an offer probability requires comparable acceptance inside and outside inspections. If workers are more likely to accept inspection-induced offers, the calculation understates the frequency of offers outside inspections. If inspected establishments offer less attractive formal jobs and workers are less likely to accept them, it overstates that frequency. The gender comparison is less sensitive to these differences if they affect men and women similarly.

The estimate is therefore an accounting measure rather than a directly observed offer probability. It summarizes the relationship between transitions and take-up across labor markets. When offer probabilities differ across cells, it reflects a weighted average of those probabilities rather than a common probability faced by all workers.

9.2 How Often Do Workers Receive Formal Offers?

Table 10 reports implied six-month offer probabilities of 38.3% for men and 29.4% for women. Both are substantially below inspection-based take-up of 78% and 68%, respectively. Under the comparable-acceptance assumption, most workers would accept formal employment, but fewer than half receive a comparable offer within six months. Women's implied offer probability is 8.9 percentage points lower than men's, or 23% below the male rate. This difference is statistically significant.

The implied offer probabilities remain below take-up under alternative definitions (Appendix Table J.7). Including transitions to other employers raises them to 52.7% for men and 38.9% for women, with a similar female-to-male ratio. The gender difference is more sensitive to the specification. It narrows when transitions are measured through the end of the inspection year or aggregated to state × year cells, and is no longer statistically

significant under either definition. The evidence that offers are less frequent than workers' willingness to accept them is thus more robust than the size of the gender gap in access.

The smaller gender gap when pooling industries is consistent with women working in industries where formal offers are less frequent relative to take-up. It does not, however, establish why these differences arise. Employers may face different registration costs or enforcement risks across industries. They may also face less pressure to offer formal contracts where workers have fewer alternatives. This possibility relates to evidence on women's lower labor supply elasticity to individual employers (Sharma, 2023) and on employer market power and workers' allocation outside wage employment (Amodio, Medina & Morlacco, 2025). Employers may also be less willing to register women if they anticipate costs associated with maternity. Without observing offers, I cannot distinguish these explanations from matching frictions or discrimination.

9.3 Comparing Changes in Acceptance and Access

Equation (16) provides a common scale for comparing changes in acceptance and access. Holding the other margin fixed, a given proportional increase in either take-up or the probability of receiving an offer raises formalization by the same proportion. I use this relationship to compare the take-up counterfactuals in Section 7 with an exercise that closes the estimated gender gap in access.

Consider women in male-dominated industries, where the counterfactuals predict relatively large increases in take-up. Assigning them men's amenity valuations raises take-up by 4.1 percentage points, or 8% relative to its baseline (Table 8). Increasing provisions on maternity and childcare, leave, harassment and discrimination, and flexibility and part-time work raises take-up by 5%. A 10% minimum-wage increase, under the preferred assumption about its effect on informal wages, raises take-up by 3%. Holding offers fixed, these changes would raise formalization by the same respective proportions.

For comparison, raising women's aggregate implied offer probability from 29.4% to 38.3%, the male rate, would increase their formalization by 30% if take-up remained unchanged. This increase is larger than those generated by the take-up counterfactuals. The exercises nevertheless differ in both scope and population: the take-up counterfactuals concern women in male-dominated industries, whereas equalizing access changes women's aggregate offer probability. They illustrate the magnitudes of the two margins, not the relative effectiveness or costs of equivalent policy interventions.

9.4 Implications for Informality

The comparison with broader labor-market transitions suggests that low acceptance alone cannot explain why workers remain informal. Most workers formalize according to the inspection-based measure, and most survey respondents accept a guaranteed formal offer. Yet same-employer formalization is much less frequent outside inspections. If workers would accept comparable offers at similar rates in both settings, this difference points to a lack of formal offers that workers would accept.

This distinction matters for policies intended to reduce informality. Higher wages and more desirable working conditions can increase acceptance among workers who receive an offer. Policies that reduce registration costs, strengthen enforcement, or ease matching frictions may instead expand access. The findings do not identify which of these barriers accounts for the inferred shortage of offers, nor do they establish the effects of specific interventions.

Acceptance and access can also change together. Improving formal job attributes may make workers more willing to accept while changing employers' willingness to offer formal contracts. Conversely, expanding offers will have a smaller effect if their terms make them unattractive to workers. The calculations hold one margin fixed to clarify the role of the other. Under this assumption, they suggest that expanding access has substantial scope to increase formal employment, while wages and working conditions shape which offers workers accept.

10. Conclusion

This paper studies whether informal workers would accept formal employment and how wages and non-wage job attributes shape that decision. Labor inspections in Brazil provide a setting in which employers are required to register workers already employed informally. Most workers formalize according to the inspection-based take-up measure, but acceptance remains incomplete and varies across workplaces. A discrete-choice model shows that men and women respond similarly to wages but differ in their valuations of non-wage attributes. These differences help explain why women formalize at higher rates than men in female-dominated industries, while the opposite holds in male-dominated industries.

A complementary survey experiment examines workers' choices under a guaranteed formal offer. Most respondents accept, and preserving take-home pay raises acceptance substantially while narrowing the gender formalization gap. Experimental valuations of

job attributes also align with the gender differences inferred from inspections. Women value several attributes of formal employment more than men, yet are less likely to accept formalization in the vignette. Within the model, this pattern is consistent with a more valuable informal alternative. Household and caregiving constraints help account for the gap, although the evidence does not establish which features of informal work explain the remaining difference.

The results also suggest that many workers remain informal because they lack formal offers they would accept. Same-employer transitions into formal employment are substantially less frequent in the broader labor market than following inspections. Combining these transitions with inspection-based take-up points to lower access to formal offers among women. Offers are not directly observed, and this calculation assumes that workers would accept comparable offers at the same rate inside and outside inspections. Under this assumption, equalizing women's access to men's would raise their formalization more than the changes in take-up considered in the counterfactuals.

These findings suggest that policies to encourage formal employment should account for the terms workers are offered. Higher wages increase take-up for both men and women, while flexibility and leave provisions can also narrow gender differences in acceptance. At the same time, improving the attractiveness of formal jobs alone may have limited effects when workers rarely receive an offer. Evaluating these policies requires understanding how employers adjust hiring, wages, and working conditions in response, beyond the partial-equilibrium changes considered here.

Two questions remain for future research. First, which features of informal employment do workers value enough to forgo formal protections, and how do household constraints shape those valuations? Second, which employer decisions and labor-market frictions limit formal offers, particularly for women? Data that jointly observe offers, acceptance decisions, and informal alternatives would help answer both questions and guide policies that expand formal employment while preserving the job attributes workers value.

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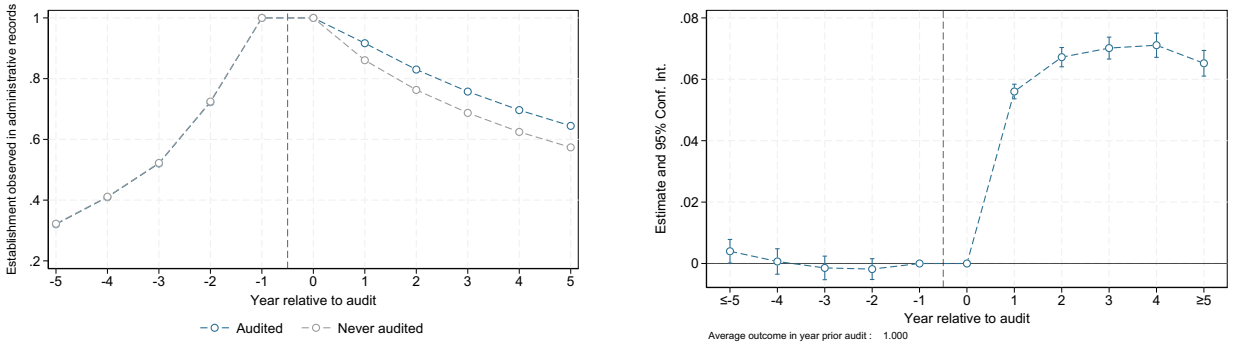
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FIGURES

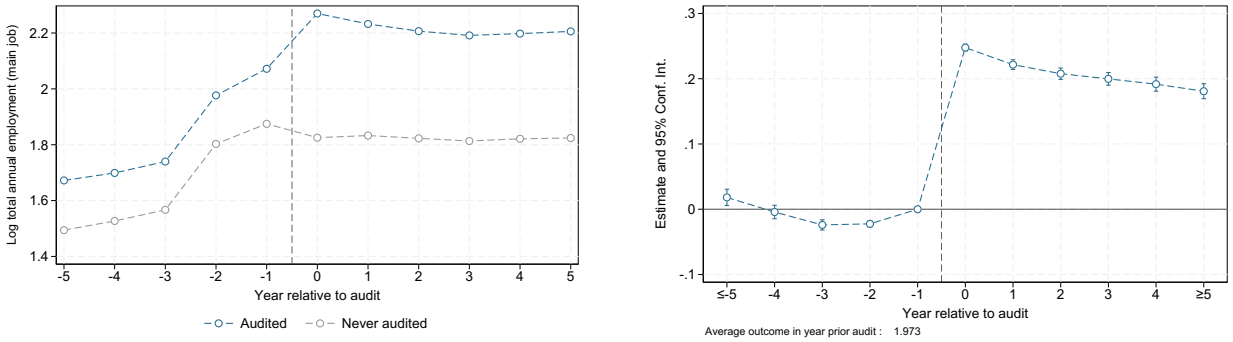
Figure 1: Formal-Sector Activity Around the Inspection



(a) Raw means **(b)** Event-study estimates

Notes: This figure reports establishment activity in the formal labor market around the inspection. The outcome is an indicator equal to one if the establishment appears in RAIS with positive employment in a given year. Panel (a) plots raw means by event time for inspected establishments and their matched never-inspected controls. Panel (b) reports the event-study coefficients θ_k estimated using equation (1). The omitted period is $k = -1$, endpoints are binned at $|k| \geq 5$, and vertical bars report 95% confidence intervals based on standard errors clustered at the firm level. The mean outcome at $k = -1$ equals one by construction.

Figure 2: Total Annual Formal Employment

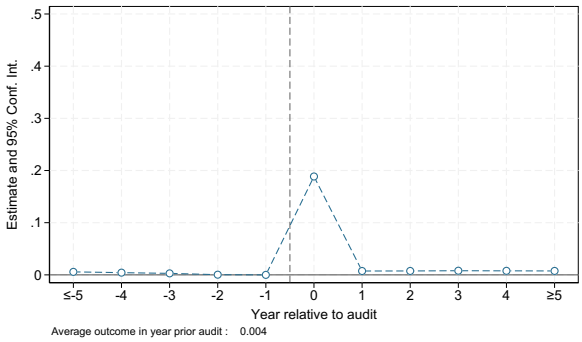


(a) Raw means

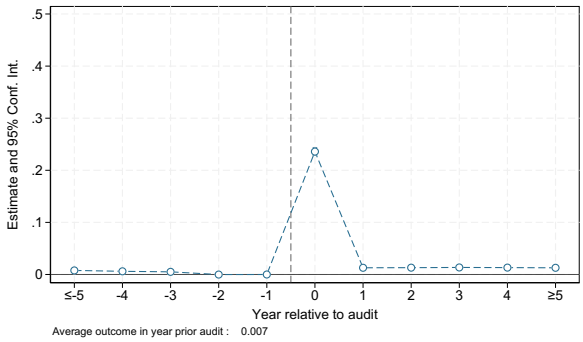
(b) Event-study estimates

Notes: This figure reports total annual formal employment among inspected establishments and their matched never-inspected controls. The outcome is the log number of workers with a registered contract at the establishment at any point during the calendar year. Panel (a) plots raw means by event time. Panel (b) reports the event-study coefficients θ_k from equation (1) for matched pairs observed at event time -3. The omitted period is $k = -1$, endpoints are binned at $|k| \geq 5$, and vertical bars denote 95% confidence intervals based on standard errors clustered at the firm level.

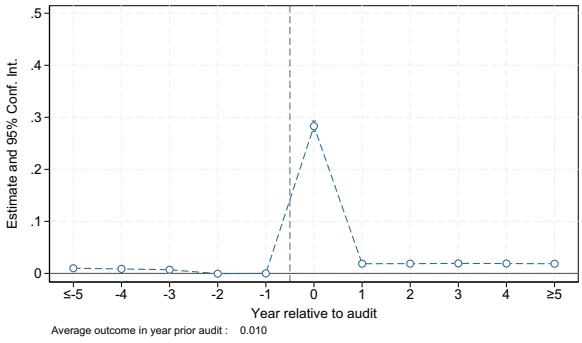
Figure 3: New Formal Hires as a Fraction of Pre-Inspection Employment, by Hiring Window



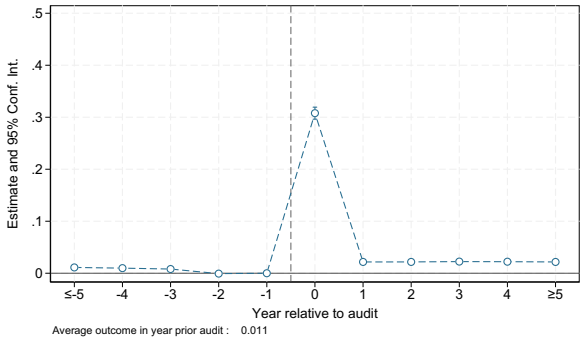
(a) Within 1 month



(b) Within 3 months



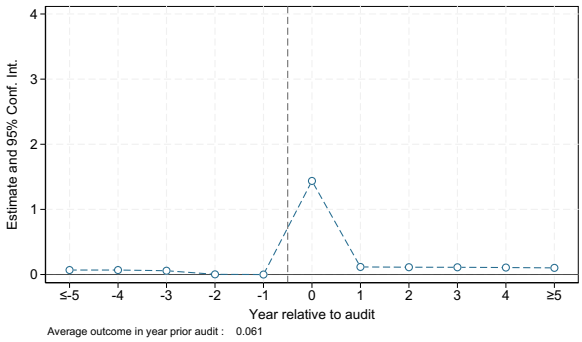
(c) Within 6 months



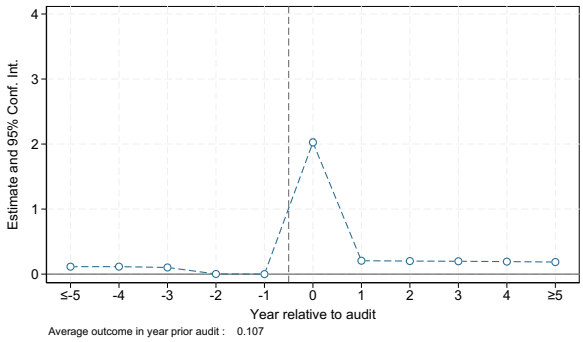
(d) Post-inspection

Notes: This figure reports new formal hires as a fraction of pre-inspection employment over four progressively wider hiring windows. The outcomes count workers hired within one, three, or six months of the inspection or during the remainder of the inspection year who had no active formal contract at any employer in the calendar month preceding the inspection, divided by establishment employment in the year before the inspection. Each panel reports the event-study coefficients θ_k estimated using equation (1) for matched pairs observed at event time -3. The omitted period is $k = -1$, endpoints are binned at $|k| \geq 5$, and vertical bars report 95% confidence intervals based on standard errors clustered at the firm level. Figure 4 reports the corresponding outcomes in counts. Appendix Figure E.8 presents the estimates separately for all inspection cohorts and for establishments first inspected in 2012 or later.

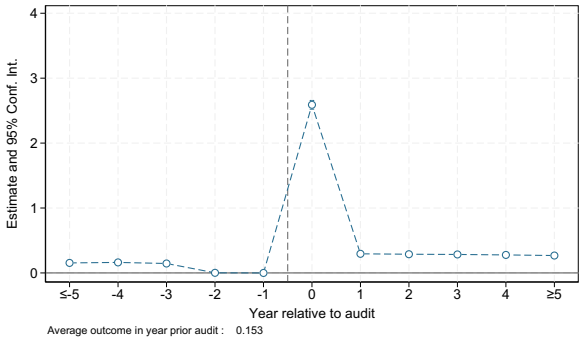
Figure 4: New Formal Hires,
by Hiring Window



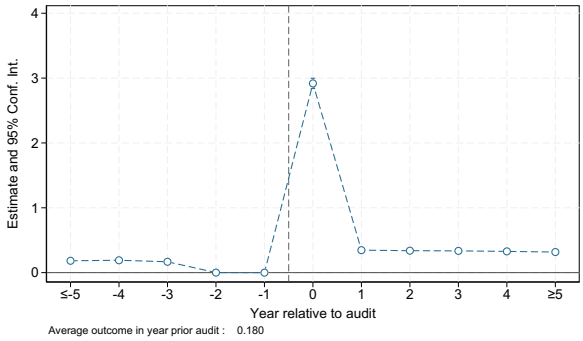
(a) Within 1 month



(b) Within 3 months



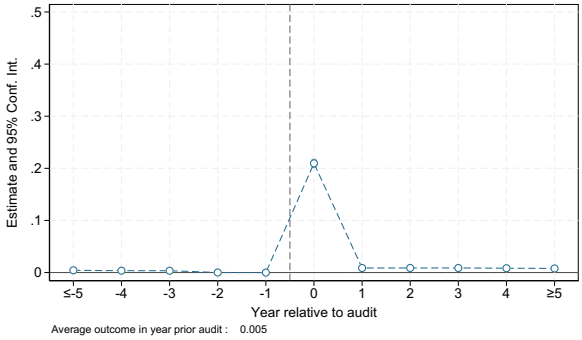
(c) Within 6 months



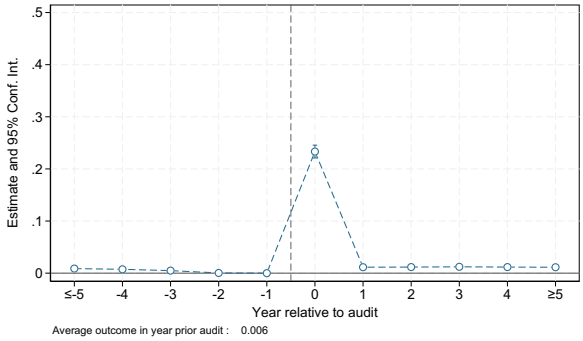
(d) Post-inspection

Notes: This figure reports new formal hires in counts over the four hiring windows defined in Figure 3. The outcomes count workers hired within one, three, or six months of the inspection or during the remainder of the inspection year who had no active formal contract at any employer in the calendar month preceding the inspection. Each panel reports the event-study coefficients θ_k estimated using equation (1) for matched pairs observed at event time -3. The omitted period is $k = -1$, endpoints are binned at $|k| \geq 5$, and vertical bars report 95% confidence intervals based on standard errors clustered at the firm level. Appendix Figure E.9 presents the estimates separately for all inspection cohorts and for establishments first inspected in 2012 or later.

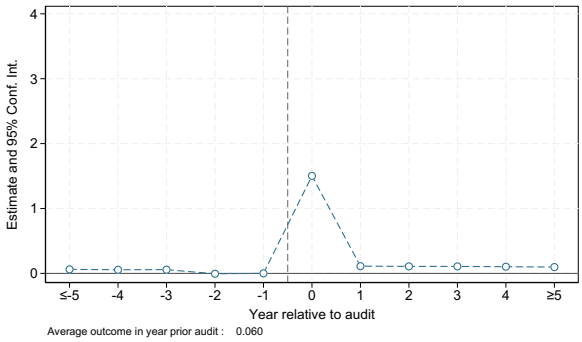
Figure 5: New Formal Hires Within Six Months, by Gender of the Detected Workers



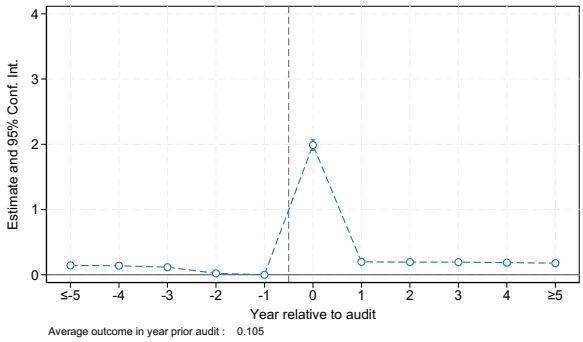
(a) Women, fraction of pre-inspection employment



(b) Men, fraction of pre-inspection employment



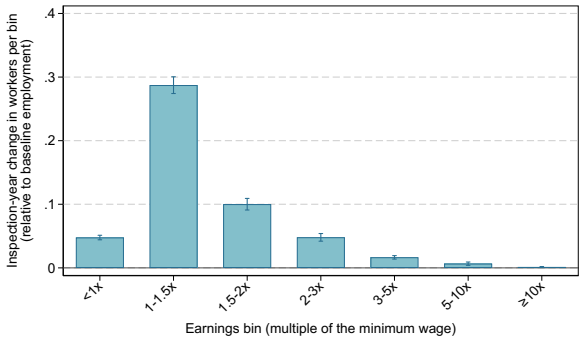
(c) Women, counts



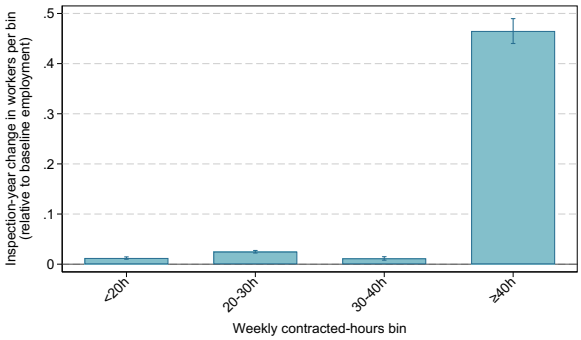
(d) Men, counts

Notes: This figure reports new formal hires within six months of the inspection separately by the gender of the informal workers detected during the inspection. The outcomes count workers of the indicated gender hired at the inspected establishment within six months of the inspection who had no active formal contract at any employer in the calendar month preceding the inspection. Panels (a) and (b) divide these counts by establishment employment in the year before the inspection; panels (c) and (d) report them in counts. The female estimates are identified from inspections in which informal women were detected and the male estimates from inspections in which informal men were detected, so the two are estimated on different and differently sized sets of establishments. The counts in panels (c) and (d) are therefore not directly comparable across genders, whereas the fractions in panels (a) and (b) are. Each panel reports the event-study coefficients θ_k estimated using equation (1) for matched pairs observed at event time -3. The omitted period is $k = -1$, endpoints are binned at $|k| \geq 5$, and vertical bars report 95% confidence intervals based on standard errors clustered at the firm level. Appendix Figures E.2 and E.3 report the one- and three-month windows and the remainder of the inspection year, as fractions of pre-inspection employment and in counts, respectively.

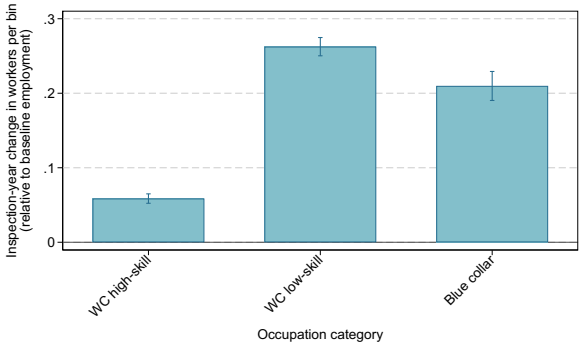
Figure 6: Distribution of the Increase in Registered Employment



(a) Multiples of the minimum wage



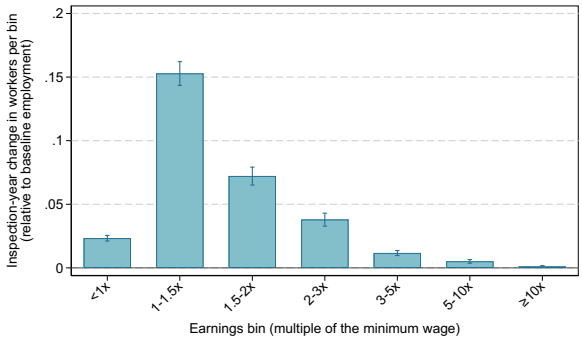
(b) Weekly contracted hours



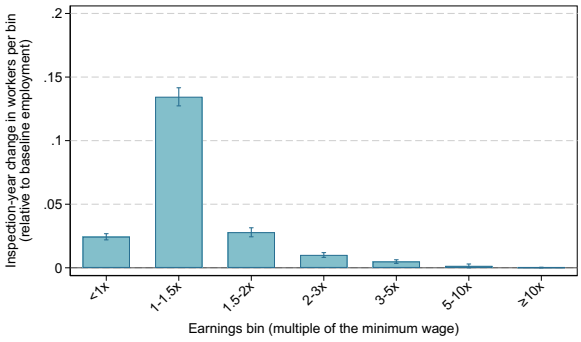
(c) Occupational group

Notes: This figure reports the inspection-year coefficients θ_{b0} from equation (2). Each coefficient measures the change in the number of workers in bin b per pre-inspection worker, relative to the year before the inspection. Panel (a) partitions workers by multiples of the minimum wage: less than one, one to 1.5, 1.5 to two, two to three, three to five, five to ten, and at least ten. Panel (b) partitions workers by weekly contracted hours: less than twenty, twenty to thirty, thirty to forty, and at least forty. Panel (c) partitions workers by occupational group. The earnings and hours partitions are exhaustive; the occupational classification is not. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

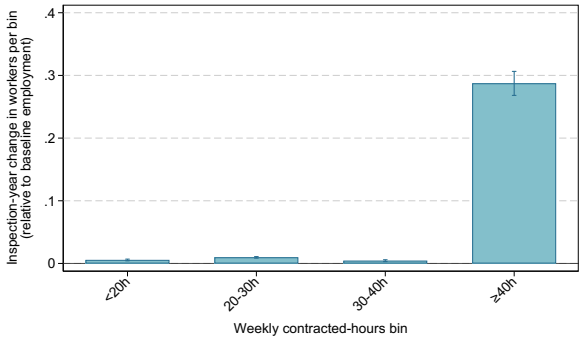
Figure 7: Distribution of the Increase in Registered Employment, by Gender



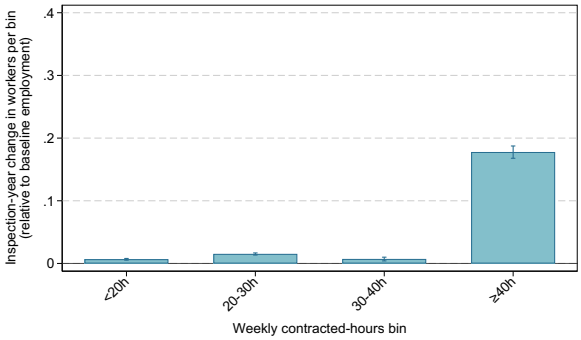
(a) Minimum-wage multiples, men



(b) Minimum-wage multiples, women



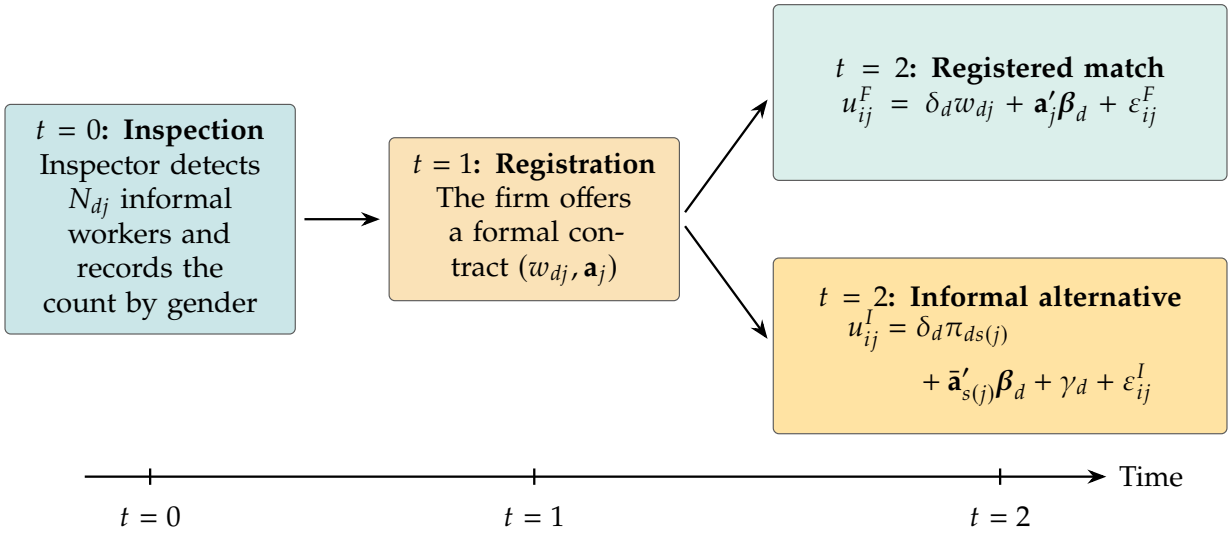
(c) Weekly hours, men



(d) Weekly hours, women

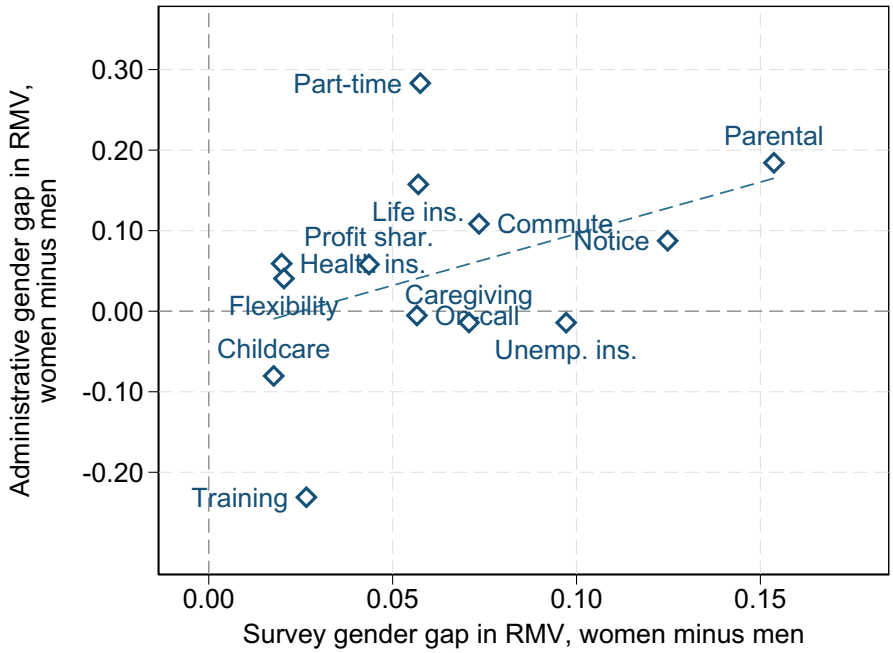
Notes: This figure reports the inspection-year coefficients θ_{b0} from equation (2) separately by gender. Panels (a) and (b) partition workers by multiples of the minimum wage; panels (c) and (d) partition workers by weekly contracted hours. Each coefficient measures the change in the number of workers of the indicated gender in bin b per unit of total pre-inspection employment, relative to the year before the inspection. The gender-specific estimates are therefore expressed on a common scale. The specification uses the same matched sample and bins as Figure 6. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

Figure 8: The Formalization Decision



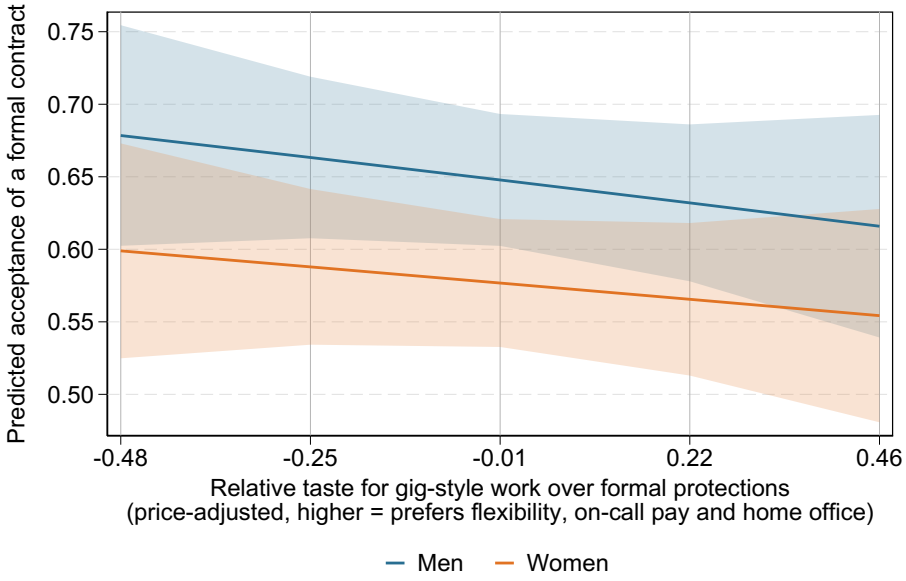
Notes: This figure summarizes the formalization decision following an inspection. Establishment j belongs to industry–state market $s(j)$, and $d \in \{m, w\}$ denotes gender. In the registered branch, the worker continues the existing match under a formal contract. In the informal branch, the worker leaves the inspected establishment and searches for an informal job elsewhere in the same industry–state labor market, which pays the market informal wage $\pi_{ds(j)}$ and provides the market amenity bundle $\bar{\mathbf{a}}_{s(j)}$; γ_d captures the residual value of this alternative, including search opportunities. The model interprets λ_{dj} as the probability of the registered branch; its empirical counterpart is the formalization ratio defined in Section 4.

Figure 9: Gender Differences in Amenity Valuations: Survey and Administrative Estimates



Notes: Each point represents one amenity. The horizontal axis reports the women–men difference in survey relative monetary values (RMVs), and the vertical axis reports the corresponding difference in administrative RMVs. The administrative estimates come from a variant of the preferred establishment-premium specification that replaces the fifteen CBA clause-group indicators with thirteen indicators for whether the establishment's collective agreement contains a provision corresponding to each survey attribute; Appendix Table K.9 reports the crosswalk. The specification is estimated separately by gender using 2SLS, instrumenting each gender's establishment pay premium with the other gender's premium at the same establishment. It includes total clause intensity, the CBA indicator, and industry × state and inspection-year fixed effects; the four amenity differences are omitted. Administrative RMVs are computed as in equation (10), $1 - \exp(-\hat{\beta}_{d\ell}/\hat{\delta}_d)$, using the wage coefficient from the same specification ($\hat{\delta}_m = 3.18$ and $\hat{\delta}_w = 3.03$). Survey RMVs are constructed analogously from the choice-experiment logit estimates. The figure includes all thirteen survey attributes with a collective-agreement counterpart. Eight have a direct crosswalk: schedule flexibility, parental leave, training, job stability and advance notice, childcare assistance, profit sharing, on-call pay, and group life insurance. The remaining five—health insurance, unemployment insurance, commuting support, paid caregiving days, and part-time hours—have less direct mappings to collective-agreement provisions. The *carteira* and home office attributes are excluded because they have no collective-agreement counterpart. The dashed line reports the OLS fit. The Pearson correlation is 0.42 ($p = 0.15$). Estimates use the ever-informal survey subsample. Appendix Figure J.1 restricts the comparison to the eight directly mapped attributes.

Figure 10: Taste for Gig and On-Demand Work and Predicted Acceptance of a Formal Contract



Notes: Predicted probability of accepting a formal contract when the employee's social-security contribution lowers take-home pay, plotted against relative preference for gig and on-demand work over formal protections. Predictions come from a logit of acceptance on the index interacted with gender, evaluated over the interdecile range of the index, with 95% confidence intervals. The index compares price-adjusted preferences for schedule flexibility, on-call pay, and working off-site with preferences for unemployment insurance, advance notice, and health and life insurance. The *carteira* task is excluded. The sample consists of ever-informal survey respondents. Appendix Table K.7 reports the underlying regressions.

TABLES

Table 1: Formal and Informal Employment in Brazil

	Formal (with *carteira de trabalho*)	**Informal**
Take-home pay	Employee social security contribution deducted from gross pay (7.5–14% by earnings bracket; about 7.5–10% at the earnings where inspections bind); income tax above the exemption threshold (7.5–27.5% by bracket)	No payroll deductions
Dismissal	Advance notice; FGTS balance and, when eligible, a 40% dismissal penalty and unemployment insurance	No statutory severance or unemployment insurance
Annual benefits	Thirteenth salary; paid annual leave plus a one-third vacation bonus	No statutory entitlement
Social insurance	Pension accrual; sickness, disability, maternity, and work-accident coverage	No employer-linked accrual or coverage
Maternity	120 days of paid leave and job protection during pregnancy and after childbirth	No statutory leave or job protection
Hours	Statutory limits on hours and overtime; formal part-time contracts subject to legal rules	Hours arranged outside the formal labor code
Enforcement	Employment rights enforceable under the CLT and through labor courts	Employment relationship is unregistered
Non-statutory amenities	May include meal vouchers, health insurance, childcare, profit sharing, and other collectively bargained or employer-provided benefits	Not guaranteed

Notes: This table summarizes the main differences between formal and informal employment in Brazil. Formal employment requires a registered labor contract (*carteira de trabalho*) and provides statutory benefits and protections, including severance pay, paid leave, and social insurance, but entails payroll deductions that reduce take-home pay. Informal employment does not provide these statutory protections or require payroll deductions. Beyond statutory benefits, formal jobs may offer additional amenities that vary across establishments and collective bargaining agreements. I exploit this variation to examine how non-wage job attributes affect workers' decisions to formalize. Appendix A provides further institutional details.

Table 2: Inspected and Never-Inspected Establishments, Before and After Matching

Sample of establishments:	(1)	(2)	(3)	(4)	(5)	(6)
		All			**Matched**	
	Inspected	Control	Difference	Inspected	Control	Difference
Firm age	7.200	9.558	−2.358	7.014	7.948	−0.934
	(7.946)	(8.905)		(7.622)	(8.059)	
Annual number of employees	23.710	5.415	18.295	11.719	9.918	1.801
	(846.635)	(13.017)		(16.238)	(14.366)	
Mean log wage	6.980	7.060	−0.080	6.909	6.899	0.010
(all workers)	(0.433)	(0.387)		(0.358)	(0.356)	
Share covered by a CBA	0.217	0.122	0.095	0.222	0.175	0.047
	(0.412)	(0.327)		(0.415)	(0.380)	
Share women	0.439	0.465	−0.026	0.458	0.432	0.026
	(0.355)	(0.414)		(0.360)	(0.373)	
Mean worker age	31.642	34.196	−2.554	30.867	31.771	−0.904
	(6.849)	(9.486)		(6.672)	(7.342)	
Mean worker tenure	21.124	31.373	−10.249	19.401	22.363	−2.962
(in months)	(24.409)	(37.000)		(20.150)	(24.426)	
Mean contracted hours	43.000	42.683	0.317	43.296	43.262	0.035
	(3.535)	(4.649)		(2.939)	(3.139)	
Share full-time workers	0.948	0.938	0.010	0.963	0.964	−0.001
	(0.177)	(0.209)		(0.147)	(0.146)	
Share below high school	0.427	0.342	0.085	0.460	0.480	−0.020
	(0.381)	(0.405)		(0.383)	(0.393)	
Share high school	0.478	0.553	−0.074	0.469	0.447	0.023
	(0.368)	(0.414)		(0.371)	(0.380)	
Share above high school	0.094	0.105	−0.011	0.071	0.074	−0.003
	(0.208)	(0.251)		(0.175)	(0.188)	
Share workers aged < 25	0.302	0.244	0.058	0.329	0.302	0.028
	(0.285)	(0.330)		(0.293)	(0.302)	
Share hires from other firms	0.193	0.150	0.043	0.192	0.174	0.018
	(0.251)	(0.271)		(0.252)	(0.256)	
Gender wage gap	−0.116	−0.095	−0.021	−0.087	−0.090	0.003
(full-time workers)	(0.350)	(0.359)		(0.298)	(0.308)	
Number of establishments	279,165	5,649,365		141,887	141,887	
Establishment-years	904,367	29,704,318		454,925	455,617	

Notes: This table reports pre-inspection characteristics of inspected and never-inspected establishments. Columns 1–3 report the full sample, while columns 4–6 restrict the sample to matched establishments. Inspected establishments are matched one-to-one, within inspection cohorts, to never-inspected establishments based on establishment size, RAIS entry cohort, industry, state, and average monthly worker earnings, all measured in the year preceding inspection. Differences are calculated as the mean for inspected establishments minus the mean for never-inspected establishments. Standard deviations are reported in parentheses. Appendix Table D.2 reports balance separately by inspection cohort.

Table 3: Formalization Rates by Gender and New-Hire Definition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Gender-Specific Samples (≥ 1 informal worker of that gender)				**Both Genders Detected** (≥ 1 informal woman and man)			
New-hire window	1 month	3 months	6 months	After insp.	1 month	3 months	6 months	After insp.
Panel A. All inspected establishments								
Men (λ_m)	0.456	0.624	0.776	0.855	0.399	0.522	0.645	0.709
Women (λ_w)	0.431	0.553	0.678	0.760	0.392	0.509	0.628	0.707
Difference ($\lambda_w - \lambda_m$)	−0.025***	−0.071***	−0.098***	−0.096***	−0.008	−0.013	−0.016	−0.003
Panel B. Inspected in 2012 or later								
Men (λ_m)	0.396	0.552	0.692	0.760	0.358	0.466	0.575	0.635
Women (λ_w)	0.400	0.523	0.642	0.718	0.339	0.448	0.559	0.631
Difference ($\lambda_w - \lambda_m$)	0.004	−0.029*	−0.050**	−0.042*	−0.019	−0.018	−0.016	−0.004

Notes: This table reports estimated formalization rates by gender. For gender d, the rate is the estimated number of new formal hires following an inspection relative to the average number of informal workers detected. New formal hires appear at the inspected establishment after the inspection and held no formal contract anywhere in Brazil in the preceding month. Columns (1)–(4) condition on detecting at least one informal worker of the relevant gender; columns (5)–(8) condition on detecting at least one informal woman and at least one informal man. The windows cover admissions within one, three, and six months and during the remaining inspection year. Panel A includes all inspections; Panel B includes inspections conducted in 2012 or later. Estimates use matched inspected and never-inspected establishments. Standard errors are clustered by establishment, and gender differences are estimated using a stacked specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Formalization Rates by Industry Gender Composition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Female-Dominated Industries				**Male-Dominated Industries**			
New-hire window	1 month	3 months	6 months	After insp.	1 month	3 months	6 months	After insp.
Panel A. All inspected establishments								
Men (λ_m)	0.366	0.470	0.570	0.634	0.530	0.750	0.945	1.037
Women (λ_w)	0.421	0.533	0.649	0.730	0.459	0.608	0.761	0.844
Difference ($\lambda_w - \lambda_m$)	0.055***	0.063***	0.079***	0.096***	−0.071***	−0.142***	−0.184***	−0.193***
Panel B. Inspected in 2012 or later								
Men (λ_m)	0.320	0.420	0.509	0.562	0.467	0.677	0.864	0.948
Women (λ_w)	0.391	0.505	0.618	0.695	0.433	0.588	0.727	0.802
Difference ($\lambda_w - \lambda_m$)	0.071***	0.085***	0.109***	0.133***	−0.035	−0.088**	−0.137***	−0.146***

Notes: This table reports estimated formalization rates by gender and industry gender composition. For gender d, the rate is the estimated number of new formal hires following an inspection relative to the average number of informal workers detected. New formal hires appear at the inspected establishment after the inspection and held no formal contract anywhere in Brazil in the preceding month. Columns (1)–(4) include two-digit industries in which the median establishment female employment share exceeds one third over 2003–2021; columns (5)–(8) include the remaining industries. The windows cover admissions within one, three, and six months and during the remaining inspection year. Each rate conditions on detecting at least one informal worker of the relevant gender. Panel A includes all inspections; Panel B includes inspections conducted in 2012 or later. Standard errors are clustered by establishment, and gender differences are estimated using a stacked specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Formalization Log-Odds: Wage Coefficients

	(1)	(2)	(3)	(4)	(5)	(6)
	Establishment-premium specification				**Wage-premium specification**	
	Men		Women		Men	Women
Establishment pay premium ψ_{dj}	3.17***	1.67***	3.08***	1.42***		
	(0.65)	(0.27)	(0.92)	(0.37)		
Wage premium Δw_{dj}					2.63***	2.84**
					(0.86)	(1.19)
Estimator	2SLS	OLS	2SLS	OLS	2SLS	2SLS
Amenity differences (4)	✓	✓	✓	✓	✓	✓
CBA clause groups (15)	✓	✓	✓	✓	✓	✓
Industry × state FE	✓	✓	✓	✓		
Industry FE, state FE					✓	✓
Inspection-year FE	✓	✓	✓	✓	✓	✓
First-stage F-statistic	2529	—	1610	—	1650	1268
N (establishments)	13,059	19,930	9,683	15,353	7,955	6,093

Notes: This table reports estimates of the wage coefficients in the formalization take-up model, where the dependent variable is the log-odds of six-month formalization, $\tilde{\lambda}_{dj}$, estimated separately by gender. Columns (1)–(4) estimate equation (14), using the establishment pay premium ψ_{dj} and absorbing the industry–state informal wage benchmark through industry × state fixed effects. Columns (1) and (3) report the preferred 2SLS estimates, instrumenting each gender's establishment pay premium with that of the other gender at the same establishment, while columns (2) and (4) report OLS estimates. Columns (5)–(6) estimate equation (13), using the formal–informal wage premium $\Delta w_{dj} = \psi_{dj} - \pi_{ds(j)}$, instrumented with the corresponding premium constructed using the other gender's establishment pay effect, and including separate industry and state fixed effects. All specifications control for four formal–informal amenity differences, fifteen CBA clause groups, total clause intensity, CBA coverage, and inspection-year fixed effects. Appendix Tables J.1 and J.2 report the corresponding amenity and CBA coefficients from the preferred specification. The sample includes establishments inspected between 2012 and 2018 with at least one informal worker of the relevant gender. Sample sizes differ because 2SLS requires an estimated establishment pay premium for both genders, while columns (5)–(6) additionally require an observed informal wage benchmark. Standard errors clustered at the establishment level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Relative Monetary Value of Formal-Contract Amenities, by Gender

	(1)	(2)	(3)	(4)	(5)
	Men ($\hat{\delta}_m = 3.17$)		**Women** ($\hat{\delta}_w = 3.08$)		
Amenity	$\hat{\beta}_{m\ell}$	$\text{RMV}_{m\ell}$	$\hat{\beta}_{w\ell}$	$\text{RMV}_{w\ell}$	$\text{RMV}_{w\ell} - \text{RMV}_{m\ell}$
Panel A. Amenity differences (formal − informal benchmark)					
Δ Part-time share	−0.03	−0.01	0.73	0.21	0.22
Δ Female manager	−0.81***	−0.29	0.29*	0.09	0.38***
Δ Interpersonal task	−0.16*	−0.05	−0.24**	−0.08	−0.03
Δ Manual task	0.21	0.06	−0.87***	−0.33	−0.39***
Panel B. Collective-agreement clause groups					
CBA negotiation	0.40	0.12	1.79**	0.44	0.32
Contract & termination	0.32	0.10	0.66	0.19	0.10
Leaves	0.13	0.04	0.65*	0.19	0.15
Harassment & discrimination	0.01	0.00	0.63	0.18	0.18
Flexibility & part-time	−0.25	−0.08	0.60*	0.18	0.26**
Other clauses	−0.32	−0.11	0.59	0.17	0.28*
Union provisions	−0.89	−0.33	0.47	0.14	0.47
Maternity & childcare	0.12	0.04	0.47	0.14	0.10
Work organization	0.20	0.06	0.00	0.00	−0.06
Employment protection	−0.04	−0.01	−0.44	−0.15	−0.14
Work shifts	0.43	0.13	−0.54	−0.19	−0.32
Wage & compensation	−0.02	−0.01	−0.66	−0.24	−0.23
Health & safety	−0.01	−0.00	−0.67	−0.24	−0.24
Bonuses & profit-sharing	−0.09	−0.03	−0.71	−0.26	−0.23
Social benefits	0.06	0.02	−0.82	−0.30	−0.32
Total clause intensity	0.01	0.00	−0.01	−0.00	−0.01

Notes: The table reports relative monetary values, $\text{RMV}_{d\ell} = 1 - \exp(-\hat{\beta}_{d\ell}/\hat{\delta}_d)$, for all non-wage regressors in the preferred 2SLS specification in columns (1) and (3) of Table 5. The $\hat{\beta}_{d\ell}$ columns reproduce the underlying coefficients from Appendix Tables J.1 and J.2, with standard errors clustered at the establishment level. RMVs use the gender-specific wage sensitivities, $\hat{\delta}_m = 3.17$ and $\hat{\delta}_w = 3.08$. Column (5) reports women–men differences in RMVs. Inference for these differences is based on an establishment-level cluster bootstrap that re-estimates the full two-stage procedure in $B = 1{,}000$ replications. Because RMVs are nonlinear in $\hat{\beta}_{d\ell}/\hat{\delta}_d$, significance of the underlying coefficient does not imply significance of the gender difference. RMV levels in columns (2) and (4) are reported as point estimates without statistical inference. Clause-group indicators are included jointly and are correlated. Panel A contains shares and standardized task measures; Panel B contains indicators for each clause group. Total clause intensity is the average number of clauses over the three years preceding the inspection. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Model-Implied and Observed Formalization Rates by Workplace Type

	(1)	(2)	(3)	(4)
	Female-Dominated		**Male-Dominated**	
	Model-implied	Observed	Model-implied	Observed
Men	61.3	50.9	73.3	86.4
Women	70.9	61.8	54.1	72.7
Difference (W − M)	+9.6	+10.9***	−19.2	−13.7***

Notes: This table compares model-implied and observed six-month formalization rates for men and women across workplace types, in percent. Industries are classified as female-dominated if their median establishment female employment share exceeds one third over 2012–2018, following Section 4. Model-implied rates are unweighted averages of establishment-level fitted probabilities, $\Lambda^{-1}(\hat{\tilde{\lambda}}_{dj})$, from the preferred specification. Observed rates are taken from Panel B of Table 4 for inspections conducted in 2012 or later. These rates are estimated as the event-study ratio $\hat{\theta}_0^d/\bar{N}_d$, as defined in Section 4.1, and therefore need not coincide in levels with the model-implied probabilities. Women–men differences in observed rates are estimated using the stacked specification in Table 4, with standard errors clustered at the establishment level. Statistical significance is reported only for observed differences. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Counterfactual Formalization Conditional on an Opportunity

	(1)	(2)	(3)	(4)	(5)	(6)
	Female-Dominated Industries			Male-Dominated Industries		
	Men	Women	Gap	Men	Women	Gap
Panel A. Baseline						
Predicted formalization	61.3	70.9	9.6	73.3	54.1	−19.2
Panel B. Equalizing Preferences						
Wage coefficient	61.5	70.9	9.5	73.3	54.1	−19.3
Establishment amenities	53.7	70.9	17.2**	73.3	58.2	−15.2**
Collective-agreement provisions	61.9	70.9	9.0	73.3	53.4	−19.9
All preferences	54.5	70.9	16.4	73.3	57.6	−15.8
Panel C. Equalizing the Formal Wage Premium						
Formal wage premium	63.1	70.9	7.8***	73.3	53.5	−19.8**
Panel D. Minimum-Wage Increase of 10%						
All establishments, $\rho = 0$	66.0	74.8	8.8	77.0	58.2	−18.9
All establishments, $\rho = 0.66$	62.9	72.3	9.3	74.6	55.5	−19.1
Bottom pay quartile, $\rho = 0$	62.6	72.0	9.4	74.2	55.0	−19.1
Bottom pay quartile, $\rho = 0.66$	61.8	71.3	9.5	73.6	54.4	−19.2
Panel E. Increasing Provision of Valued Job Attributes						
Maternity, leaves, harassment, and flexibility	61.3	73.8	12.5***	73.3	56.9	−16.4***
CBA negotiation	62.4	75.0	12.6	74.2	58.0	−16.2*

Notes: The table reports model-predicted formalization rates conditional on a formalization opportunity, in percent. The gap is the women's minus men's formalization rate, in percentage points. Female-dominated industries are two-digit industries whose median establishment female employment share exceeds one third over 2012–2018. Panel A reports baseline predictions. Panel B replaces the indicated preference parameters with those of the reference gender, holding job characteristics fixed. Panel C assigns the minority group the reference group's formal wage premium, holding preferences and non-wage attributes fixed. Panel D raises formal wages by 10% under alternative assumptions about informal-wage pass-through, following Derenoncourt et al. (2025). Panel E raises the indicated collective-agreement provisions to the sample mean at establishments initially below it. The CBA-negotiation group captures rules governing the negotiation process. All exercises hold formal-job opportunities and other components of the estimated environment fixed. Statistical significance for the reported gender gaps refers to the change in the gap relative to its corresponding baseline and is based on 1,000 establishment-level bootstrap replications: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: What Explains the Gender Gap in Stated Formalization Take-Up

	(1) Endowments	(2) Coefficients	(3) Interaction	(4) Total
Household and care	−0.024	−0.032	0.010	−0.046
	(0.020)	(0.061)	(0.024)	(0.067)
Children	−0.005	−0.004	0.006	−0.003
	(0.008)	(0.046)	(0.011)	(0.047)
Knowledge and trust	−0.002	−0.153	0.000	−0.155
	(0.007)	(0.154)	(0.005)	(0.153)
Formal experience	−0.002	0.065	−0.004	0.059
	(0.007)	(0.094)	(0.010)	(0.090)
Demographics	−0.001	0.064	−0.011	0.052
	(0.008)	(0.151)	(0.012)	(0.146)
Sorting: state and industry	0.003	0.144	−0.025	0.122
	(0.022)	(0.090)	(0.030)	(0.090)
Constant		−0.118		−0.118
		(0.233)		(0.233)
Total (raw gap W−M)	−0.031	−0.034	−0.024	−0.089
	(0.032)	(0.041)	(0.042)	(0.034)
% of raw gap	35.1	38.0	26.9	100.0

Notes: The table reports a three-fold Oaxaca–Blinder decomposition of the women–men gap in stated acceptance of a formal contract when gross pay is unchanged, estimated using a linear probability model with gender-specific coefficients. Endowments are $(X_w - X_m)'b_m$, coefficients are $X_m'(b_w - b_m)$, and the interaction is $(X_w - X_m)'(b_w - b_m)$. The sample consists of ever-informal respondents, and the gap is women minus men. Covariates are grouped as follows. *Household and care* includes inability to accept a better-paying job with fixed hours from 8 a.m. to 6 p.m. because of childcare, other family responsibilities, or a second job; weekly hours of care work and housework; a household financial buffer of at most one month; being the only household earner; receipt of government benefits; and spouse formal-employment status. Indicators for unknown financial buffer and missing partner or spouse status are included separately. *Children* includes having children, having a child aged 0–5, being the primary caregiver for one's children, and number of children. *Knowledge and trust* includes a twelve-item score measuring knowledge of *carteira* entitlements, knowledge of dismissal payments and parental leave, and trust that employers provide these benefits. *Formal experience* includes ever having held a *carteira*, currently holding one, and having worked at an inspected establishment, with a separate indicator for not remembering inspection status. *Demographics* includes age and tertiary education. *Sorting* includes state and two-digit industry indicators, normalized to sum to zero, with separate categories for unreported state and industry. Robust standard errors are reported in parentheses. Appendix Table J.8 reports the corresponding two-fold decomposition, and Appendix Table J.9 reports the three-fold decomposition for the full survey sample.

Table 10: Formal-Job Opportunities and the Demand Margin

	(1) Men		(2) Women
Formal-opportunity slope, $\hat{q}_d$	0.383		0.294
	(0.014)		(0.023)
Women − men		−0.089	
		(0.027)	
Women / men		0.77	
Cells	281		199

Notes: This table reports estimates of the implied formal-job opportunity rate by gender, based on equation (17). The analysis uses industry × state × year cells, where p_{dc} is the probability that an informal worker transitions to formal employment with the same employer within six months, measured using PNAD-C, and λ_{dc} is the corresponding inspection-based take-up rate. Regressions are estimated separately by gender, without a constant, and weighted by the number of informal worker-quarter observations used to construct p_{dc}. Under the decomposition in equation (16), the estimated coefficient $\hat{q}_d$ captures the implied formal-job opportunity rate. The women–men difference is estimated from a pooled regression allowing gender-specific slopes. Robust standard errors are reported in parentheses. Appendix J.9 reports estimates within years and using transitions across all employers.

Appendix and Supplementary Material

A. Formal and Informal Employment in Brazil

This appendix documents the statutory differences between registered and unregistered salaried employment summarized in Table 1.

A.1 Registration and Statutory Coverage

A salaried employment relationship is formal when the employer registers it in the worker's *Carteira de Trabalho e Previdência Social* (CTPS) and the corresponding administrative systems. Registration brings the relationship under the *Consolidação das Leis do Trabalho* (CLT) and associated social-insurance legislation. An employee working without this registration is informal.

Registration activates several statutory entitlements. The employer deposits 8% of monthly pay into the worker's FGTS severance account; dismissal without cause generally releases the balance and triggers a 40% employer penalty on accumulated deposits (Lei 8.036/1990). Eligible workers may receive three to five unemployment-insurance payments (Lei 7.998/1990). Formal workers also receive a thirteenth salary, paid annual leave with a one-third vacation bonus, pension accrual, sickness and disability coverage, and work-accident protection.

The CLT also regulates hours and employment conditions. Standard hours are capped at eight per day and 44 per week, with overtime paid at a premium of at least 50% (CLT Arts. 58–59). Formal part-time employment is permitted subject to statutory hour limits. The national minimum wage applies, and registered workers can enforce employment rights through the labor courts.

Family protections include 120 days of paid maternity leave and job protection from confirmation of pregnancy until five months after childbirth. Additional provisions include paternity leave and nursing breaks. These protections do not apply to unregistered employment.

Meal or food vouchers, private health insurance, childcare assistance, profit sharing, and similar amenities are not universal statutory entitlements. They may instead be provided by the employer or through collective agreements. I use their variation across establishments to estimate heterogeneity in the value of the formal job.

A.2 Payroll Deductions and Take-Home Pay

Registration also changes current compensation. Employee social-security contributions (INSS) are deducted from gross pay. Before 2020, contribution rates of 8, 9, or 11% applied by earnings bracket. Since 2020, contributions have been calculated using marginal rates of 7.5, 9, 12, and 14% on successive earnings brackets. Income tax (IRRF) is withheld above the applicable exemption threshold at marginal rates of 7.5, 15, 22.5, and 27.5%.

For earnings relevant to the inspection sample, registration at an unchanged gross wage reduces take-home pay by approximately 7.5–10%. At the 2021 minimum wage of R$1,100, the employee contribution is 7.5%; at R$1,500 it is approximately 7.9%; and at R$2,500 the employee contribution and income tax together are approximately 9.8%. This payroll deduction is the basis for the formalization vignette in the survey.[44]

A.3 Employer Payroll Obligations

Registration also generates employer-side payroll costs. These include employer social-security contributions, occupational-risk contributions, third-party payroll levies, and an 8% FGTS contribution, together with accruals for the thirteenth salary and paid vacation. Appendix Table A.1 reports the individual components.

[44]The vignette offers the *carteira* at an unchanged gross wage, so that the social-security contribution reduces take-home pay by roughly 7 to 11%. Under this offer, 61% of respondents say they would accept. When the gross wage is raised so that take-home pay is unchanged, 79% would accept. Appendix K documents the wording.

Table A.1: The Formal Employment Bundle Under the CLT

Entitlement	Content	Legal basis
Panel A: What the worker gains		
Registration in the CTPS	The employment relationship becomes legally enforceable	CLT Arts. 29, 41
Minimum wage	Floor on monthly pay	CF/88 Art. 7 IV
Regulated hours and overtime	44 hours weekly; overtime at +50%	CLT Arts. 58, 59
Paid annual leave	30 days plus a one-third constitutional bonus	CLT Arts. 129–130; CF/88 Art. 7 XVII
Thirteenth salary	One additional month's pay per year	Lei 4.090/1962
Severance fund (FGTS)	8% of pay deposited monthly; released on dismissal without cause	Lei 8.036/1990
Dismissal penalty	Employer pays 40% of the accumulated FGTS balance	Lei 8.036/1990 Art. 18
Advance notice	30 days plus 3 days per year of service, capped at 90	CLT Art. 487; Lei 12.506/2011
Unemployment insurance	3–5 monthly payments after dismissal without cause	Lei 7.998/1990
Social security (INSS)	Contributory pension, sickness and maternity benefits	Lei 8.212/1991
Maternity leave	120 days paid, with job stability	CLT Art. 392; CF/88 Art. 7 XVIII
Part-time contract	Permitted; capped at 25 hours/week until 2017, 30 thereafter	CLT Art. 58-A
Panel B: What the employer pays, as a share of the wage		
INSS employer contribution	20%	Lei 8.212/1991 Art. 22 I
Occupational risk (RAT/FAP)	1–3%, adjusted by a firm-level multiplier	Lei 8.212/1991 Art. 22 II
Third-party levies	≈5.8% (SESI, SENAI, SEBRAE, education salary, INCRA)	Lei 8.212/1991 Art. 3
FGTS	8%	Lei 8.036/1990
Direct charges	*≈35.8%*	
Accrued provisions	13th (8.33%) + vacation (8.33%) + bonus (2.78%) + charges on those (≈7%)	
Total	*≈62%*	

Notes: Firms under the simplified *Simples Nacional* regime are exempt from the 20% employer contribution and the third-party levies, so their marginal cost of a formal hire is closer to 28% of the wage. The ≈62% total is consistent with the 65% markup that Almeida & Carneiro (2012) report for Brazil ("for a net wage of R$100 the employer disburses about R$165"). The additional 10% FGTS contribution introduced by LC 110/2001 was abolished by Lei 13.932/2019, effective January 2020.

A.4 The Trade-Off Between Formal and Informal Employment Regimes

Table A.2 provides the full accounting of how registration changes the employment relationship, separating the immediate effect on take-home pay from statutory benefits, employment protections, and non-statutory amenities.

Table A.2: Formal Versus Informal Employment: The Full Trade-Off

Dimension	**Formal (*carteira assinada*)**	**Informal**
Panel A: Pay		
Take-home pay	Gross less social security (7.5–14%) and, above the threshold, income tax (7.5–27.5%)	Full gross, no deductions
Minimum wage	Binding	Not enforceable
Overtime	Premium of at least 50% beyond 44 hours	Negotiated individually
Thirteenth salary	One extra month's pay per year	None
Panel B: Insurance and savings		
Severance fund	8% of pay deposited monthly by the employer; 40% penalty added on dismissal without cause	None
Unemployment insurance	3–5 monthly payments after dismissal without cause	None
Retirement	Contributions credit toward the pension	No accrual unless the worker contributes voluntarily
Sickness and disability	Social security benefits	None
Work accidents	Covered; employer additionally liable if negligent	Out of pocket
Panel C: Job security		
Notice of dismissal	30 days, rising to 90 with tenure	None
Dismissal for cause	Employer must prove an enumerated cause	Unconstrained
Pregnancy protection	Cannot be dismissed from confirmation to 5 months post-birth	None
Legal recourse	Labor courts; 2-year limitation period	Effectively unenforceable
Panel D: Time and family		
Paid annual leave	30 days plus a one-third bonus	None
Maternity leave	120 days paid, 180 at participating firms	None
Paternity leave	5 days, 20 at participating firms	None

continued on the next page

Table A.2, continued from the previous page

Dimension	**Formal (*carteira assinada*)**	**Informal**
Hours	Capped at 8 a day, 44 a week	Unregulated
Part-time	Available but capped at 30 hours (25 before 2017), with scaled vacation	Effectively unconstrained
Schedule flexibility	Requires a collective agreement or employer initiative	Arranged directly with the employer
Multiple jobs	Permitted but constrained by contracted hours	Easily combined
Panel E: Provided but not guaranteed		
Meal voucher, health plan, life insurance, profit sharing, childcare	Common, through collective agreements or employer discretion; *not* statutory	Rare

Notes: Statutory basis: CLT (Decreto-Lei 5.452/1943) as amended by Lei 13.467/2017; Lei 8.036/1990 (severance fund); Lei 7.998/1990 (unemployment insurance); Lei 4.090/1962 (thirteenth salary); Lei 8.212/1991 and EC 103/2019 (social security); Lei 12.506/2011 (notice); Lei 11.770/2008 (extended parental leave); Lei 7.418/1985 (transport). Panel E lists the provisions that vary across formal establishments and are therefore the source of the identifying variation in Section 6.

B. Labor-Market Institutions

This appendix provides additional institutional detail on the penalty for failing to register a worker, on the costs establishments face under each response to an inspection, and on collective bargaining.

B.1 Penalty for Non-Registration

Article 47 of the CLT imposes a fine for each unregistered worker. Before November 2017, the fine was R$402.57 per worker (do Prado, Santos & Van Doornik, 2025). Lei 13.467/2017 increased the standard fine to R$3,000 per worker, with a reduced fine of R$800 for microenterprises and small firms. The fine does not discharge the employer's obligation to register the worker. Repeated non-compliance may generate additional sanctions.

Table B.1: The Article 47 Fine Relative to the Minimum Wage

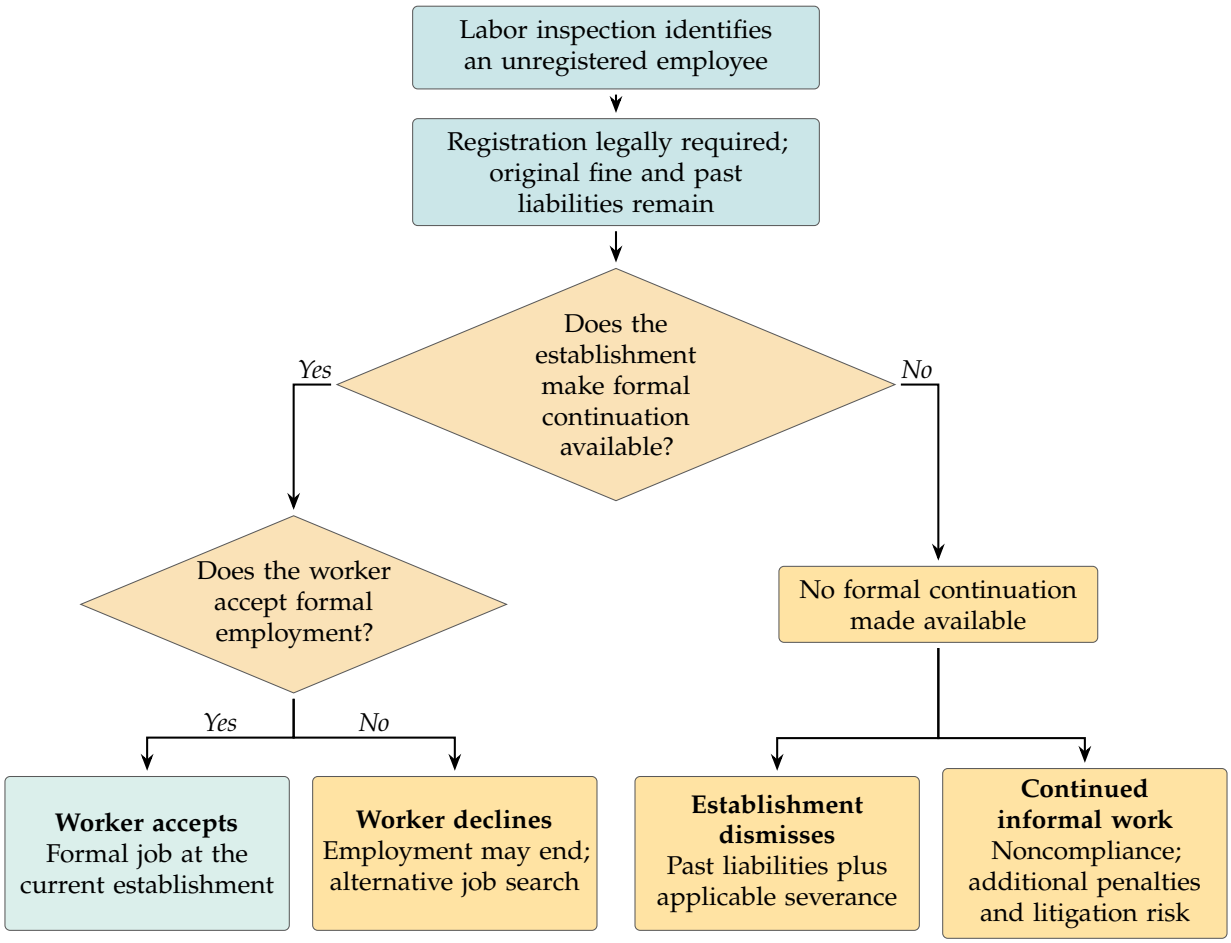
Year	Monthly minimum wage (R$)	Fine per worker (R$)	Fine / minimum wage
2003	240	402.57	1.68
2007	380	402.57	1.06
2011	545	402.57	0.74
2014	724	402.57	0.56
2017	937	402.57	0.43
2018	954	3,000	3.14
2021	1,100	3,000	2.73

Notes: Minimum-wage values are nominal monthly minimum wages. The fine is the Article 47 penalty for failure to register a worker: R$402.57 before Lei 13.467/2017 and R$3,000 thereafter for firms subject to the standard rate.

B.2 Establishment Costs by Response to an Inspection

Section 2.2 describes the three responses available to an establishment after inspectors identify an unregistered employee: formalizing and retaining the worker, dismissing the worker, or continuing the employment relationship informally. Figure B.1 summarizes the establishment's and the worker's decisions, and Table B.2 compares the costs associated with each response. The original fine and the liabilities from past informal employment are common to all three, so the responses differ in future payroll obligations, dismissal costs, the loss of the existing match, and exposure to further enforcement.

Figure B.1: Establishment and Worker Decisions Following Labor Inspections



Notes: The diagram illustrates the economic decisions following the identification of an unregistered employment relationship. All branches remain subject to the original inspection penalty and applicable liabilities from past informal employment. Formal continuation is legally compliant only if the employment relationship is properly registered. Dismissal does not eliminate past liabilities, and continued informal employment exposes the establishment to additional enforcement risks. A worker who declines formal employment leaves the inspected establishment and searches for a job elsewhere in the labor market; Section 5 treats this branch as informal employment in the same industry–state market. A worker's refusal of formal conditions does not automatically eliminate the employer's termination obligations. Table B.2 compares the costs associated with each establishment response.

Table B.2: Establishment Costs Following the Detection of Informal Employment

	Formalize and retain	**Dismiss**	**Continue informally**
Original administrative fine	Yes	Yes	Yes
Past FGTS and social security liabilities	Yes	Yes	Yes
Past unpaid labor benefits	Yes	Yes	Yes
Future formal payroll obligations	Yes	No	Not paid; liability accrues
Dismissal-related costs	No immediate dismissal	Applicable severance	If subsequently dismissed
Loss of existing employment match	No	Yes	No
Additional administrative penalties	Avoidable through compliance	Avoidable through compliance	Possible
Additional labor claims	Possible for past violations	Possible for past or termination violations	Greater ongoing exposure
MPT referral or further enforcement	Reduced through compliance	Depends on circumstances	Possible
Risk to *Simples Nacional* eligibility	Reduced through compliance	Depends on compliance	Possible with repeated violations

Notes: The table compares the consequences of alternative establishment responses after inspectors identify an unregistered employee. The original fine and liabilities associated with past informal employment are not avoided through dismissal. Future payroll costs depend on the applicable tax and employment regime. Termination costs depend on the type of separation. Additional enforcement and litigation costs are contingent rather than automatic. The table assumes that the employment relationship falls under the CLT. FGTS is the severance fund (*Fundo de Garantia do Tempo de Serviço*); MPT is the Public Labor Prosecutor's Office (*Ministério Público do Trabalho*).

B.3 Collective Bargaining

Collective bargaining provides the establishment-level variation in non-statutory amenities used in the paper. Brazil operates under a single-union system: within a given economic category and geographic area, one union represents all covered workers, regardless of union membership.[45] There are two collective bargaining instruments. A *convenção coletiva* is negotiated between a worker union and an employers' association and applies to all covered employers and workers in the relevant category and territory. An *acordo coletivo* is negotiated between a worker union and an individual firm and applies to that firm's covered workers. Firm-level agreements can supplement sectoral agreements and commonly provide additional wages and non-wage benefits (Lagos, 2026; Corradini, Lagos & Sharma, 2025; Horn, 2009; Liukkunen, 2019).

Collective agreements regulate both wages and non-wage working conditions. Negotiations typically occur annually, although agreements may remain in force for up to two years. Bargained provisions include wage floors and adjustments as well as working hours, leave, job security, food allowances, and other workplace amenities. Once negotiated, agreements are filed with the Ministry of Labor and recorded in the *Sistema Mediador*. I use 571,937 documents from 2009 to 2021 and match them to RAIS establishments using firm identifiers and the geographic scope of each agreement. The document match rate is approximately 86–87%. In the matched analysis sample, collective-agreement coverage is about 22% among inspected establishments and 18% among never-inspected controls.

The distinction between statutory and bargained provisions is important for the empirical analysis. The CLT establishes the baseline rights associated with formal registration, whereas collective bargaining generates variation in additional attributes of formal jobs across establishments and over time. I classify the text of the agreements into 137 provisions and map these provisions to the 15 amenity groups used in the regressions. Appendix Table C.1 reports the mapping.

[45] Representation is generally determined by the employer's main economic activity, although some occupations are represented by occupation-specific unions. The municipality is the smallest geographic unit of representation, but unions may cover larger territories.

C. Data Appendix

This appendix provides detailed documentation of the data sources, sample construction, and variable definitions used in the empirical analysis, including the information needed to reproduce the analysis from the underlying data.

C.1 RAIS

RAIS is an annual administrative census of formal employment in Brazil. Filing is compulsory for every establishment that employed at least one registered worker at any point during the year. Because reporting is linked to the administration of severance payments and social security benefits, the data cover nearly the universe of formal employment. Each record corresponds to a worker–establishment–year spell and reports the worker's identifier, gender, date of birth, education, race, six-digit occupation, contracted weekly hours, monthly earnings, contract type, and admission and separation dates and reasons. The data also identify the establishment's municipality, legal status, and five-digit industry.

I aggregate the worker-level records to the establishment–year level. The resulting panel contains total employment, overall and by gender; hires and separations by month; mean log monthly earnings for all workers, full-time workers, workers of each gender, and workers with tenure above and below twelve months; occupational and educational composition; the number of distinct six-digit occupations; average age and tenure; the share of workers hired from other formal jobs; and two measures of the establishment-level gender pay gap. The first is the difference in mean log monthly earnings between women and men among workers with more than twelve months of tenure; the second is the corresponding difference without the tenure restriction. The tenure-restricted measure provides a more stable proxy for the establishment's pay policy, whereas the unrestricted measure responds more quickly to changes in workforce composition.

I define a new hire as a worker admitted after the inspection who did not hold an active formal contract at any employer in the calendar month immediately preceding the inspection. A contract is active in a given month if its admission date is no later than that month and its separation date is no earlier. For January inspections, I determine prior formal employment using contracts active in December of the preceding year. This restriction aligns the numerator with the inspection records by excluding workers who merely move between formal employers and isolating admissions of workers who were

outside formal employment when the inspection occurred. I construct hiring outcomes over one-, three-, and six-month windows following the inspection, as well as over the remainder of the inspection year.

C.2 Labor Inspection Records

The inspection release comprises two administrative regimes: a legacy regime running from August 1995 to November 2016 and a current regime from December 2016 onward, joined on the inspection-report identifier. Establishments are identified by their tax identifier, and the records also report the industry, the municipality, and the administrative unit that conducted the inspection.

What "irregular workers found" means. The data report four worker counts, each split by gender: total workers at the establishment; workers reached by the inspection; *workers found in an irregular situation during the inspection*; and workers *registered during the inspection*. I build the treatment variable from the third: an establishment is treated if its first inspection recorded a positive count of irregular workers.

The count of irregular workers refers to workers reached by the inspector at the time of the inspection. Informal workers who are not reached are not included. As a result, the number of workers detected may understate the total number of informal workers employed at the establishment, which would mechanically raise the formalization rate constructed using this count as the denominator.

Gender. The inspection records report the number of workers found in an irregular employment relationship separately for women and men. This gender breakdown allows me to construct gender-specific measures of formalization.

Classification of inspection outcomes. I classify inspections according to whether they identify workers in an irregular employment relationship. An establishment enters the treatment group when its first inspection identifies at least one such worker. Inspections that identify other labor violations but no informal workers are analyzed separately in the placebo analysis.

C.3 Analysis Samples

I construct four samples to distinguish the effects of inspections that detect informal employment from general inspection effects, within-firm spillovers, and differences in the composition of inspected establishments.

Matched sample. For the event-study analysis, I match each treated establishment one-to-one to a never-inspected establishment within treatment cohort, state, industry, lagged employment-size bin, and RAIS entry cohort. The employment bins are fewer than 5 workers, 5–9, 10–19, 20–49, 50–99, 100–249, and 250 or more workers. Within these cells, matches are selected based on similarity in mean log earnings during the year preceding treatment. Controls belong to firms with no establishment ever inspected and found to employ informal workers, limiting contamination from within-firm responses.

Ever-treated sample. The ever-treated sample includes establishments in which an inspection detected at least one informal worker. Treatment occurs at the first such inspection; subsequent inspections do not generate additional treatment events.

Placebo sample. The placebo sample includes establishments whose first inspection identified a violation involving disability quotas, discrimination, or rest periods, but detected neither informal employment nor any other labor violation. The first qualifying inspection defines the placebo event. I match placebo establishments to never-inspected establishments using the same procedure as in the main analysis, including the treatment-cohort cells, exact matching characteristics, employment-size bins, and pre-event earnings criterion. This comparison isolates inspection and enforcement activity without a requirement to formalize workers.

Spillover sample. The spillover sample includes non-inspected establishments belonging to firms with another establishment in which an inspection detected informal employment. The firm's first such inspection defines the event year. I match spillover establishments to never-inspected establishments using the same procedure as in the main analysis, including the treatment-cohort cells, exact matching characteristics, employment-size bins, and pre-event earnings criterion. This comparison tests whether the response extends to other establishments within the inspected firm.

C.4 The Formalization Rate

In the descriptive tables of Section 4, λ is an event-study coefficient divided by a denominator mean: $\hat{\lambda}_d = \hat{\theta}_0^d / \bar{N}_d$, where $\hat{\theta}_0^d$ is the inspection-year coefficient from equation (1) run on new hires of gender d with standard errors clustered at the establishment level, and $\bar{N}_d$ is the mean number of informal workers of gender d recorded at treated establishments at event time zero within the same estimation sample. The gender difference is tested on a stacked specification in which each gendered outcome is pre-scaled by its own denominator, with establishments clustered jointly across genders, so that the covariance between the two rates is accounted for.

In the structural estimation of Section 5.4, $\hat{\lambda}_{dj}$ is instead the establishment-level single difference of equation (12), winsorized at 0.001 and 0.999 before the logit transform.

C.5 The Amenity Measures

Establishment-level amenities from RAIS. The RAIS amenity measures used in the estimation are the share of workers contracted for fewer than forty hours per week, an indicator for whether the establishment's highest-paid worker is a woman, and the mean nonroutine-manual and nonroutine-interpersonal task content of the establishment's occupational mix. The task measures are constructed from O*NET work activities and mapped to Brazilian occupations through a CBO crosswalk.

The female-leadership measure is defined by pay rank rather than by occupation code. The natural alternative, the share of workers in managerial occupations who are women, is unavailable for most of the sample: only 20% of RAIS establishment-years contain any worker in a managerial occupation, and coverage falls to 9% among establishments with fewer than five workers. Since the median inspected establishment employs roughly seven workers, that measure would be missing for about two-thirds of the estimation sample and would retain the largest establishments disproportionately. The top-earner indicator is defined for every establishment with recorded earnings and is strongly related to the occupational measure where both exist, with a correlation of 0.66 and nearly the same mean (0.40 against 0.44). Both are coarse markers of the same underlying feature: where the occupational measure is available, 82% of establishments report a value of exactly zero or one, because most establishments have only one managerial worker.

A fifth measure, an indicator for whether any worker at the establishment recorded a parental-leave spell exceeding the statutory 120 days, is constructed but is *not* used in the estimation. It is unavailable before 2007, fewer than 1% of establishment-years record such

a spell, and because the indicator fires on any single worker it rises more than sevenfold across establishment-size quartiles—so at a small establishment it reflects whether the firm happened to employ a woman of childbearing age rather than the generosity of its leave policy.

Informal counterparts from PNAD-C. I construct the corresponding informal-market measures by industry and state using PNAD-C. These are the share of informal workers employed part-time, the share of informal managers who are women, and the worker-weighted mean of each task measure. For each amenity, the variable entering the estimation is the difference between the establishment-level RAIS measure and its industry-by-state informal counterpart, as defined in equation (8).

The female-leadership difference is the one case in which the two sides are not the same statistic. PNAD-C identifies occupations but not employers, so an establishment-level notion of the highest-paid worker has no survey analogue, and the informal benchmark must be the occupational share. The difference therefore nets an establishment-level indicator against a market-level share. Because the informal benchmark is constant within an industry–state cell, this affects only the level of the difference and not the within-cell variation that identifies $\beta_{d\ell}$; the corresponding coefficient should nonetheless be read as pricing the presence of a woman at the top of the establishment's pay distribution relative to the prevalence of women in management in the same informal market, rather than as a comparison of two identically constructed shares.

Collective agreements and clause groups. Collective agreements filed in the *Sistema Mediador* between 2009 and 2021 measure the amenities provided by formal jobs. I match agreements to RAIS through firm identifiers and assign them to establishments based on sector and municipality. The agreements cover approximately 20% of private establishments and half of formal workers. I classify their text into 137 provisions and aggregate these provisions into the fifteen groups reported in Table C.1. For each group, the regression measure is an indicator equal to one if the applicable agreement contains at least one provision from that group in any of the three pre-inspection years.

Table C.1: Collective-Bargaining Clause Groups

#	Group	Clauses	Content
1	Wage and compensation	9	Wage adjustments, payment rules, and other wage provisions
2	Bonuses and profit-sharing	14	Profit sharing, bonuses, supplements, and premia
3	Social benefits	13	Meal, transportation, education, health, and childcare allowances
4	Contract and termination	4	Contract types, separation procedures, and advance notice
5	Maternity and childcare	6	Maternity and adoption leave, day-care allowances, and job protection for mothers
6	Leaves	6	Family, unpaid, and other leave entitlements
7	Harassment and discrimination	4	Harassment, equal opportunity, and women's-work provisions
8	Flexibility and part-time	6	Fixed-term contracts, reduced hours, and special or weekend schedules
9	Work organization	11	Staffing, working conditions, and task assignment
10	Employment protection	10	Job-stability provisions
11	Work shifts	7	Shift patterns, rotation, and night work
12	Health and safety	19	Injury prevention, protective equipment, and occupational health
13	Union provisions	12	Union organization and union–firm relations
14	Negotiation process	6	Rules governing collective bargaining
15	Other	10	Vacation and residual provisions
	Total	**137**	

Notes: This table reports the fifteen collective-bargaining clause groups used in the regressions. The 137 provisions are classified from agreements filed in the *Sistema Mediador* and assigned to the mutually exclusive groups shown. For each group, the regression measure is an indicator equal to one if the applicable agreement contains at least one provision from that group in any of the three years preceding the inspection.

D. Labor Inspections: Coverage and Enforcement

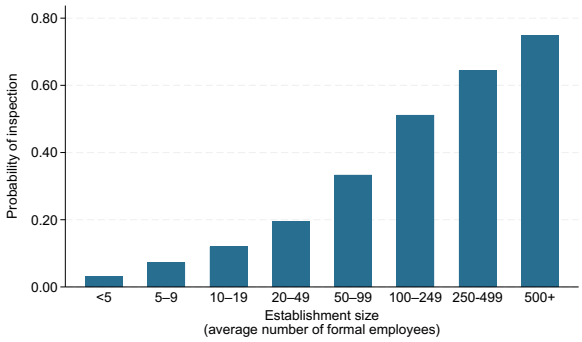
The inspection records cover enforcement actions conducted by the Ministry of Labor between 1995 and 2023. They identify the inspected establishment, the inspection date, the violations detected, and the workers covered. Before 2016, worker counts are reported separately by registration status—registered or unregistered—and, within each status, for women, children, and all workers. From 2016 onward, the records report unregistered workers separately for women and men. The records also identify repeat inspections. I define treatment as the establishment's first inspection, and the treated sample consists of establishments in which that inspection detects at least one unregistered worker.

D.1 Scale and Coverage

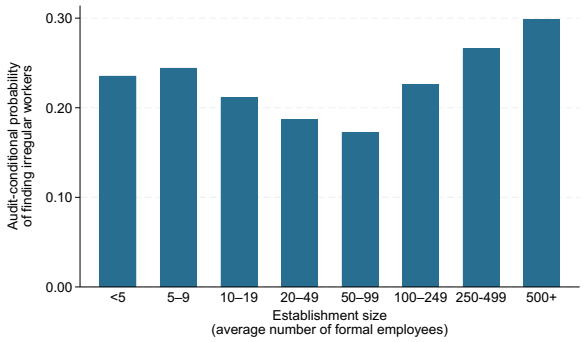
The analysis window contains 279,165 establishments in which an inspection detected informal workers between 2005 and 2018, corresponding to 904,367 establishment-years. The comparison pool contains 5,649,365 never-inspected establishments. Across the inspection records through 2021, 817,342 establishments were found employing informal workers at their first inspection.

Inspected establishments differ systematically from the broader population of establishments: they are larger, younger, pay slightly lower wages, and are more likely to be covered by a collective agreement. Figure D.1 shows that both inspection incidence and the probability of detecting informal employment increase sharply with establishment size. Conditional on detection, however, informal workers account for a larger share of the workforce at smaller establishments.

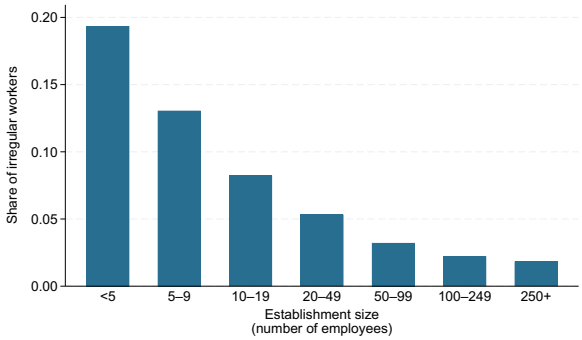
Figure D.1: Inspections and Informality by Establishment Size



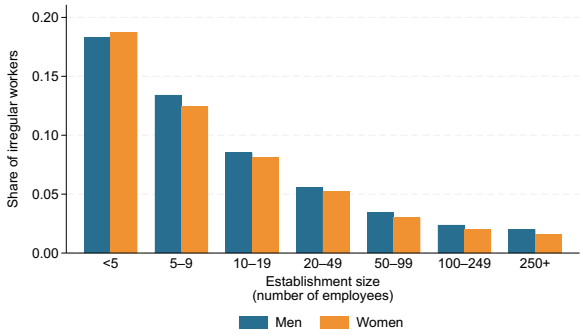
(a) Probability of inspection



(b) Probability of detecting informal workers



(c) Informal-worker share



(d) Informal-worker share, by gender

Notes: This figure describes inspections and detected informal employment by establishment size over 2003–2021. Panels (a) and (b) pool private-sector establishment–years in RAIS: Panel (a) reports the probability that an establishment is inspected and Panel (b) the probability that an inspection detects informal workers, with size measured as average formal employment and grouped into eight bins. Panels (c) and (d) pool inspections over the same years: Panel (c) reports detected informal workers as a share of all workers covered by the inspection and Panel (d) the corresponding shares by gender, with size measured by the number of workers recorded at the inspection and grouped into the seven bins used in the matching procedure.

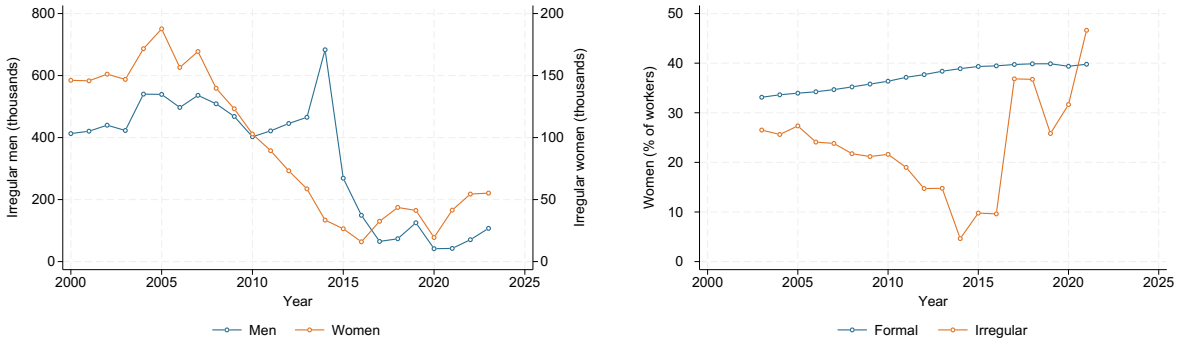
D.2 Workers Detected during Inspections

The number of informal workers detected is strongly right-skewed and forms the denominator of the estimated take-up rate. Across all first inspections detecting informal workers between 1995 and 2021, the mean number detected is 4.9 and the median is 2; 37.8% of these inspections detect exactly one worker and 58.1% detect at most two. In the matched sample the median inspection detects two informal workers, 43.9% detect exactly one, and 65.7% detect no more than two. The mean is 3.29 workers per inspection. Conditioning on the gender for which take-up is defined, inspectors detect an average of 3.27 informal men at the 109,275 establishments where at least one informal man is found and 2.23 informal

women at the 84,699 establishments where at least one informal woman is found. The take-up rates exclude matched pairs above the 99th percentile of informal workers detected; in the resulting event-study sample, the denominators are 2.37 informal men and 1.93 informal women.

Figure D.2 describes the gender composition of detected informal workers. Women remain a minority throughout the sample, but their share increases after 2014, consistent with the shift toward inspections in female-dominated industries documented in Section 4.4.

Figure D.2: Gender Composition of Detected Informal Workers



(a) Informal workers detected, by gender

(b) Female share of detected workers

Notes: This figure describes the gender composition of unregistered workers detected during labor inspections. Panel (a) reports the numbers of unregistered men (left axis) and women (right axis) detected in each year of inspection, 2000–2023, in thousands, and is computed on the inspection records alone. Panel (b) reports women as a percentage of formal employment and of all unregistered workers detected in each year, 2003–2021, the window over which the inspection records are linked to RAIS. Before 2016, the records distinguish registered from unregistered workers and report, within each group, the numbers of women, children, and all workers.

D.3 Regularization at the Next Inspection

Nearly half of establishments in which unregistered workers are detected are inspected again: 48.5% in the full records and 48.0% in the matched sample. The median interval between the first and second inspections is two years. Table D.1 reports outcomes at the second inspection.

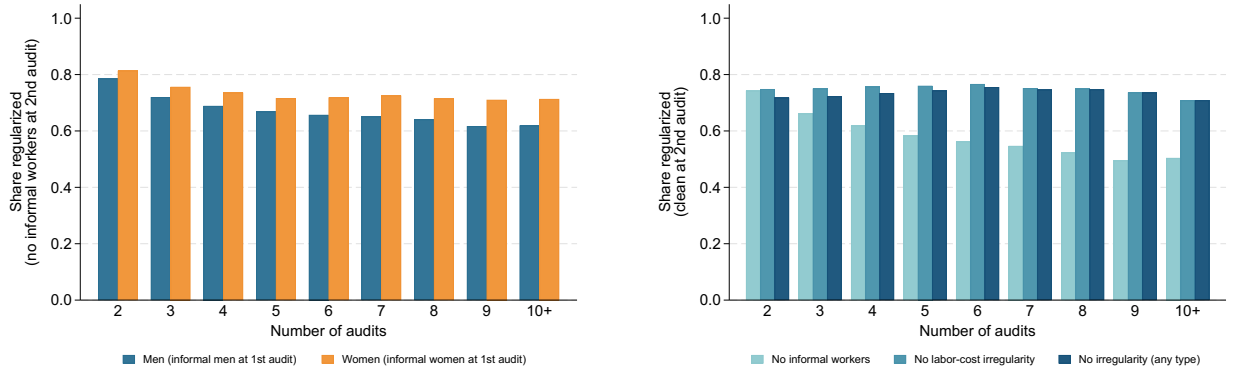
Table D.1: Regularization at the Second Inspection

Sample of inspected establishments:	(1) **All** (1995–2021)	(2) **Matched** (2005–2018)
Establishments (informal at 1st inspection)	817,342	141,887
Inspected ≥ 2 times (%)	48.5	48.0
Number inspected ≥ 2 times	396,144	68,074
Regularized at 2nd inspection (% of those inspected ≥ 2 times):		
No informal workers	66.3	70.0
No labor-cost irregularity	75.0	76.7
No irregularity (any type)	72.6	71.7
Regularized informal workers at 2nd inspection, by gender (% of those at risk for that gender):		
No informal women (women found at 1st inspection)	77.0	80.1
No informal men (men found at 1st inspection)	72.6	78.5
Informal workers as % of the workforce, where informal workers are found again at 2nd inspection:		
Number of establishments	133,532	20,404
All workers, 1st inspection	48.2	31.9
All workers, 2nd inspection	34.9	25.5
Women, 1st inspection	62.6	47.5
Women, 2nd inspection	47.2	38.9
Men, 1st inspection	55.9	42.0
Men, 2nd inspection	41.4	34.5
Informal workers as % of the workforce, all establishments inspected ≥ 2 times:		
Number of establishments	396,144	68,074
All workers, 1st inspection	47.4	32.5
All workers, 2nd inspection	11.7	7.6
Women, 1st inspection	61.5	47.5
Women, 2nd inspection	10.6	7.4
Men, 1st inspection	53.7	41.4
Men, 2nd inspection	11.2	7.3
Median years between 1st and 2nd inspection	2	2
Mean years between 1st and 2nd inspection	3.1	3.3

Notes: This table reports regularization at the second inspection among establishments where informal workers were initially detected. Column (1) covers all inspected establishments (1995–2021); column (2) restricts the sample to the matched event-study sample (2005–2018). Dates refer to first inspections; subsequent inspections may occur later. Regularization measures are not nested: informal-worker counts and regulatory irregularities are recorded separately. Gender-specific regularization rates condition on detecting informal workers of that gender at the first inspection. Informal-worker shares divide unregistered workers by the workforce recorded at each inspection, separately by gender. The first share block includes establishments with informal workers detected at both inspections, requiring detection of the corresponding gender in gender-specific rows. The second includes all re-inspected establishments, restricting gender-specific rows to those with informal workers of that gender initially detected and assigning zero at the second inspection when none are detected. Within each group, shares at both inspections use the same establishments. Percentages exclude missing outcomes. Inspection counts refer to distinct inspection years.

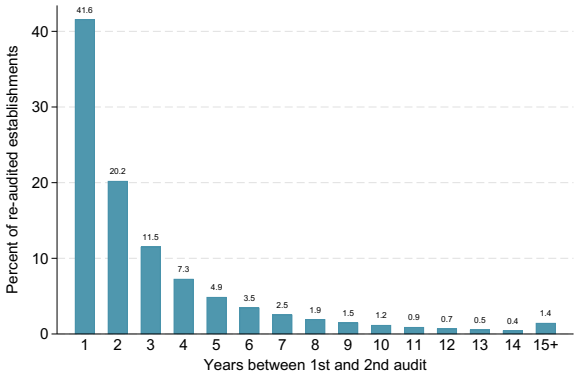
Subsequent inspections indicate substantial compliance. Approximately 70% of re-inspected establishments have no informal workers at the second inspection. Regularization is slightly more common for women: 80.1% of establishments in which informal women were initially detected have no informal women at the next inspection, compared with 78.5% for men. Figure D.3 relates detected irregularities to inspection history.

Figure D.3: Regularization by Inspection History



(a) By number of inspections and gender

(b) Alternative regularization measures



(c) Time between inspections

Notes: This figure reports outcomes at the second inspection for establishments whose first inspection detected informal workers, pooling first inspections from 1995 to 2021; every panel is computed among the establishments that were inspected again. In Panels (a) and (b) the horizontal axis is the total number of inspections the establishment receives, capped at ten or more. Panel (a) reports the share with no informal workers of the relevant gender at the second inspection, separately for establishments where informal men and where informal women were found at the first. Panel (b) reports the share regularized at the second inspection under three definitions: no informal workers, no labor-cost irregularity, and no irregularity of any type. Panel (c) shows the distribution of the number of years between the first and second inspections.

D.4 Balance by Inspection Cohort

Table 2 reports balance in the pooled matched sample, while Table D.2 reports the same comparison separately for the 2005–2010, 2011–2014, and 2015–2018 inspection cohorts.

Table D.2: Balance by Inspection Cohort

Inspection Cohort:	(1)	(2) **2005–2010**	(3)	(4)	(5) **2011–2014**	(6)	(7)	(8) **2015–2018**	(9)
	Inspected	Control	Diff.	Inspected	Control	Diff.	Inspected	Control	Diff.
Firm age	6.887	8.341	−1.454	6.896	7.633	−0.737	7.398	7.773	−0.375
	(7.408)	(8.156)		(7.722)	(8.028)		(7.809)	(7.920)	
Annual number of employees	11.146	9.512	1.634	13.048	10.890	2.158	10.851	9.253	1.597
	(15.819)	(13.871)		(17.149)	(15.632)		(15.496)	(13.188)	
Mean log wage	6.811	6.805	0.006	6.963	6.948	0.015	7.048	7.037	0.012
(all workers)	(0.342)	(0.345)		(0.356)	(0.351)		(0.332)	(0.327)	
Share covered by a CBA	0.157	0.133	0.024	0.246	0.191	0.055	0.202	0.162	0.040
	(0.364)	(0.340)		(0.431)	(0.393)		(0.402)	(0.369)	
Share women	0.452	0.427	0.025	0.463	0.430	0.034	0.461	0.447	0.014
	(0.365)	(0.376)		(0.352)	(0.365)		(0.363)	(0.377)	
Mean worker age	30.506	31.542	−1.035	30.781	31.653	−0.873	31.831	32.484	−0.653
	(6.585)	(7.392)		(6.548)	(7.104)		(6.973)	(7.563)	
Mean worker tenure	18.766	22.913	−4.148	19.409	21.096	−1.686	20.839	23.181	−2.342
(in months)	(19.950)	(25.430)		(20.457)	(23.136)		(20.026)	(24.058)	
Mean contracted hours	43.300	43.251	0.049	43.326	43.324	0.002	43.241	43.185	0.055
	(3.068)	(3.258)		(2.744)	(2.927)		(2.948)	(3.198)	
Share full-time workers	0.964	0.964	−0.000	0.963	0.967	−0.004	0.959	0.958	0.001
	(0.148)	(0.148)		(0.144)	(0.138)		(0.151)	(0.157)	
Share below high school	0.536	0.552	−0.016	0.421	0.450	−0.029	0.351	0.364	−0.013
	(0.382)	(0.390)		(0.374)	(0.386)		(0.365)	(0.378)	
Share high school	0.403	0.382	0.021	0.500	0.471	0.029	0.568	0.553	0.015
	(0.364)	(0.370)		(0.365)	(0.375)		(0.368)	(0.380)	
Share above high school	0.061	0.066	−0.005	0.079	0.079	−0.000	0.080	0.082	−0.002
	(0.164)	(0.178)		(0.180)	(0.193)		(0.188)	(0.200)	
Share workers aged < 25	0.342	0.311	0.031	0.332	0.303	0.030	0.295	0.278	0.017
	(0.299)	(0.309)		(0.286)	(0.294)		(0.288)	(0.296)	
Share hires from other firms	0.165	0.144	0.021	0.216	0.200	0.016	0.214	0.200	0.014
	(0.240)	(0.241)		(0.257)	(0.263)		(0.263)	(0.270)	
Gender wage gap	−0.084	−0.090	0.006	−0.089	−0.092	0.004	−0.089	−0.086	−0.003
(full-time workers)	(0.306)	(0.321)		(0.293)	(0.299)		(0.291)	(0.290)	
Number of establishments	83,157	83,157		39,446	39,446		19,284	19,284	
Establishment-years	211,108	211,249		151,295	151,495		92,522	92,873	

Notes: This table reports pre-inspection characteristics of inspected establishments and their matched never-inspected controls by inspection cohort. Inspection cohorts are 2005–2010, 2011–2014, and 2015–2018 and together make up the main event-study sample. "Diff." denotes the inspected mean minus the control mean, and standard deviations appear in parentheses.

D.5 From the Inspected Universe to the Estimation Samples

Table D.3 traces inspected establishments through the nested samples used in the analysis. Column (1) includes all inspected establishments, column (2) retains those with a suitable matched control, column (3) imposes the event-study restrictions, and column (4) imposes

the additional data requirements of the structural model. Because each column reports the same pre-inspection characteristics, differences across columns reflect sample selection.

Table D.3: Inspected Establishments at Each Stage of the Analysis

Sample of establishments:	(1) **All inspected**	(2) **Matched**	(3) **Event study**	(4) **Model**
Firm age	7.200	6.844	7.014	6.547
	(7.946)	(7.520)	(7.622)	(6.997)
Annual number of employees	23.710	11.158	11.729	12.358
	(846.635)	(15.933)	(16.250)	(15.491)
Mean log wage	6.980	6.901	6.909	6.960
(all workers)	(0.433)	(0.358)	(0.358)	(0.298)
Share covered by a CBA	0.217	0.212	0.222	0.190
	(0.412)	(0.408)	(0.415)	(0.393)
Share women	0.439	0.459	0.458	0.463
	(0.355)	(0.365)	(0.360)	(0.304)
Mean worker age	31.642	30.852	30.869	30.575
	(6.849)	(6.785)	(6.672)	(5.984)
Mean worker tenure	21.124	18.875	19.396	17.084
(in months)	(24.409)	(19.962)	(20.148)	(14.892)
Mean contracted hours	43.000	43.298	43.297	43.403
	(3.535)	(2.985)	(2.939)	(2.510)
Share full-time workers	0.948	0.963	0.963	0.967
	(0.177)	(0.148)	(0.147)	(0.135)
Share below high school	0.427	0.460	0.460	0.395
	(0.381)	(0.388)	(0.383)	(0.363)
Share high school	0.478	0.470	0.469	0.538
	(0.368)	(0.376)	(0.371)	(0.358)
Share above high school	0.094	0.069	0.071	0.067
	(0.208)	(0.175)	(0.175)	(0.165)
Share workers aged < 25	0.302	0.331	0.329	0.338
	(0.285)	(0.299)	(0.293)	(0.269)
Share hires from other firms	0.193	0.194	0.192	0.232
	(0.251)	(0.257)	(0.252)	(0.249)
Gender wage gap	−0.116	−0.103	−0.105	−0.093
(full-time workers)	(0.350)	(0.345)	(0.343)	(0.299)
Establishment-years	904,367	532,952	455,177	81,485

Notes: This table reports pre-inspection characteristics of inspected establishments at successive stages of the analysis. Column (1) includes all establishments in which an inspection detected informal workers between 2005 and 2018. Column (2) restricts the sample to establishments matched one-to-one to never-inspected controls. Column (3) reports the sample used to estimate equation (1), and column (4) reports the structural sample used in Section 6. Entries are means, with standard deviations in parentheses, calculated from the same pre-inspection establishment panel in every column.

Table D.4 reports the industry composition and size of the same four samples. Of the 279,165 inspected establishments, 172,974, or 62.0%, are matched; 141,887, or 50.8%, enter the main event study; and 17,908, or 6.4%, enter the structural analysis.

Table D.4: Industry Composition at Each Stage of the Analysis

Sample of establishments:	(1) **All inspected**	(2) **Matched**	(3) **Event study**	(4) **Model**
Share of establishments (%)				
Retail trade	37.8	43.2	43.4	44.8
Manufacturing	13.1	13.3	13.7	12.4
Accommodation and food services	12.8	12.1	11.8	17.7
Wholesale trade	9.3	9.4	9.5	7.2
Construction	5.1	4.7	4.1	2.6
Administrative and support services	3.5	3.3	3.3	3.4
Educational services	3.3	2.7	2.8	3.2
Transportation and warehousing	2.7	2.0	2.0	1.5
Other industries	12.3	9.3	9.4	7.3
Number of establishments	279,165	172,974	141,887	17,908
% of all inspected establishments	100.0	62.0	50.8	6.4
in the men's regression				13,059
in the women's regression				9,683

Notes: This table reports the industry composition and establishment counts for the nested samples used in the analysis. Column (1) includes all establishments in which an inspection detected informal workers between 2005 and 2018. Column (2) restricts the sample to establishments matched one-to-one to never-inspected controls, column (3) reports the event-study sample, and column (4) reports the structural sample. Industry is measured in the final year before inspection. The structural model is estimated separately by gender; column (4) counts establishments appearing in either gender-specific sample, and the final two rows report the corresponding counts.

Matching produces the largest change in composition. Relative to the full inspected population, the matched sample contains smaller and slightly lower-wage establishments: mean annual employment falls from 23.7 to 11.2 workers, while the retail share rises from 37.8 to 43.2%. Large establishments in thin matching cells are less likely to have suitable controls. The event-study restrictions produce little additional change in observable composition.

The structural sample is more selected. Relative to the event-study sample, it contains somewhat larger and higher-wage establishments, more workers with a high-school education, and a larger share of establishments in accommodation and food services. Inclusion requires an inspection in 2012 or later, detected informal workers of the relevant

gender, an estimable gender-specific wage premium, and the amenity and collective-agreement measures used in the model. The post-2012 restriction accounts for most of the reduction in sample size.

The men's structural sample contains 13,059 establishments, the women's sample contains 9,683, and 4,834 appear in both. Because most inspections detect informal workers of only one gender, the structural comparison is primarily across establishments. The structural estimates apply to establishments satisfying the timing, wage, amenity, collective-agreement, and gender-specific outcome requirements.

E. Additional Results from the Matched Sample

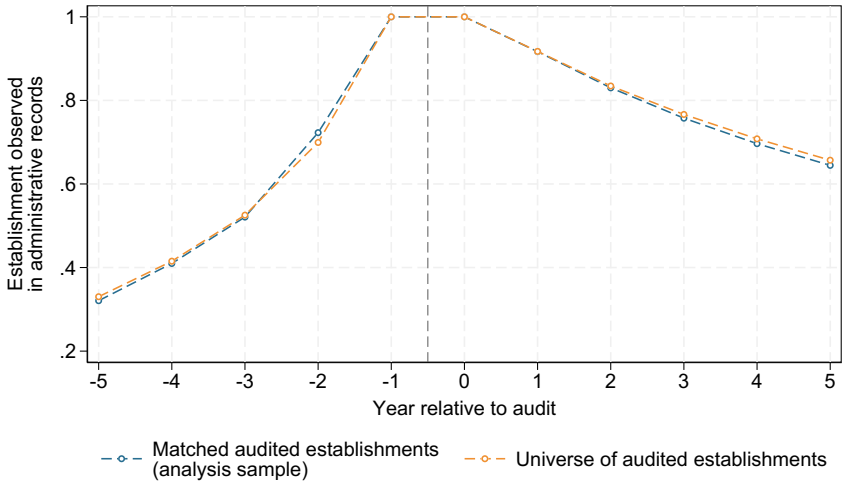
Section 3 presents the main results from the matched event-study design. This appendix provides six complementary sets of results. First, I compare the formal-sector activity of inspected establishments retained by the matching procedure with that of the full population of establishments ever found to employ informal workers. Second, I report gender-specific hiring responses over the one- and three-month windows and the remainder of the inspection year, which the main text does not show. Third, I report the gender pay-gap estimates that include part-time workers, which the main text reports for full-time workers only. Fourth, I compare the main event-study estimates across all inspection cohorts with those obtained when the sample is restricted to establishments first inspected in 2012 or later. Fifth, I examine whether inspected establishments relocate. Sixth, I report the occupational diversity of registered employment.

The cohort comparison addresses changes in the composition of enforcement over time. Section 4.4 documents that, beginning around 2014, inspections shifted toward industries in which women account for a larger share of informal employment. This change motivates the structural analysis in Section 6, which focuses on inspections conducted in 2012 or later. In the event-study figures below, one series includes all inspection cohorts and the other restricts the sample to establishments first inspected in 2012 or later. Both use the matched design in equation (1). The similarity of the estimates across the two samples indicates that the main event-time patterns are not driven by earlier inspection cohorts.

E.1 Representativeness of the Matched Sample

The matching procedure in Section 3.2 retains an inspected establishment only when a never-inspected establishment with similar pre-inspection characteristics is available. Figure E.1 assesses whether this restriction selects inspected establishments with systematically different patterns of formal-sector activity. It compares the probability of appearing in RAIS with positive employment in the matched inspected sample and in the full sample of establishments ever found to employ informal workers, imposing the same pre-inspection presence requirement on both groups. The two series track each other closely throughout the event window. Thus, along this observable margin, the matched establishments resemble the broader population of establishments in which inspections detect informal workers.

Figure E.1: Formal-Sector Activity in the Matched and Ever-Inspected Samples



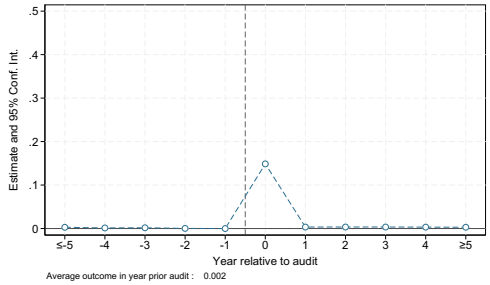
Notes: This figure compares formal-sector activity among inspected establishments retained by the matched event-study design and among all establishments ever found to employ at least one informal worker. The outcome equals one if an establishment appears in RAIS with positive formal employment in a given year and zero otherwise. To make the samples comparable, the ever-inspected sample is restricted to establishments observed in RAIS during every pre-inspection period required by the matching procedure. Event time is measured in years relative to the first inspection in which informal workers were detected. The figure presents sample means; it does not report regression estimates or confidence intervals.

E.2 New Formal Hires by Gender, Remaining Hiring Windows

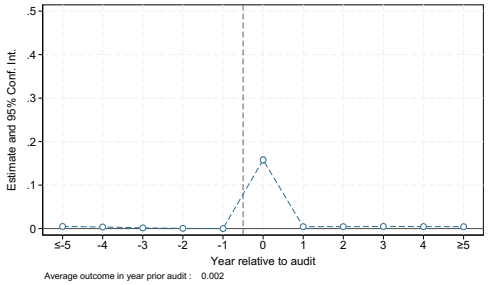
Section 3.4 reports gender-specific hiring within six months of the inspection. This appendix reports the remaining windows: one month, three months, and the remainder of the inspection year.[46] Figure E.2 expresses new formal hires in those three windows as a fraction of pre-inspection employment, and Figure E.3 reports the corresponding counts. In both figures the left column restricts the sample to inspections detecting at least one informal woman and reports new formal hires of women; the right column applies the analogous definition to men. In every window, the increase in registrations is concentrated in the inspection year, and the male coefficient exceeds the female coefficient by roughly one to two and a half percentage points of pre-inspection employment. Because the female and male estimates use different sets of establishments, their levels in counts should not be interpreted as a within-establishment gender comparison; the employment shares in Figure E.2 are comparable across genders.

[46]The take-up measures of Section 4 use the six-month window as the baseline and report these windows as robustness checks.

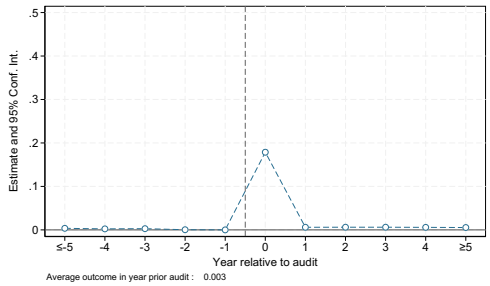
Figure E.2: Gender-Specific New Formal Hires as a Fraction of Pre-Inspection Employment, by Hiring Window



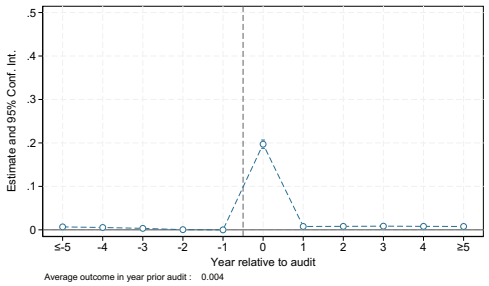
(a) Women, within 1 month



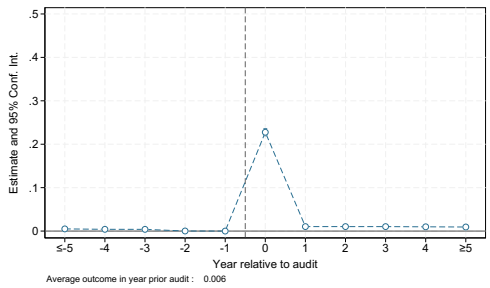
(b) Men, within 1 month



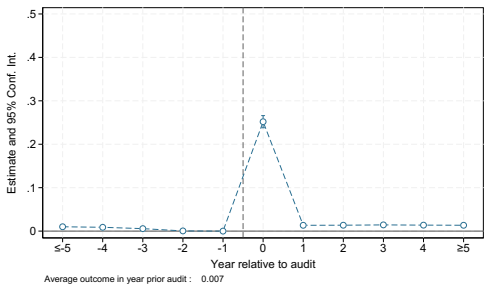
(c) Women, within 3 months



(d) Men, within 3 months



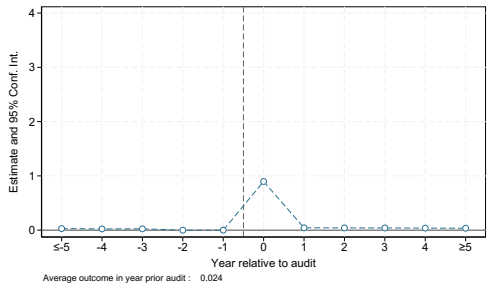
(e) Women, remainder of the inspection year



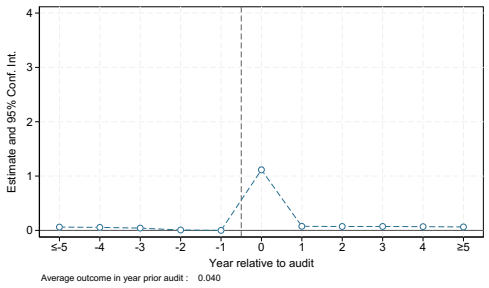
(f) Men, remainder of the inspection year

Notes: This figure reports gender-specific new formal hires, divided by establishment employment in the year before the inspection, over the three hiring windows not shown in Figure 5. A new formal hire is a worker registered at the inspected establishment after the inspection who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a), (c), and (e) restrict the sample to inspections in which at least one informal woman was detected and count women registered within one month, within three months, and over the remainder of the inspection year; panels (b), (d), and (f) apply the analogous definitions to men. Because both series are scaled by pre-inspection employment, the female and male panels are comparable. The inspection-year coefficients are 0.149 and 0.158 within one month, 0.179 and 0.197 within three months, and 0.228 and 0.252 over the remainder of the year, for women and men respectively. Each panel plots the event-time coefficients θ_k from equation (1), estimated using matched pairs observed at event time -3. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

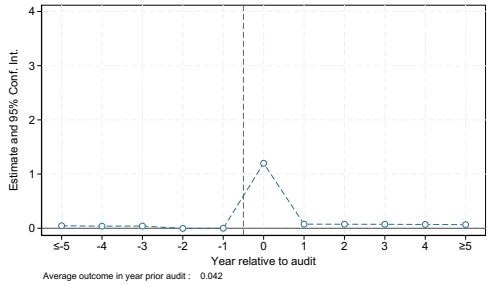
Figure E.3: Gender-Specific New Formal Hires in Counts, by Hiring Window



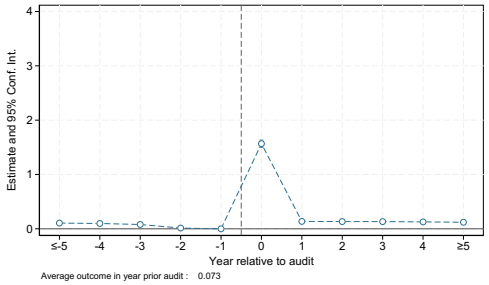
(a) Women, within 1 month



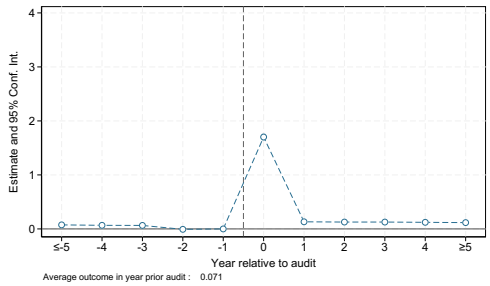
(b) Men, within 1 month



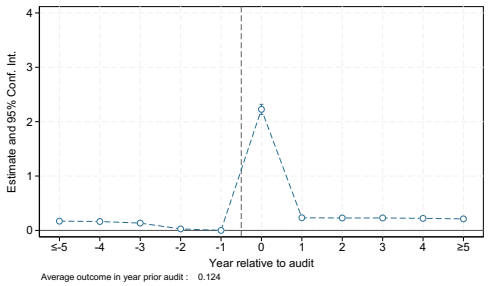
(c) Women, within 3 months



(d) Men, within 3 months



(e) Women, remainder of the inspection year



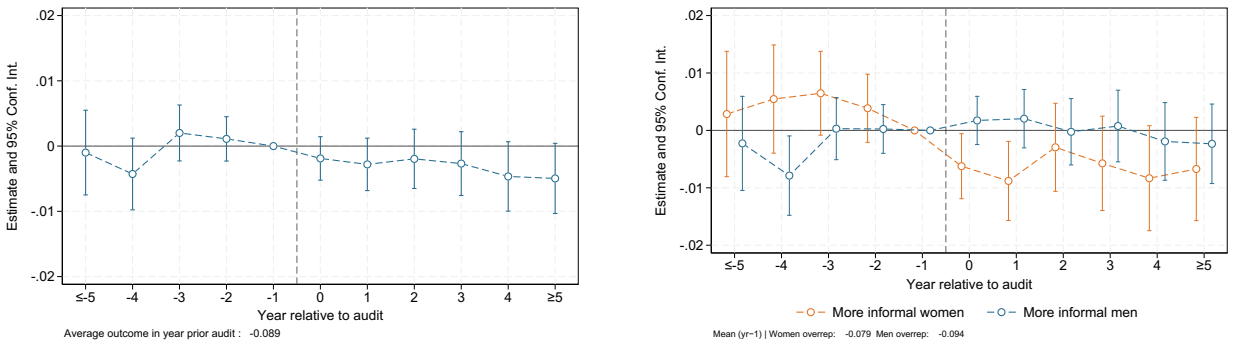
(f) Men, remainder of the inspection year

Notes: This figure reports gender-specific new formal hires, in counts, over the three hiring windows not shown in Figure 5. A new formal hire is a worker registered at the inspected establishment after the inspection who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a), (c), and (e) restrict the sample to inspections in which at least one informal woman was detected and count women registered within one month, within three months, and over the remainder of the inspection year, respectively. Panels (b), (d), and (f) apply the analogous definitions to men. Because the female and male specifications use different sets of inspected establishments, the coefficient levels are not directly comparable across genders; Figure E.2 reports the same windows scaled by pre-inspection employment, which are comparable. Each panel plots the event-time coefficients θ_k from equation (1), estimated using matched pairs observed at event time -3. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

E.3 The Gender Pay Gap

This appendix reports the effect of inspections on the establishment-level gender pay gap summarized in Section 3.5. If newly registered women earn closer to the minimum wage than men, their entry into the formal labor force can lower women's observed average wage relative to men's. Figure E.4 examines the establishment-level gender pay gap among full-time workers, measured as women's mean log wage minus men's. In the full matched sample, estimates are negative but statistically indistinguishable from zero (panel a). However, when detected informal workers are disproportionately female relative to the establishment's pre-inspection formal workforce, the gender pay gap widens in the inspection year and further in the following year (panel b). No comparable effect appears when men are overrepresented. These estimates reflect changes in the composition of formal employment rather than individual wage effects, since detected workers cannot be linked directly to RAIS. Figure E.5 reports the same estimates with part-time workers included; the pattern is unchanged, so restricting attention to full-time workers removes hours composition from the comparison without changing the conclusion.

Figure E.4: Inspections and the Gender Pay Gap, Full-Time Workers

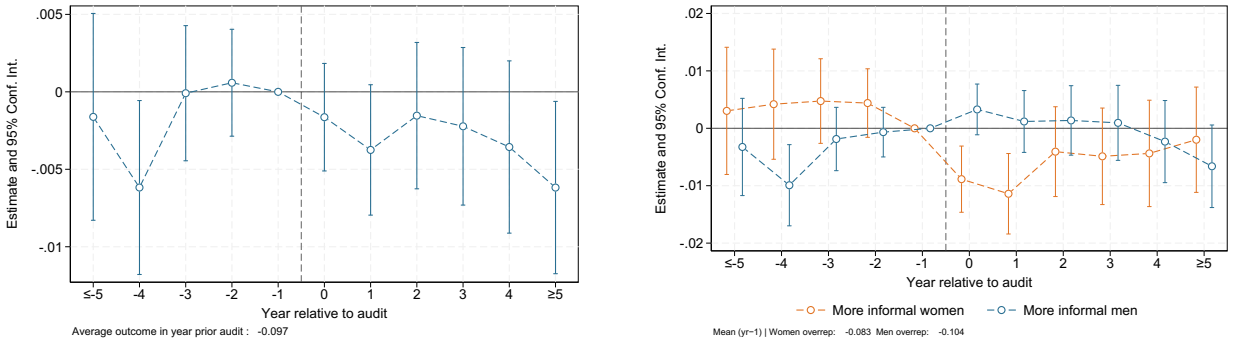


(a) Full matched sample

(b) By overrepresented gender

Notes: This figure reports event-study estimates of the effect of inspections that detect informal workers on the establishment-level gender pay gap among formally registered full-time workers, defined as women's mean log wage minus men's mean log wage. Panel (a) uses the full matched sample. Panel (b) divides inspected establishments by which gender is overrepresented among detected informal workers relative to the establishment's formal workforce in the year before the inspection. Estimates follow equation (1) using matched inspected and never-inspected establishments observed at event time −3. Event time is measured relative to the inspection year, $k = -1$ is omitted, and endpoints are binned at $k \leq -5$ and $k \geq 5$. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm. Values below each panel report the pre-inspection mean among inspected establishments. Figure E.5 reports the corresponding estimates when part-time workers are included.

Figure E.5: Inspections and the Gender Pay Gap, All Workers



(a) Full matched sample

(b) By overrepresented gender

Notes: This figure reports event-study estimates of the effect of inspections that detect informal workers on the establishment-level gender pay gap among all formally registered workers, defined as women's mean log wage minus men's mean log wage. Panel (a) uses the full matched sample. Panel (b) divides inspected establishments by which gender is overrepresented among detected informal workers relative to the establishment's formal workforce in the year before the inspection. Estimates follow equation (1) using matched inspected and never-inspected establishments observed at event time −3. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm. Values below each panel report the pre-inspection mean among inspected establishments. Figure E.4 reports the corresponding estimates for full-time workers.

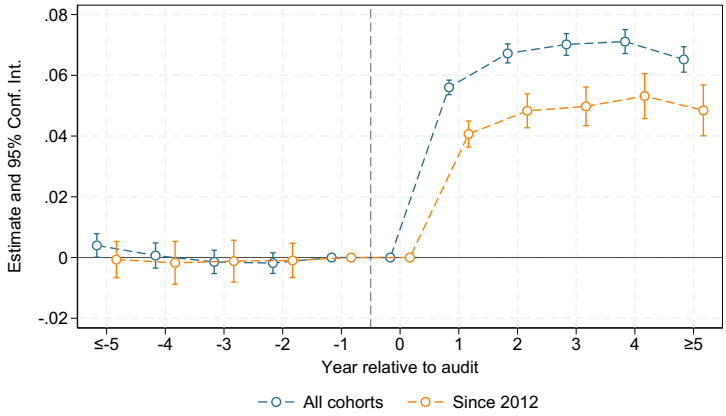
E.4 Event Paths by Inspection Cohort

The remaining figures assess whether changes in the composition of inspections over time affect the main results. Each panel reports the event-time coefficients θ_k from equation (1) for two samples: all inspection cohorts and establishments first inspected in 2012 or later.

Formal-sector activity, employment, and hiring. Figures E.6–E.9 report formal-sector activity, total annual formal employment, and new formal hiring. Hiring is expressed both in levels and relative to pre-inspection employment. Restricting the sample to establishments first inspected in 2012 or later leaves the timing and approximate magnitude of the estimated responses largely unchanged.

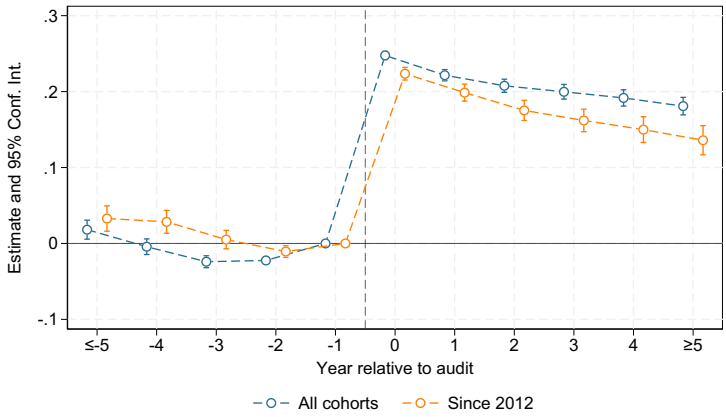
Hiring by gender. Figures E.10 and E.11 report new formal hires separately for inspections detecting informal women and informal men. The concentration of registrations in the inspection year is present in both cohort samples and for both genders. Because the gender-specific estimates use different establishments and denominators, the take-up measures in Section 3.4, rather than the coefficient levels reported here, provide the relevant gender comparison.

Figure E.6: Formal-Sector Activity, by Inspection Cohort



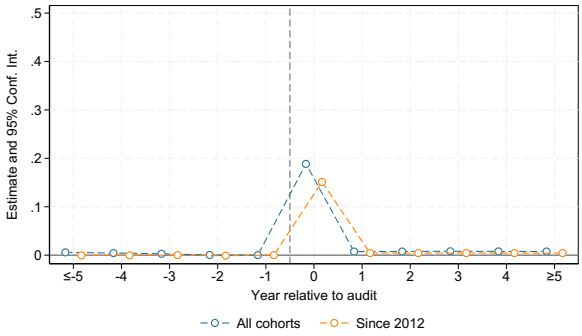
Notes: This figure reports establishment activity in the formal labor market around an inspection that detected at least one informal worker. The outcome equals one if the establishment appears in RAIS with positive formal employment in a given year and zero otherwise. One series includes all inspection cohorts; the other restricts the sample to establishments first inspected in 2012 or later. Both plot the event-time coefficients θ_k from equation (1), estimated on the balanced 2003–2021 panel of matched inspected and never-inspected establishments. Years after an establishment ceases to appear in RAIS with positive employment are coded as zero. Event time is measured in years relative to the inspection year. The coefficient for the inspection year is zero by construction, $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

Figure E.7: Total Annual Formal Employment, by Inspection Cohort

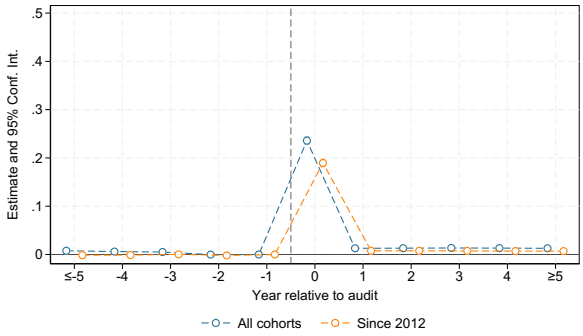


Notes: This figure reports total annual formal employment around an inspection that detected at least one informal worker. The outcome is the log number of workers who held a registered contract with the establishment at any point during the calendar year. One series includes all inspection cohorts; the other restricts the sample to establishments first inspected in 2012 or later. Both plot the event-time coefficients θ_k from equation (1), estimated using matched inspected and never-inspected establishments observed at event time -3. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

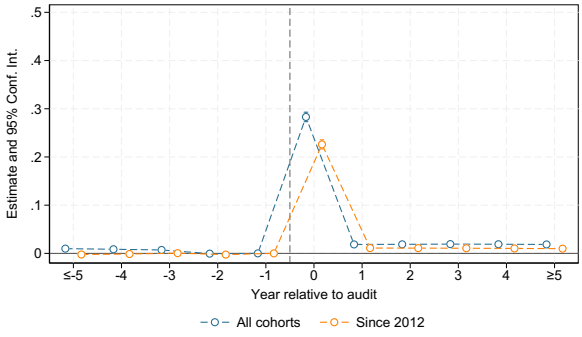
Figure E.8: New Formal Hires Relative to Pre-Inspection Employment, by Inspection Cohort



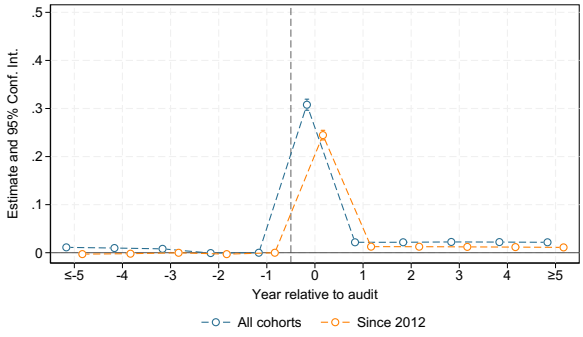
(a) Within 1 month



(b) Within 3 months



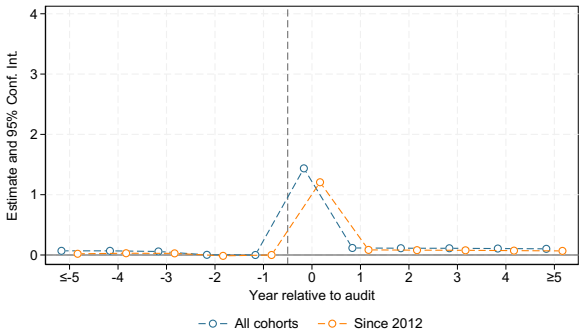
(c) Within 6 months



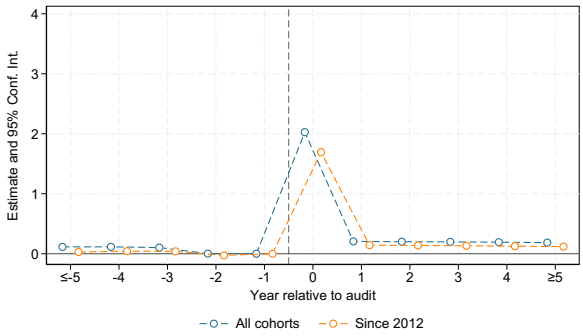
(d) Remainder of the inspection year

Notes: This figure reports new formal hires relative to pre-inspection employment following an inspection that detected at least one informal worker. A new formal hire is a worker registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. Each outcome is divided by the establishment's formal employment in the year preceding the inspection. In every panel, one series includes all inspection cohorts and the other restricts the sample to establishments first inspected in 2012 or later. Both plot the event-time coefficients θ_k from equation (1), estimated using matched inspected and never-inspected establishments observed at event time -3. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

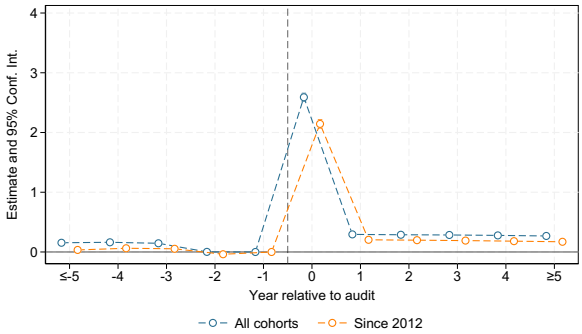
Figure E.9: New Formal Hires, by Inspection Cohort



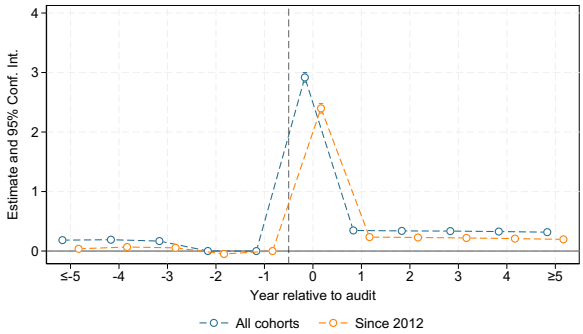
(a) Within 1 month



(b) Within 3 months



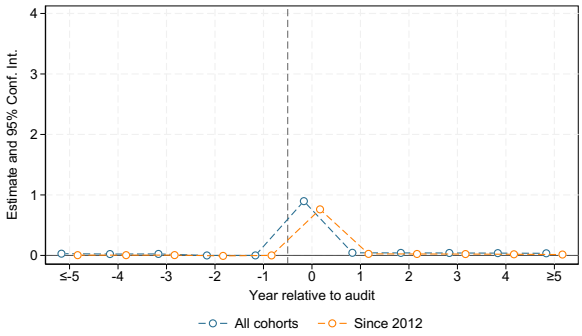
(c) Within 6 months



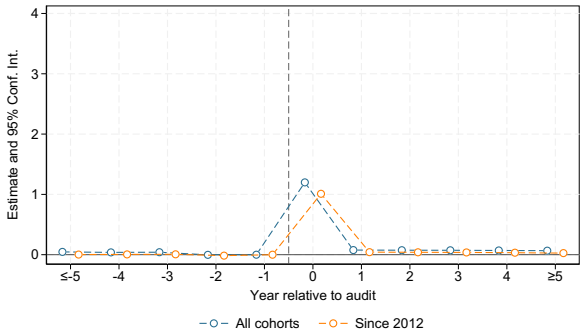
(d) Remainder of the inspection year

Notes: This figure reports new formal hires following an inspection that detected at least one informal worker. A new formal hire is a worker registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. The outcomes are expressed as numbers of workers. In every panel, one series includes all inspection cohorts and the other restricts the sample to establishments first inspected in 2012 or later. Both plot the event-time coefficients θ_k from equation (1), estimated using matched inspected and never-inspected establishments observed at event time -3. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

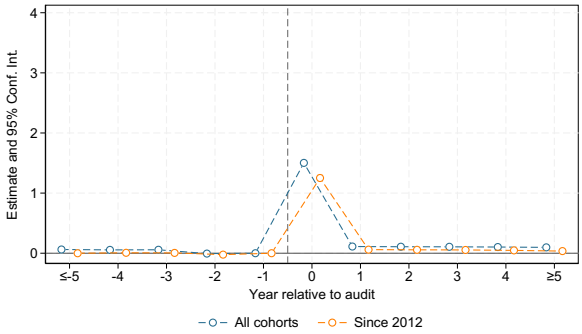
Figure E.10: New Formal Hires of Women, by Inspection Cohort



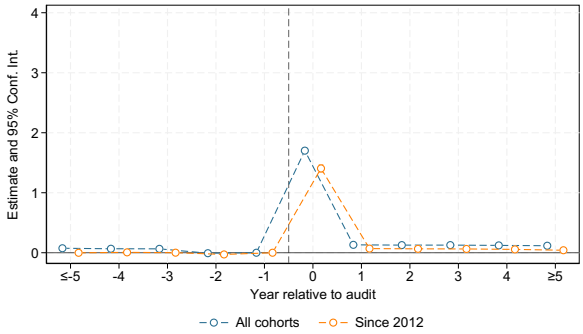
(a) Within 1 month



(b) Within 3 months



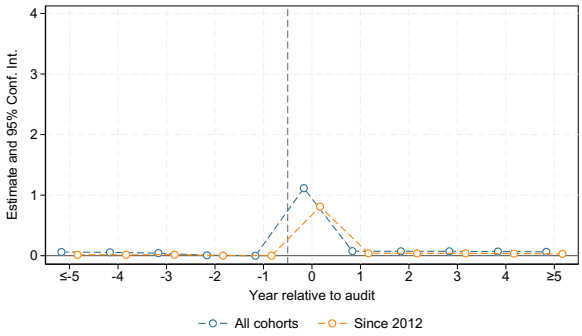
(c) Within 6 months



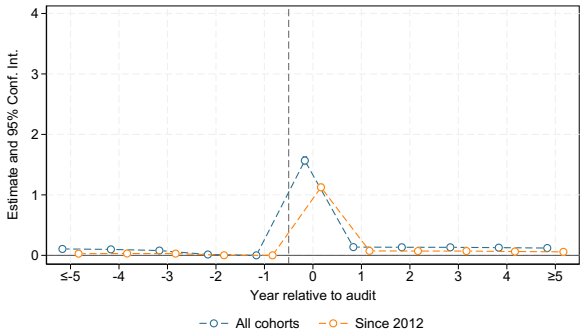
(d) Remainder of the inspection year

Notes: This figure reports new formal hires of women following inspections in which at least one informal woman was detected. A new formal hire is a woman registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. In every panel, one series includes all inspection cohorts and the other restricts the sample to establishments first inspected in 2012 or later. Both plot the event-time coefficients θ_k from equation (1), estimated using matched inspected and never-inspected establishments observed at event time -3. The sample includes only matched pairs whose inspected establishment had at least one informal woman detected. Because the corresponding estimates for men use a different set of establishments, coefficient levels are not directly comparable across genders. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

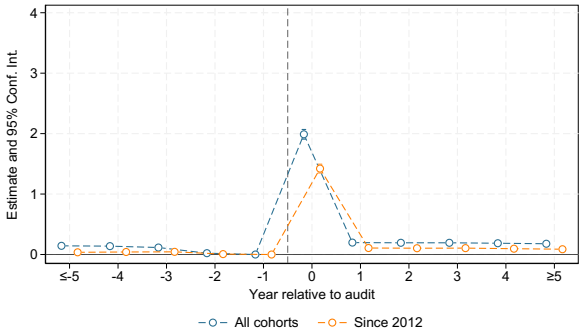
Figure E.11: New Formal Hires of Men, by Inspection Cohort



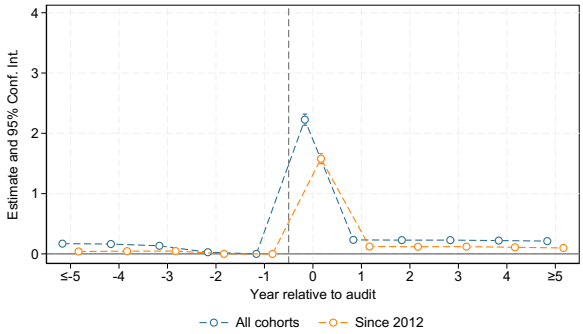
(a) Within 1 month



(b) Within 3 months



(c) Within 6 months



(d) Remainder of the inspection year

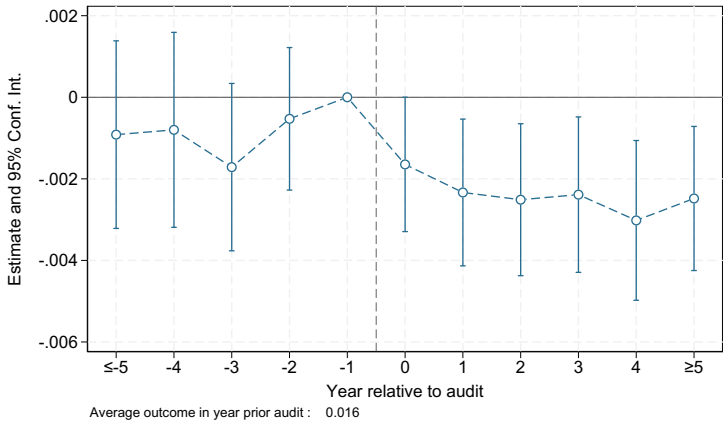
Notes: This figure reports new formal hires of men following inspections in which at least one informal man was detected. A new formal hire is a man registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. In every panel, one series includes all inspection cohorts and the other restricts the sample to establishments first inspected in 2012 or later. Both plot the event-time coefficients θ_k from equation (1), estimated using matched inspected and never-inspected establishments observed at event time -3. The sample includes only matched pairs whose inspected establishment had at least one informal man detected. Because the corresponding estimates for women use a different set of establishments, coefficient levels are not directly comparable across genders. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

E.5 Establishment Relocation

A potential concern is that inspected establishments respond by relocating, which could affect both measured establishment survival and workforce composition. Figure E.12 estimates equation (1) using an indicator for whether an establishment's municipality in RAIS differs from its municipality in the previous year. Relocation is uncommon: 1.8%

of inspected establishments change municipality in the year before the inspection. The estimates show no differential pre-trends and no increase in relocation following inspection. If anything, the post-inspection coefficients are slightly negative. These results suggest that establishment relocation does not account for the post-inspection changes in activity and employment.

Figure E.12: Establishment Relocation Around the Inspection

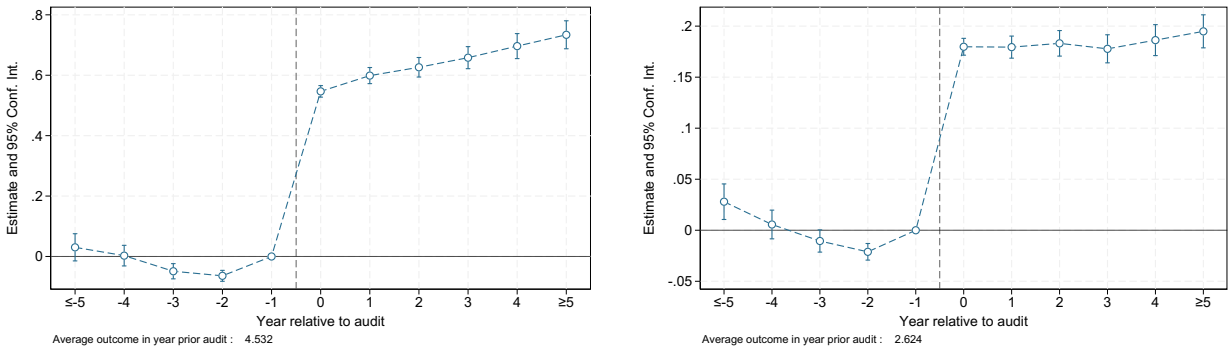


Notes: The figure reports event-study coefficients θ_k from equation (1), where the outcome equals one if the establishment's municipality in RAIS differs from its municipality in the previous year and zero otherwise. The sample consists of matched inspected and never-inspected establishments observed at event time -3, for inspections that detected at least one informal worker. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

E.6 Occupational Diversity of Registered Employment

Figure E.13 reports event-study estimates for the number of distinct occupations held by registered workers at the establishment, measured at the six-digit level of the Brazilian occupational classification (CBO) and at the level of its one-digit major groups. In the year before inspection, inspected establishments register workers in an average of 4.7 distinct six-digit occupations and 2.7 major groups. The number of distinct six-digit occupations increases by 0.55 in the inspection year and the number of major groups by 0.18, and both increases persist over the following years.

Figure E.13: Distinct Occupations Among Registered Workers Around the Inspection



(a) Distinct six-digit occupations

(b) Distinct major occupational groups

Notes: The figure reports event-study coefficients θ_k from equation (1), where the outcome is the number of distinct six-digit CBO occupations in panel (a) and one-digit CBO major groups in panel (b) held by workers registered at the establishment during the year. The sample consists of matched inspected and never-inspected establishments observed at event time -3, for inspections that detected at least one informal worker. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

F. Additional Evidence on Post-Inspection Formal Hiring

This appendix presents three exercises that assess the interpretation of the increase in formal registrations documented in Section 3.3. First, I estimate a treated-only event study using all establishments ever found to employ informal workers. This exercise expands coverage beyond the establishments for which a suitable never-inspected match is available. Second, I examine whether an inspection generates hiring responses at other, uninspected establishments belonging to the same firm. Third, I study establishments whose inspections identified violations unrelated to informal employment. This comparison helps distinguish the response to an inspection itself from the additional response associated with detecting informal workers. A final subsection reports how long the newly registered workers remain at the inspected establishment, compared with its incumbent workers.

Throughout, a new formal hire is a worker registered at the relevant establishment after the inspection who held no active formal contract with any employer in the calendar month immediately preceding the inspection. I measure these registrations within one, three, and six months of the inspection and between the inspection date and the end of the inspection year. The figures report the outcomes both in levels and as fractions of formal employment in the year preceding the inspection.

F.1 Treated-Only Estimates for the Ever-Inspected Sample

The matched design excludes inspected establishments for which no suitable never-inspected comparison can be found. To assess whether this restriction drives the main result, I estimate a treated-only event study using all establishments ever found to employ at least one informal worker. The specification is

$$y_{it} = \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \theta_k \mathbf{1}\{e_{it} = k\} + \alpha_i + \eta_{z(i)t} + \nu_{r(i)t} + \varepsilon_{it}, \tag{F.1}$$

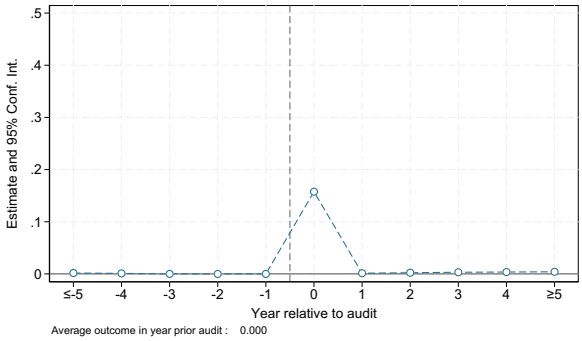
where $e_{it} = t - T_i$ denotes event time relative to establishment i's first inspection in which informal workers were detected. The terms α_i, $\eta_{z(i)t}$, and $\nu_{r(i)t}$ denote establishment, industry-by-year, and state-by-year fixed effects, respectively. Event year $k = -1$ is omitted, endpoints are binned at $|k| \geq 5$, and standard errors are clustered by establishment. I restrict the sample to establishments with positive formal employment in each of the three

years preceding the inspection.

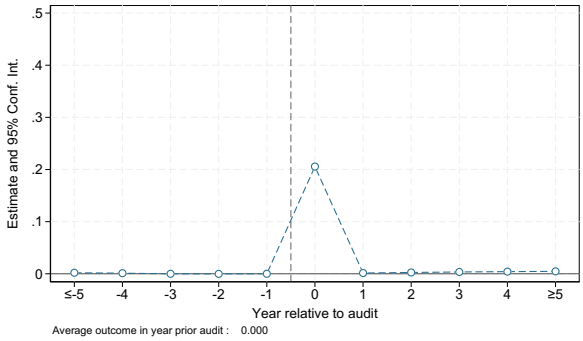
This design identifies changes around the inspection from differences in inspection timing across ever-inspected establishments. Because it does not include a never-inspected comparison group, it does not provide the same counterfactual as the matched design. I therefore interpret it as complementary evidence on the timing and external validity of the main response.

Figures F.1 and F.2 reproduce the main temporal pattern. New formal registrations increase sharply in the inspection year, and the magnitude of the response rises as the post-inspection window expands. The estimates in levels are larger than those from the matched sample, consistent with this specification retaining larger establishments for which no suitable match is available. Thus, the concentration of registrations around the inspection is not specific to the establishments retained by the matching procedure.

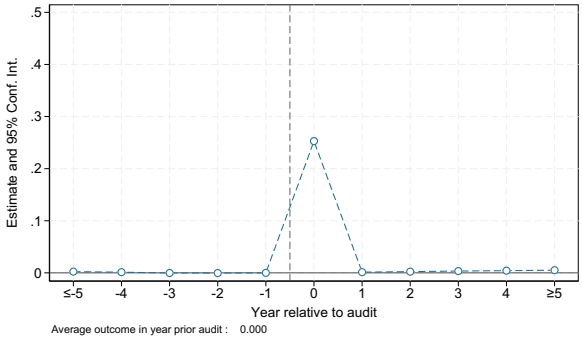
Figure F.1: New Formal Hires in the Ever-Inspected Sample, Relative to Pre-Inspection Employment



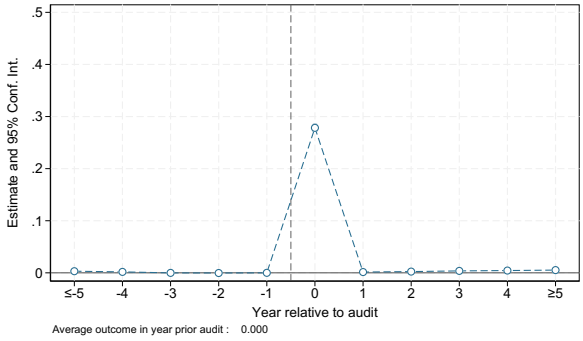
(a) Within 1 month



(b) Within 3 months



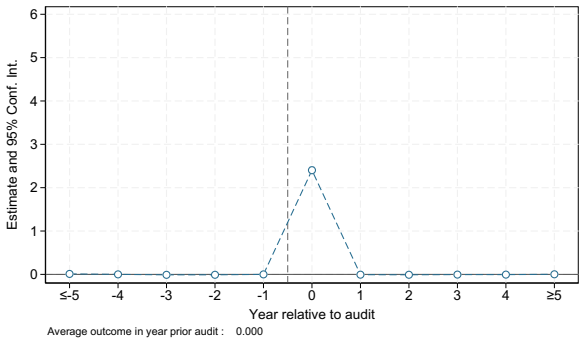
(c) Within 6 months



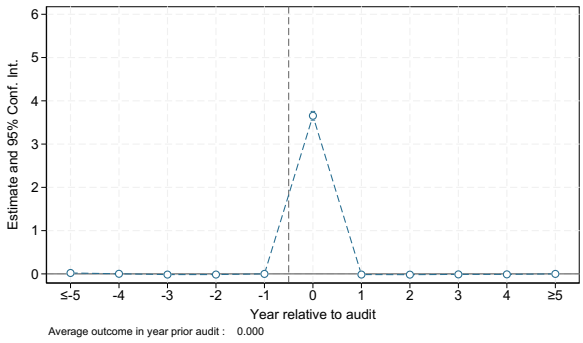
(d) Remainder of the inspection year

Notes: This figure reports treated-only event-study estimates for establishments in which an inspection detected at least one informal worker. A new formal hire is a worker registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. Each outcome is divided by the establishment's formal employment in the year preceding the inspection. The sample includes all ever-inspected establishments with positive formal employment in each of the three years preceding their first inspection in which informal workers were detected. Each panel plots the coefficients θ_k from equation (F.1). All specifications include establishment, industry-by-year, and state-by-year fixed effects. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by establishment.

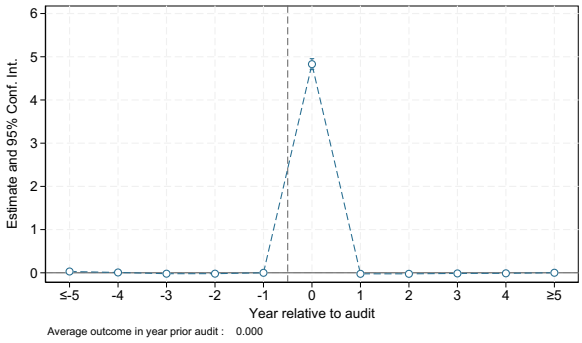
Figure F.2: New Formal Hires in the Ever-Inspected Sample



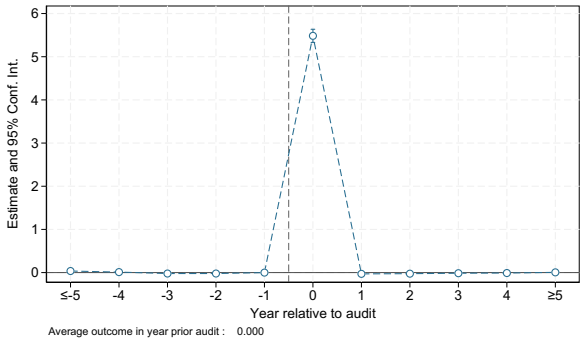
(a) Within 1 month



(b) Within 3 months



(c) Within 6 months



(d) Remainder of the inspection year

Notes: This figure reports treated-only event-study estimates for establishments in which an inspection detected at least one informal worker. A new formal hire is a worker registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. The outcomes are expressed as numbers of workers. The sample includes all ever-inspected establishments with positive formal employment in each of the three years preceding their first inspection in which informal workers were detected. Each panel plots the coefficients θ_k from equation (F.1). All specifications include establishment, industry-by-year, and state-by-year fixed effects. Event time is measured in years relative to the inspection year; $k = -1$ is omitted, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by establishment.

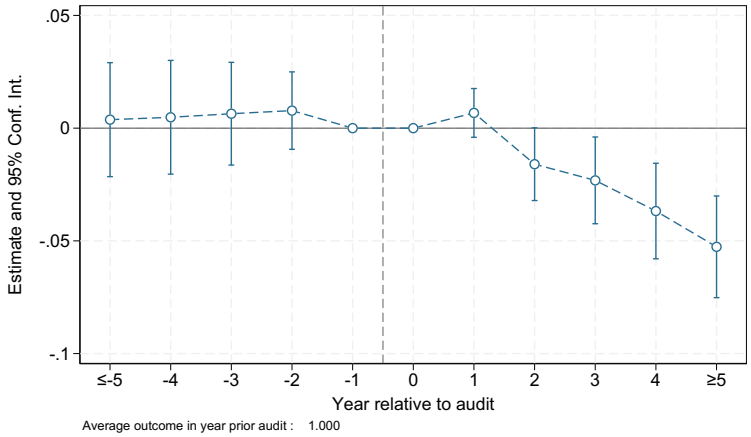
F.2 Within-Firm Spillovers

An inspection may affect establishments other than the site that inspectors visit. In particular, a multi-establishment firm may respond by registering workers at its other establishments, whether because the inspection changes firm-wide compliance or because employment is reallocated across sites. Restricting the outcome to the inspected establishment could therefore miss part of the firm's response.

To examine this possibility, I construct a spillover sample consisting of establishments that were never inspected but belong to a firm with at least one establishment in which an inspection detected informal workers. I compare these establishments with observably similar establishments belonging to firms with no inspected sites, matched by the procedure of Section 3.2: exact matches on pre-inspection size, RAIS entry cohort, industry, and state, with a caliper on the establishment's pre-inspection mean log wage. Event time is measured in years relative to the firm's first inspection in which informal workers were detected.

Figures F.3–F.5 show that the inspection is followed by an increase in new formal hires at other establishments within the same firm, although the response is smaller than at the inspected site. The effects therefore extend beyond the establishment directly exposed to the inspection. This finding is consistent with a firm-wide compliance response, but it may also reflect the reallocation of workers or registrations across establishments within the firm. The probability that these uninspected establishments remain active in RAIS does not change immediately, but declines beginning approximately two years after the inspection. This longer-run response suggests that firms may subsequently consolidate or reorganize activity across establishments.

Figure F.3: Within-Firm Spillovers: Establishment Activity in the Formal Sector



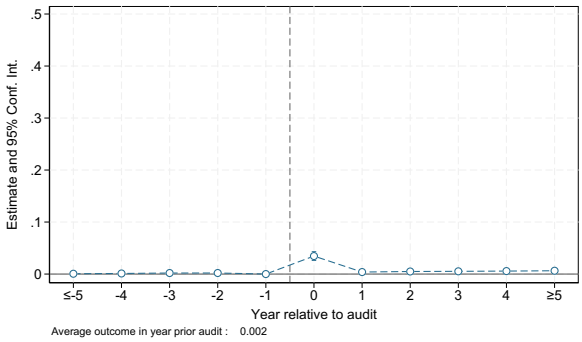
Notes: This figure reports event-study estimates for establishments that were not inspected but belong to firms with at least one establishment in which inspectors detected informal workers. Treated establishments are compared with matched establishments belonging to firms with no inspected sites. Event time is measured in years relative to the firm's first inspection in which informal workers were detected. The outcome equals one if the establishment appears in RAIS with positive formal employment in the corresponding year and zero otherwise. The omitted period is $k = -1$, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

F.3 Comparison Inspections Unrelated to Informal Employment

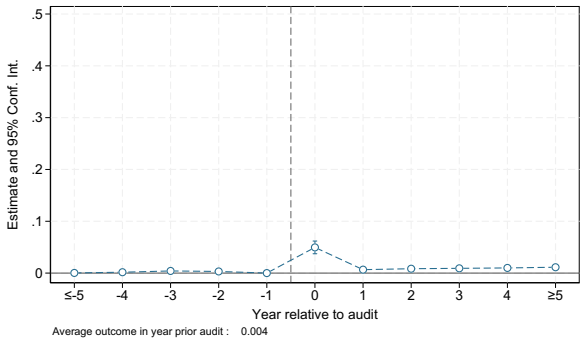
The increase in formal registrations could partly reflect a general response to enforcement contact rather than the specific requirement to register workers detected as informal. To assess this possibility, I examine establishments whose first inspection identified a violation involving disability quotas, discrimination, or mandated rest periods, but did not record any informal workers. These establishments were exposed to labor enforcement without a recorded informality violation. I compare them with never-inspected establishments matched by the same procedure as in the main analysis (Section 3.2): exact matches on pre-inspection size, RAIS entry cohort, industry, and state, with a caliper on the pre-inspection mean log wage.

Figures F.6–F.8 show that new formal hiring also increases following these comparison inspections, but by substantially less than after inspections that detect informal workers. This pattern suggests that enforcement contact may itself induce some establishments to register workers—for example, by increasing the perceived probability of future scrutiny. The larger response following inspections that directly identify informal employment is nevertheless consistent with an additional effect of the requirement to register the detected workers. Establishment activity in RAIS also increases after comparison inspections, although the effect attenuates later in the event window.

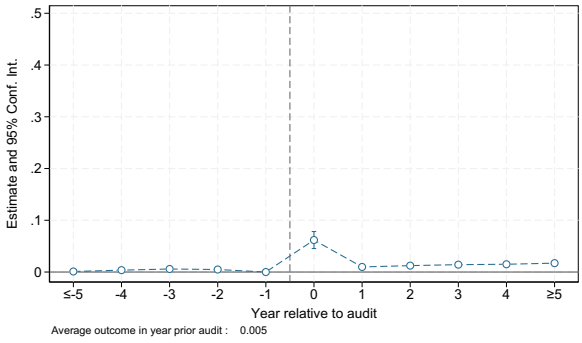
Figure F.4: Within-Firm Spillovers: New Formal Hires Relative to Pre-Inspection Employment



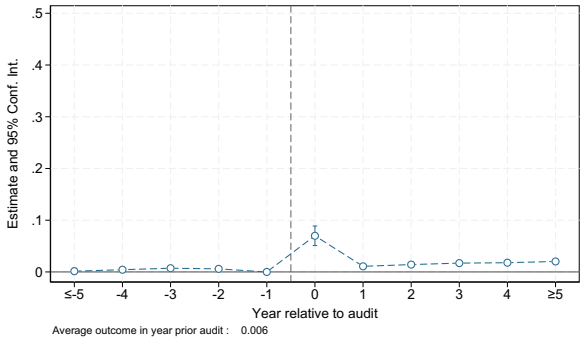
(a) Within 1 month



(b) Within 3 months



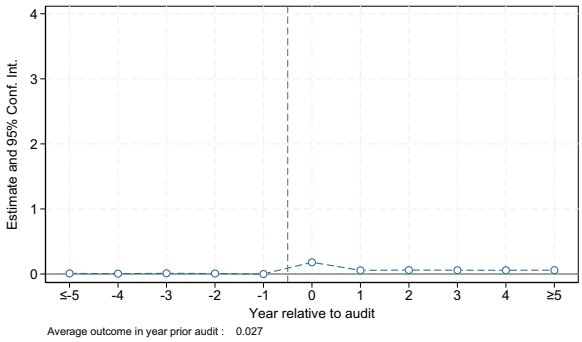
(c) Within 6 months



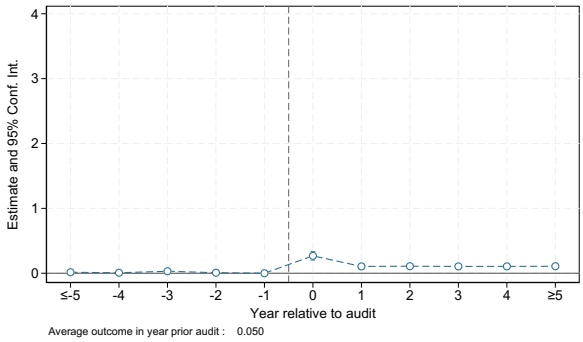
(d) Remainder of the inspection year

Notes: This figure reports within-firm spillover estimates for establishments that were not inspected but belong to firms with at least one establishment in which inspectors detected informal workers. Treated establishments are compared with matched establishments belonging to firms with no inspected sites. Event time is measured in years relative to the firm's first inspection in which informal workers were detected. A new formal hire is a worker registered at the uninspected establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the firm's inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. Each outcome is divided by the uninspected establishment's formal employment in the year preceding the inspection. The omitted period is $k = -1$, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

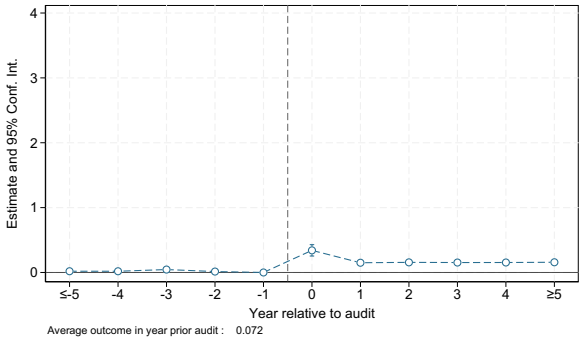
Figure F.5: Within-Firm Spillovers: New Formal Hires



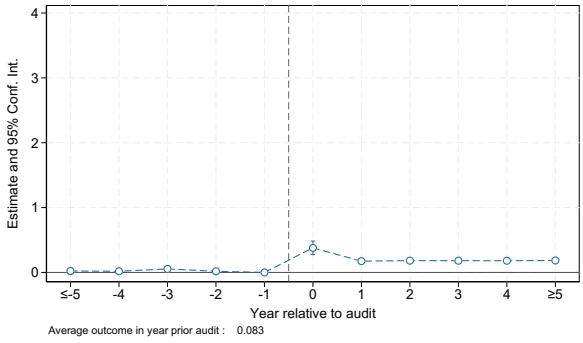
(a) Within 1 month



(b) Within 3 months



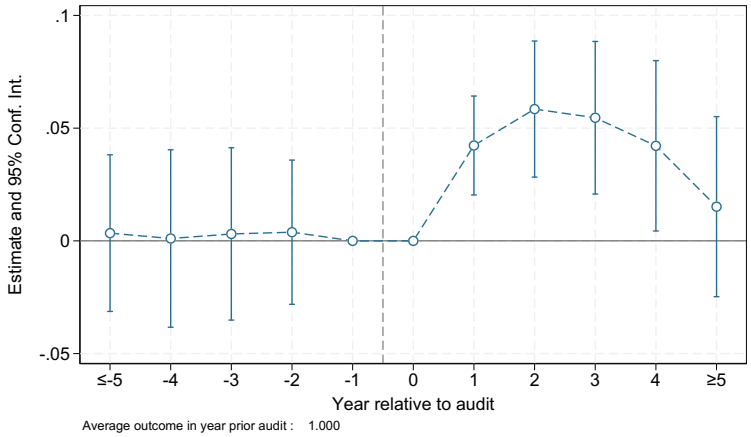
(c) Within 6 months



(d) Remainder of the inspection year

Notes: This figure reports within-firm spillover estimates for establishments that were not inspected but belong to firms with at least one establishment in which inspectors detected informal workers. Treated establishments are compared with matched establishments belonging to firms with no inspected sites. Event time is measured in years relative to the firm's first inspection in which informal workers were detected. A new formal hire is a worker registered at the uninspected establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the firm's inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. The outcomes are expressed as numbers of workers. The omitted period is $k = -1$, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

Figure F.6: Comparison Inspections: Establishment Activity in the Formal Sector



Notes: This figure reports event-study estimates for establishments whose first inspection identified a violation involving disability quotas, discrimination, or mandated rest periods but did not detect informal workers. These establishments are compared with matched establishments that were never inspected. Event time is measured in years relative to the first qualifying inspection. The outcome equals one if the establishment appears in RAIS with positive formal employment in the corresponding year and zero otherwise. The omitted period is $k = -1$, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

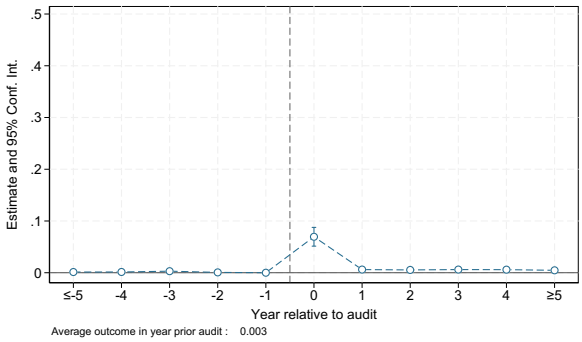
F.4 Retention of Newly Formalized Workers and Incumbents

Figure F.9 compares how long newly formalized hires and incumbent workers remain at the inspected establishment. Each column refers to one gender and each row to one horizon. Within each panel, the first four bars group newly formalized hires by the number of months between the inspection and their admission, the fifth bar pools all of them, and the last bar reports incumbents. The figure covers 300,440 newly formalized men, 210,727 newly formalized women, 855,958 incumbent men, and 615,103 incumbent women. Because the inspections in the sample took place between 2005 and 2018 and RAIS extends through 2021, every contract is observed over the full three-year horizon.

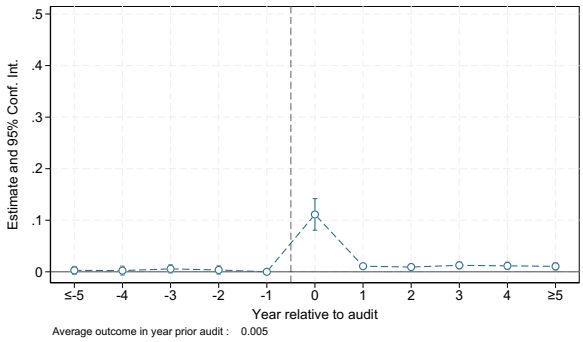
I compare newly formalized workers with incumbents, defined as workers already registered at the establishment in the month before inspection. Newly formalized jobs are relatively short-lived. After one year, 33% of newly formalized men and 37% of newly formalized women remain at the establishment; after three years, these shares fall to 10 and 12%. Retention is substantially higher among incumbents: 47% of men and 50% of women remain after one year, and 20 and 21% remain after three years. The incumbent advantage is similar across genders, at about 14 percentage points after one year and 10 percentage points after three years.

Lower retention is not specific to workers formalized following an inspection. Post-

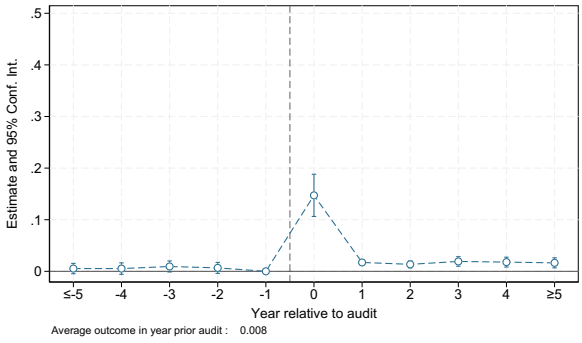
Figure F.7: Comparison Inspections: New Formal Hires Relative to Pre-Inspection Employment



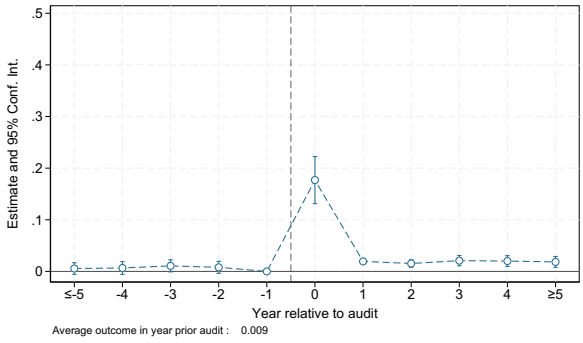
(a) Within 1 month



(b) Within 3 months



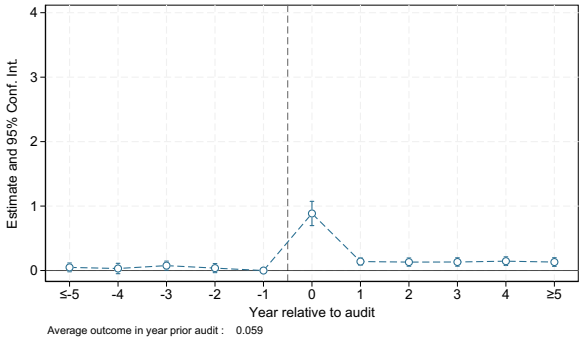
(c) Within 6 months



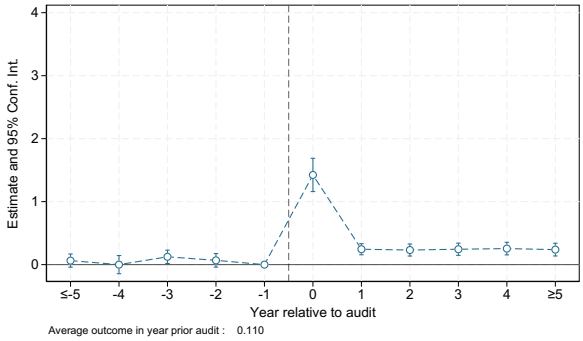
(d) Remainder of the inspection year

Notes: This figure reports event-study estimates for establishments whose first inspection identified a violation involving disability quotas, discrimination, or mandated rest periods but did not detect informal workers. These establishments are compared with matched establishments that were never inspected. Event time is measured in years relative to the first qualifying inspection. A new formal hire is a worker registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. Each outcome is divided by the establishment's formal employment in the year preceding the inspection. The omitted period is $k = -1$, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

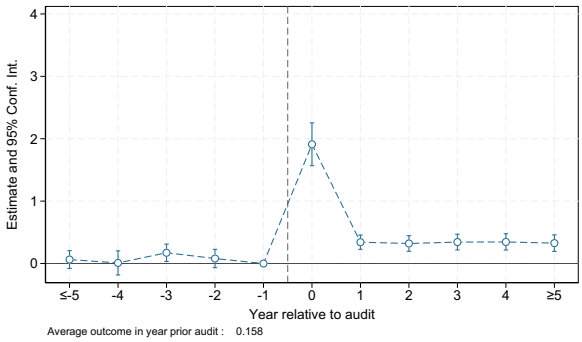
Figure F.8: Comparison Inspections: New Formal Hires



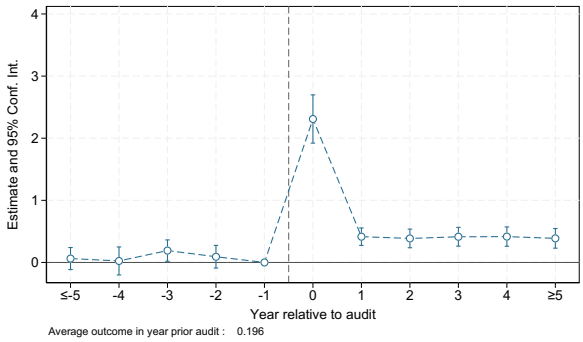
(a) Within 1 month



(b) Within 3 months



(c) Within 6 months

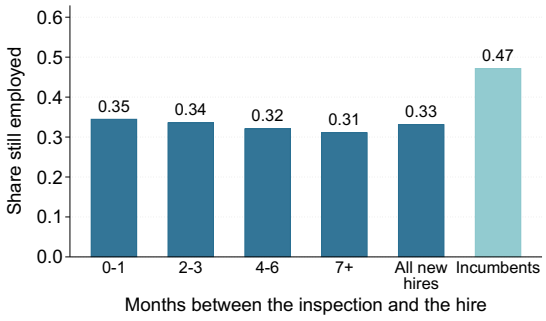


(d) Remainder of the inspection year

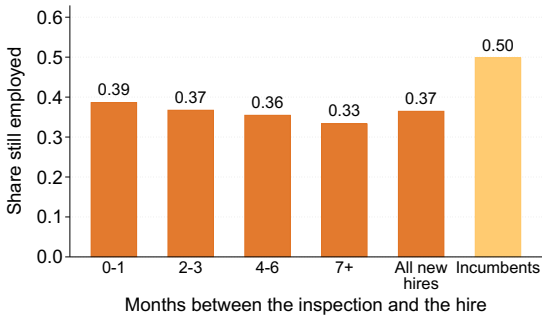
Notes: This figure reports event-study estimates for establishments whose first inspection identified a violation involving disability quotas, discrimination, or mandated rest periods but did not detect informal workers. These establishments are compared with matched establishments that were never inspected. Event time is measured in years relative to the first qualifying inspection. A new formal hire is a worker registered at the establishment during the indicated post-inspection window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Panels (a)–(c) count registrations occurring within one, three, and six months of the inspection, respectively. Panel (d) counts registrations occurring between the inspection date and the end of the inspection year. The outcomes are expressed as numbers of workers. The omitted period is $k = -1$, and coefficients at $k \leq -5$ and $k \geq 5$ are binned. Vertical bars denote 95% confidence intervals based on standard errors clustered by firm.

inspection hires arriving from another formal job have similar retention rates: 32% of men and 38% of women remain after one year. The difference between incumbents and new hires is therefore consistent with separation rates declining with tenure, rather than with unusually high separation among newly formalized workers. Gender differences in retention are comparatively small, with women 1 to 4 percentage points more likely than men to remain among both newly formalized workers and incumbents.

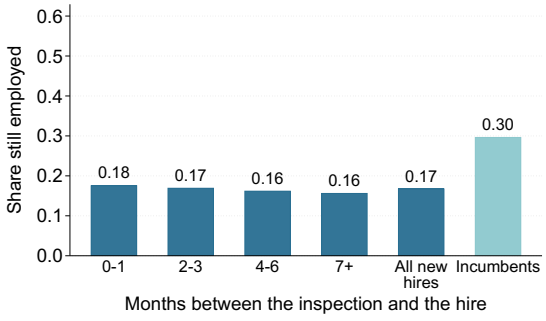
Figure F.9: Retention of Newly Formalized Hires and Incumbents at Inspected Establishments



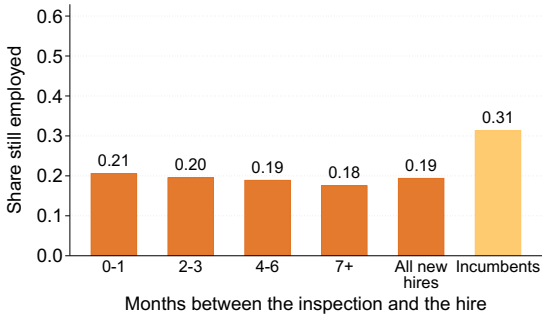
(a) Men, one year



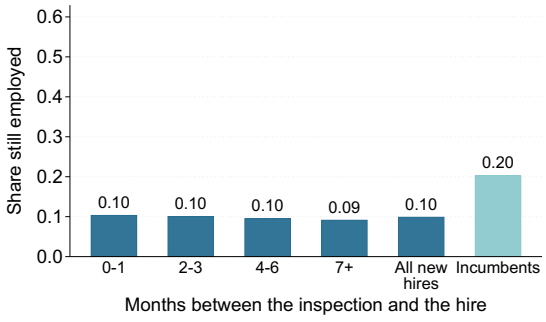
(b) Women, one year



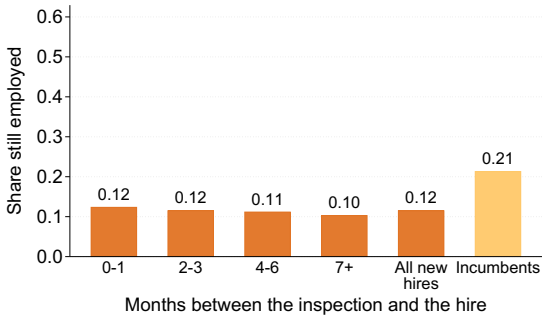
(c) Men, two years



(d) Women, two years



(e) Men, three years



(f) Women, three years

Notes: The figure reports the share of registered contracts still active at the inspected establishment one, two, and three years later (rows), separately for men and women (columns). The sample includes establishments in the matched sample inspected between 2005 and 2018, excluding pairs above the 99th percentile of informal workers detected. Newly formalized hires are workers admitted in or after the inspection month who held no active formal contract with any employer in the calendar month preceding the inspection; these are the registrations used to construct the take-up measures in Section 4. The first four dark bars separate these hires by time between inspection and admission, and the fifth pools them. The light bar reports incumbents, defined as workers already registered at the establishment in the month before inspection. Contract spells are followed in RAIS for three years and are considered retained if they remain active for at least the indicated horizon, measured from admission for newly formalized hires and from inspection for incumbents. Workers who separate and are later rehired are not considered retained. Each bar reports retained spells as a share of all spells in the corresponding group, pooled across establishments.

G. Take-Up Rates: Additional Evidence

Section 4 shows that most workers formalize following an inspection, while take-up remains incomplete and differs systematically across establishments. This appendix provides additional evidence on both results. I first show that incomplete take-up is not mechanically generated by establishment closure, worker turnover, inspector measurement, or extreme detections. I then examine how take-up evolves with the targeting of enforcement, varies with establishments' pre-inspection gender pay gaps, differs at establishments that are inspected again, and changes when the analysis is extended beyond the matched sample. Finally, I document geographic variation in take-up.

G.1 Measurement of Incomplete Take-Up

The incomplete take-up documented in Section 4.2 is not explained by mechanical features of the measure. Four potential concerns are relevant.

Establishment closure. If inspected establishments disappeared after an inspection, detected workers could not subsequently appear as registered at the establishment, mechanically lowering measured take-up. The data show the opposite: inspected establishments are 5.6 to 7.1 percentage points more likely than their matched controls to appear in the register in each of the five years following inspection (Section 3.3). Differential closure therefore cannot account for incomplete take-up.

Worker turnover. Workers who leave the inspected establishment for another formal job before registration are excluded from the numerator by the prior-contract restriction but remain in the denominator. This can lower measured take-up. The measure should therefore be interpreted as whether the inspected informal employment relationship becomes formal, rather than whether the worker subsequently obtains any formal job.

Inspector measurement. Inspectors record informal workers present when the establishment is inspected. Workers absent at the time of the visit may therefore be missed, which reduces the denominator and raises measured take-up. This measurement error works against finding incomplete take-up.

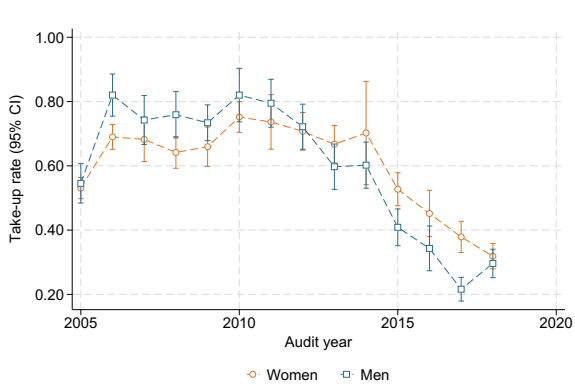
Extreme detections. A small number of inspections detect unusually large numbers of informal workers. Because these observations can disproportionately affect aggregate take-up, the baseline excludes matched pairs in which either establishment lies above the 99th percentile of total informal workers detected in the inspection year, approximately 2% of pairs. This restriction matters particularly for the pooled gender comparison because

the upper tail of detections is substantially heavier for men. Including these observations disproportionately increases the male denominator and can narrow or reverse the aggregate gender gap. I therefore treat the pooled gender difference as sensitive to the extreme tail and place greater weight on the systematic heterogeneity across establishments and industries documented in the main analysis.

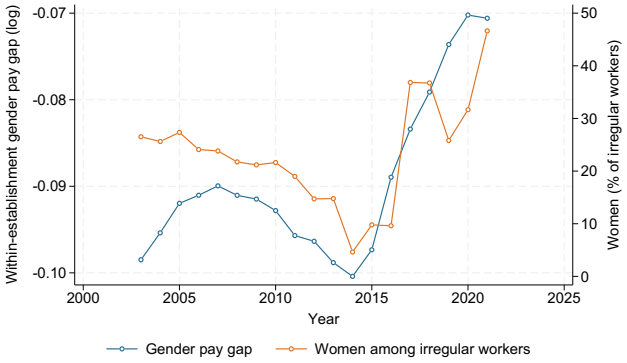
G.2 Take-Up Over Time and the Targeting of Enforcement

Figure G.1 places the take-up measure in time. Panel (a) reports the six-month rate by inspection year separately for women and men; Panel (b) plots two annual series: the share of detected informal workers who are women and the average within-establishment gender pay gap. The share of women declines until 2014 and then rises sharply, as enforcement moves toward female-dominated industries, while the gender pay gap narrows over the same period. Because the gender gap in take-up reverses across industries, this shift in targeting can change the aggregate gap without any change in behavior within industries, which is why inspection-year fixed effects are included throughout.

Figure G.1: Take-Up Over Time and the Changing Targeting of Enforcement



(a) Formalization rate by inspection year



(b) Women among detected informal workers and the gender pay gap

Notes: The two panels are computed on different samples because they answer different questions. Panel (a) reports mean six-month formalization rates by inspection year separately for women and men, with 95% confidence intervals, on the matched sample of inspected and never-inspected establishments used throughout the paper. The formalization rate is the estimated number of new formal hires following an inspection relative to the average number of informal workers detected during the inspection. New formal hires are workers who appear at the inspected establishment within six months of the inspection and did not hold a formal contract at any establishment in the month preceding the inspection. Panel (b) reports, by year, the share of informal workers detected during inspections who are women (right axis, percent) and the average within-establishment gender pay gap (left axis, log points). It describes whom inspectors found rather than how those workers responded, so it is computed on the full universe of inspected establishments, 2003–2021, without matching and without a comparison group; restricting it to the matched sample would describe the estimation sample rather than the allocation of enforcement.

G.3 Take-Up by the Establishment's Pre-Inspection Gender Pay Gap

Section 4.3 shows that the gender gap in take-up arises across establishments and varies across industries. Table G.1 instead divides establishments by whether their average pre-inspection gender pay gap exceeds the national average of 11 log points.

Among high-gap establishments, six-month take-up is 65.9% for women and 87.7% for men, a gap of 21.8 percentage points (Panel A). Among low-gap establishments, take-up is 76.2% for women and 78.9% for men, and the 2.7-percentage-point gap is only marginally significant. The gap is larger at high-gap establishments in every window and in the sample inspected in 2012 or later (Panel B). An establishment's pre-inspection gender pay gap may capture other dimensions of how it treats women, including non-wage job attributes. Establishment characteristics therefore help explain why women and men respond differently to formalization within the same broad enforcement environment.

Table G.1: Formalization Rates by the Establishment's Pre-Inspection Gender Pay Gap

	(1)	(2)	(3)	(4)	(5)	(6)
	High Gender Pay Gap (gap > 11 log points)			**Low Gender Pay Gap** (gap ≤ 11 log points)		
New-hire window	1 month	3 months	6 months	1 month	3 months	6 months
Panel A. All inspected establishments						
Men (λ_m)	0.503	0.703	0.877	0.451	0.627	0.789
Women (λ_w)	0.408	0.526	0.659	0.458	0.610	0.762
Difference ($\lambda_w - \lambda_m$)	−0.095***	−0.177***	−0.218***	0.007	−0.017	−0.027*
Panel B. Inspected in 2012 or later						
Men (λ_m)	0.510	0.682	0.801	0.391	0.557	0.708
Women (λ_w)	0.376	0.493	0.612	0.427	0.576	0.718
Difference ($\lambda_w - \lambda_m$)	−0.134**	−0.188*	−0.189*	0.036***	0.019	0.011

Notes: This table reports formalization rates by gender and the inspected establishment's average pre-inspection gender pay gap. Establishments with a gap above 11 log points are classified as high-gap, and each inspected establishment's classification is assigned to its matched control. Columns define new formal hires within one, three, and six months of the inspection. New hires had no active formal contract in the month before the inspection. For each gender, the rate is the estimated number of new formal hires relative to the average number of informal workers detected, conditional on detecting at least one. Panel A includes all inspections; Panel B includes inspections conducted since 2012. Standard errors are clustered by establishment. Gender differences are estimated using a stacked specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G.4 Take-Up at Establishments Inspected Again

A detected worker may remain informal because the employer does not offer a formal contract or because the worker declines it. Repeat inspections provide evidence on the employer margin. I compare six-month take-up at establishments where no informal workers are found at the second inspection with take-up at those where informal workers are detected again (Table G.2).

Among establishments with no informal workers at the second inspection, take-up is 97.9% for men and 78.6% for women, a gap of 19.4 percentage points (Panel A, column 1). The gap is similar where informal workers are detected again (22.0 points, column 2) and persists among establishments inspected since 2012 (20.7 points, Panel B, column 1). Women's lower take-up therefore persists even where establishments subsequently comply. However, restricting the sample to establishments where inspectors detected both informal women and men eliminates the gap: take-up differs by only 2.1 percentage points among establishments with no informal workers at the second inspection and 2.0 points where informal workers are detected again (columns 3–4). Neither difference is statistically significant, including in the post-2012 sample.

As in the full sample (Section 4.3), the gender gap arises across rather than within establishments. This pattern provides no evidence that subsequently complying establishments selectively formalize men over women.

Table G.2: Formalization Rates at Establishments Inspected at Least Twice

	(1)	(2)	(3)	(4)
	Gender-Specific Samples (≥ 1 informal worker of that gender)		**Both Genders Detected** (≥ 1 informal woman and man)	
Informal at second inspection	None	Found	None	Found
Panel A. All inspected establishments				
Men (λ_m)	0.979	1.090	0.739	0.856
Women (λ_w)	0.786	0.870	0.718	0.835
Difference ($\lambda_w - \lambda_m$)	−0.194***	−0.220***	−0.021	−0.020
Panel B. Inspected in 2012 or later				
Men (λ_m)	1.043	1.007	0.768	0.874
Women (λ_w)	0.836	0.892	0.748	0.781
Difference ($\lambda_w - \lambda_m$)	−0.207***	−0.115	−0.020	−0.093

Notes: This table reports six-month formalization rates for matched inspected establishments subsequently inspected in a later year. Columns (1) and (3) include establishments where no informal workers are detected at the second inspection; columns (2) and (4) include those where informal workers are detected again. Columns (1)–(2) condition on detecting at least one informal worker of the relevant gender at the first inspection; columns (3)–(4) require informal workers of both genders. Each inspected establishment's classification is assigned to its matched control. Formalization rates are estimated as in column (3) of Table 3: the inspection-induced increase in new formal hires within six months, relative to the average number of informal workers of the corresponding gender detected. New hires had no formal contract anywhere in Brazil in the month preceding the inspection. Rates may exceed one because detected workers cannot be linked directly to subsequent registrations. Panel A includes all inspections; Panel B restricts the sample to inspections conducted in 2012 or later. Re-inspection is not randomly assigned. Standard errors are clustered by establishment, and gender differences are estimated using a stacked specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G.5 Take-Up in the Ever-Inspected Sample

The matched design excludes inspected establishments for which no comparable never-inspected establishment is available. As a robustness check on the gender-by-industry pattern, Table G.3 instead uses the broader ever-inspected sample and the treated-only estimates from Section F.1. These estimates are not directly comparable with the matched-sample take-up rates. Without a comparison group, the change in qualifying formal hires cannot be isolated from other changes in hiring around the inspection, so I use this exercise only to assess whether the relative pattern across genders and industries persists outside the matched sample.

The pattern is similar. In female-dominated industries, the treated-only measure is

higher for women than for men at every hiring horizon, with women–men differences ranging from 8.8 to 15.3 percentage points in the full sample and from 13.9 to 23.2 points among establishments inspected since 2012. In male-dominated industries, the differences are smaller and generally negative. Thus, although the levels of the treated-only and matched estimates have different interpretations, the reversal in relative formalization across male- and female-dominated industries is also present in the broader ever-inspected sample.

Table G.3: Formalization Rates, Ever-Inspected Sample, by Industry Gender Composition

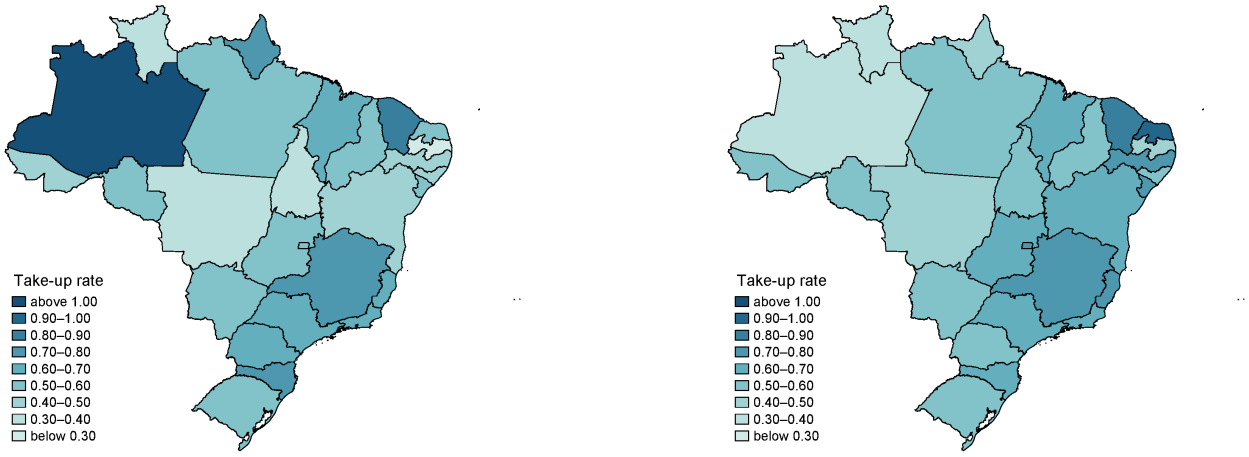
	(1)	(2)	(3)	(4)	(5)	(6)
	Female-Dominated Industries			**Male-Dominated Industries**		
New-hire window	1 month	3 months	6 months	1 month	3 months	6 months
Panel A. All ever-inspected establishments						
Men (λ_m)	0.427	0.623	0.811	0.521	0.799	1.049
Women (λ_w)	0.515	0.736	0.963	0.499	0.726	0.950
Difference ($\lambda_w - \lambda_m$)	**0.088*****	**0.113*****	**0.153*****	**−0.023**	**−0.073**	**−0.100***
Panel B. Inspected in 2012 or later						
Men (λ_m)	0.321	0.474	0.604	0.333	0.520	0.690
Women (λ_w)	0.461	0.647	0.836	0.337	0.507	0.657
Difference ($\lambda_w - \lambda_m$)	**0.139*****	**0.173*****	**0.232*****	**0.004**	**−0.013**	**−0.033**

Notes: This table reports gender-specific take-up measures for establishments in which an inspection detected at least one informal worker. For gender $d \in \{m, w\}$, the measure is $\hat{\lambda}_d = \hat{\theta}_0^d / \bar{N}_d$, as in equation (3), where $\hat{\theta}_0^d$ is the inspection-year coefficient from equation (F.1) for new formal hires of gender d and $\bar{N}_d$ is the average number of informal workers of that gender detected during the inspection. A new formal hire is a worker registered at the establishment within the indicated window who held no active formal contract with any employer in the calendar month immediately preceding the inspection. Because inspection records cannot be linked to RAIS at the worker level, $\hat{\lambda}_d$ is an implied take-up measure rather than the directly observed fraction of detected workers subsequently registered; it may exceed one if the estimated increase in qualifying registrations exceeds the number of informal workers detected. A two-digit industry is classified as female-dominated if its median establishment female employment share exceeds one third over 2003–2021. Panel A includes all ever-inspected establishments, and Panel B restricts the sample to establishments first inspected in 2012 or later. Both panels require positive formal employment in each of the three years preceding the inspection. Gender-specific measures are estimated using establishments in which at least one informal worker of the corresponding gender was detected and therefore need not use identical samples. The reported difference is $\hat{\lambda}_w - \hat{\lambda}_m$, and p-values come from a covariance-corrected stacked specification: the male and female outcomes are stacked, each pre-scaled by its own denominator, every regressor and fixed effect is interacted with gender, and standard errors are clustered by establishment, so the covariance between $\hat{\lambda}_w$ and $\hat{\lambda}_m$ induced by establishments where informal workers of both genders were detected is accounted for. This is the same test used for the matched-sample take-up tables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G.6 Take-Up Across States

Figure G.2 maps the six-month formalization rate by state, separately for women and men. The geographic gradient is visible for both genders. The maps use the matched sample and pool all inspection years.

Figure G.2: Formalization Rate by State, Six-Month Window



(a) Women **(b)** Men

Notes: This figure maps six-month formalization rates by state for the matched sample, pooling all inspection years. New formal hires are workers registered within six months of the inspection who held no active formal contract with any employer in the month before it. Each state is shaded by the mean rate among inspected establishments located there, separately for women in panel (a) and men in panel (b).

H. Constructing the Formal Wage Premium

The wage premium $\Delta w_{dj} = \psi_{dj} - \pi_{ds(j)}$ of equation (7) is assembled from two separate estimations on two separate datasets: an AKM wage regression on RAIS, 2010–2021, which yields the establishment premium ψ_{dj} entering the main specification in equation (14), and a worker fixed-effects regression on PNAD-C, 2012–2019, which yields the informal benchmark $\pi_{ds(j)}$ subtracted in the complementary specification in equation (13). This appendix documents both and reports the diagnostics.

H.1 The Establishment Pay Premium

Sample. I estimate establishment pay premia using RAIS 2010–2021, restricting the sample to private-sector employees aged 18–65 on indeterminate-duration CLT contracts paid a monthly salary, with positive contracted hours and earnings and non-missing education. Before restricting to connected sets, the sample contains 436.9 million worker-year observations, 76.6 million workers, and 7.57 million establishments.

Specification. Following Morchio & Moser (2026), I estimate a two-way fixed-effects wage regression separately by gender. For worker i of gender d employed at establishment $j(i,t)$ in year t,

$$\ln w_{it} = \alpha_i + \psi_{d\,j(i,t)} + \mathbf{X}'_{it}\mathbf{b}_d + \varepsilon_{it}, \tag{H.1}$$

where w_{it} is real monthly earnings in 2016 reais, α_i is a worker fixed effect, ψ_{dj} is the gender-specific establishment pay premium, and $\mathbf{X}_{it}$ is a vector of time-varying controls with gender-specific coefficients $\mathbf{b}_d$. The controls include (i) education × age-group fixed effects, interacting four education groups (up to completed fifth grade; sixth grade to incomplete high school; completed high school; and some college or more) with five age groups (18–24, 25–34, 35–44, 45–54, and 55–65); (ii) education × year fixed effects, interacting the same four education groups with calendar years 2010–2021, thereby absorbing common year effects while allowing returns to education to vary over time; (iii) fixed effects for four bins of contracted weekly hours (below 20, 20–30, 31–39, and 40 or more); (iv) one-digit occupation fixed effects based on the Brazilian occupational classification (CBO 2002); (v) formal-sector experience and its square, where experience is measured as years since the worker's first observed spell in RAIS, using employment histories beginning in 2003; and (vi) tenure at the current employer and its square, where tenure is measured in completed years and capped at formal-sector experience. The establishment effect ψ_{dj}

captures the establishment-specific component of pay for workers of gender d, conditional on worker fixed effects and these time-varying controls. The main specification uses real monthly earnings net of employee social-security contributions and income taxes; alternative earnings measures are considered below.

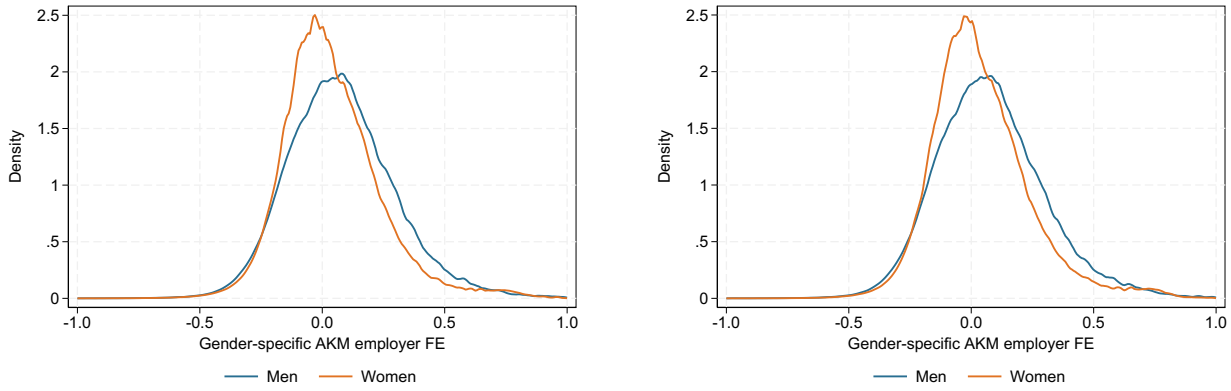
Because equation (H.1) is estimated separately by gender, ψ_{mj} and ψ_{wj} are estimated from nonoverlapping sets of workers. This feature motivates the cross-gender instrument in Section 5.4.3: the other gender's establishment premium can capture common establishment pay policy without sharing the mechanical estimation error in the own-gender premium.

Connected set. Fixed effects are identified only within a connected set of establishments linked by worker mobility. I take the largest weakly connected component of the mobility graph, separately for each gender, requiring an establishment to be observed for at least three years. The male connected set contains 180.5 million observations, 25.7 million workers, and 2.79 million establishments; the female set 109.3 million observations, 16.3 million workers, and 2.42 million establishments. This retains 67.8% of male and 64.0% of female observations. The observations lost are dropped by the filters applied before the connected set is formed, which remove establishments without mobility links and establishments observed for fewer than three years. Among the establishments that survive these filters, 99.8% belong to the largest connected set for each gender, so the connectivity requirement itself removes almost no establishments.

Normalization. Because the regression carries gender-specific establishment effects, two normalizations are required, one per gender, and the choice of reference sector is not innocuous. I impose that the employment-weighted mean establishment effect is zero among restaurants and fast-food outlets (CNAE 56112), separately by gender and within each connected set, following Morchio & Moser (2026). The sector must satisfy three conditions: it must rank near the bottom of the utility ladder for both genders, it must treat men and women as near-perfect substitutes in production, and it must provide near-zero amenity value to both. The first is standard since Card, Cardoso & Kline (2016) and is verified in Brazilian data by Morchio & Moser (2026).

Diagnostics. The variance of the estimated establishment effects is 0.060 for men and 0.049 for women on gross monthly earnings, and 0.054 and 0.045 on earnings net of employee contributions and income taxes. The correlation between worker and establishment effects is about 0.35 for both genders, in the range reported for Brazil elsewhere. Figure H.1 plots the distributions.

Figure H.1: Establishment Pay Premia by Gender



(a) Net monthly earnings

(b) Net hourly earnings

Notes: The figure shows kernel densities of gender-specific establishment pay premia, ψ_{dj}, estimated separately for men and women using RAIS 2010–2021. Establishment effects are normalized to zero among restaurants and fast-food outlets within each connected set. Panel (a) reports premia based on net monthly earnings, the wage measure used in the main analysis, and panel (b) reports premia based on net hourly earnings. Net earnings subtract employee social-security contributions and income taxes. Densities are weighted by establishment–gender worker-years.

Wage concepts. The establishment premium is estimated on monthly and on hourly earnings, each gross and net of the employee social-security contribution and income tax. The main specification uses *net monthly* earnings, paired with the PNAD-C take-home earnings concept for the informal benchmark, because the object in the worker's utility function is take-home pay. Columns (4)–(6) of Appendix Table J.3 re-estimate the 2SLS specification, without the amenity battery, under alternative wage concepts. The wage coefficient is 3.03 for men and 2.98 for women with the gross monthly premium and 3.15 and 3.05 with the net hourly premium, close to the main estimates of 3.17 and 3.08 in Table 5. Weighting the net monthly specification by establishment employment leaves the male coefficient at 3.11 and lowers the female coefficient to 2.42, which is less precisely estimated. The wage concept therefore does not drive the estimated wage coefficients. The female-to-male ratio of the coefficients stays close to one under every unweighted wage concept and falls to about 0.78 only in the employment-weighted specification; as discussed in Section 5, these ratios are informative about relative choice behavior rather than relative wage preferences.

H.2 The Informal Wage Benchmark

I estimate the informal wage benchmark from the PNAD-C using a worker fixed-effects specification, separately by gender:

$$\ln w_{it} = \theta_i + \sum_{z \neq z^*} \delta_z^F \cdot \mathbf{1}\{z_{it} = z,\ f_{it} = F\} + \sum_{z,r} \delta_{zr}^I \cdot \mathbf{1}\{z_{it} = z,\ r_{it} = r,\ f_{it} = I\} + \mathbf{X}_{it}'\boldsymbol{\kappa} + u_{it}, \quad \text{(H.2)}$$

where θ_i is a worker fixed effect, z_{it} and r_{it} denote the worker's industry and state, and $f_{it} \in \{F, I\}$ denotes formality status. For gender d, I define $\pi_{ds} \equiv \hat{\delta}^I_{z(s)r(s)}$ as the informal wage component in market $s = (z, r)$, where the coefficients are obtained from the gender-d regression. The specification is estimated on private-sector employees aged 18–65 in the PNAD-C from 2012Q1 to 2019Q4. Identification comes from within-worker changes in formality status, with worker fixed effects absorbing time-invariant worker heterogeneity. Formal employment in restaurants and fast food (z^*) is the omitted category, matching the normalization used for the establishment pay premia. For the main specification, I estimate the model using take-home earnings to match the net earnings measure used to construct the formal establishment premium.

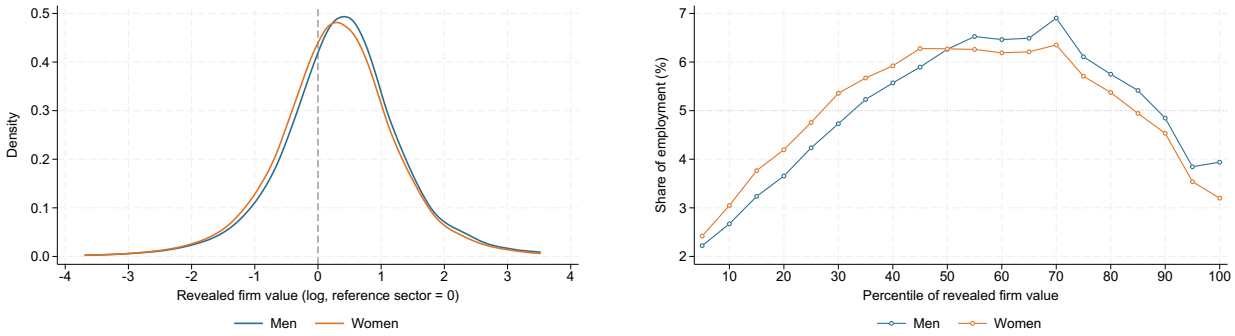
H.3 Firm Quality from Worker Mobility

The observed amenity measures capture only a subset of the non-wage attributes workers may value. I therefore construct a revealed-preference measure of establishment value following Sorkin (2018). The measure uses job-to-job transitions to infer which establishments workers prefer: establishments that systematically attract workers from other employers receive higher revealed values. Because the measure is constructed from worker mobility rather than wages or observed amenities, it provides an independent proxy for dimensions of job quality omitted from the baseline specification.

I construct the index separately by gender using employer-to-employer transitions in RAIS from 2003 to 2021. I normalize the resulting establishment values to zero in restaurants and fast food, matching the normalization of the establishment pay premia. Figure H.2 shows the resulting distributions. Women are somewhat more concentrated in establishments with lower revealed value: the employment-weighted difference in the index between men and women is 0.079 log points.

For this measure to provide a useful robustness check, it must capture variation in establishment value beyond the pay premium itself. Figure H.3 compares the revealed-preference index, $\hat{v}_{dj}$, with the AKM establishment pay premium, $\hat{\psi}_{dj}$, in the estimation

Figure H.2: Revealed-Preference Firm Value



(a) Distribution by gender

(b) Employment share by firm-value ventile

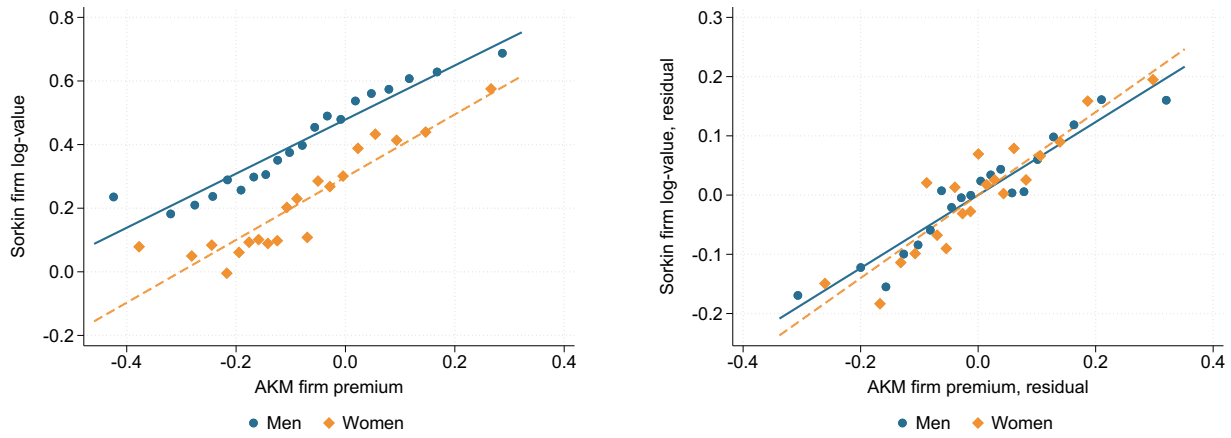
Notes: The figure describes revealed establishment value constructed from job-to-job transitions following Sorkin (2018). The index is computed separately by gender using employer-to-employer transitions in RAIS, 2003–2021, and normalized to zero in restaurants and fast food. Panel (a) shows the distribution of the all-worker index across establishments, weighted by male and by female employment, between the 0.5th and 99.5th percentiles; the dashed line marks the reference sector. Panel (b) shows male and female employment shares across ventiles of the all-worker index. The employment-weighted gender difference in revealed value is 0.079 log points.

sample. The two measures are positively but weakly related. In levels, the correlation is 0.16 for men and 0.15 for women. After residualizing both measures on the industry × state and inspection-year fixed effects and the CBA coverage indicator used in the estimating specification, the correlations fall to 0.11 and 0.10, respectively. Thus, revealed establishment value contains substantial variation not captured by the establishment pay premium.

The weak relationship is not specific to inspected establishments. Among all establishments for which both measures can be constructed, the correlation between $\hat{\psi}_{dj}$ and $\hat{v}_{dj}$ is 0.21 for men and 0.23 for women, rising to 0.26 and 0.33 when establishments are weighted by employment. The revealed-preference measure therefore provides a complementary proxy for establishment desirability.

The last two columns of Appendix Table J.6 assess whether omitted establishment desirability confounds the main estimates. I re-estimate equation (14) controlling for the component of $\hat{v}_{dj}$ orthogonal to $\hat{\psi}_{dj}$, standardized to unit variance. The wage coefficients decline modestly, from 3.17 to 2.90 for men and from 3.08 to 2.78 for women, while the relative monetary values are largely unchanged. Their rank correlation with the baseline estimates is 0.88 for men and 0.96 for women. Thus, controlling for revealed establishment desirability has little effect on the estimated relative valuations of wages and amenities.

Figure H.3: Revealed-Preference Firm Value and the Pay Premium



(a) Levels **(b)** Within estimation fixed effects

Notes: The figure compares the gender-specific revealed-preference establishment value, $\hat{v}_{dj}$, with the AKM establishment pay premium, $\hat{\psi}_{dj}$, in the model estimation sample. Points are means within twenty gender-specific quantile bins of $\hat{\psi}_{dj}$; fitted lines and reported correlations use establishment–gender observations. Panel (a) reports the relationship in levels. Panel (b) residualizes both measures on industry × state and inspection-year fixed effects and the CBA coverage indicator. The corresponding correlations are 0.164 and 0.111 for men and 0.149 and 0.097 for women.

I. Counterfactual Implementation

This appendix details the implementation of the counterfactuals in Section 7. For establishment j and gender d, write the baseline fitted log-odds of the estimating equation as

$$\Lambda_{dj} = \delta_d \psi_{dj} + \boldsymbol{\beta}_d' \Delta \mathbf{a}_j + \boldsymbol{\beta}_d^{c\prime} \mathbf{c}_j + R_{dj}, \tag{I.1}$$

where ψ_{dj} is the formal wage component, $\Delta \mathbf{a}_j = \mathbf{a}_j - \bar{\mathbf{a}}_{s(j)}$ collects the formal–informal amenity differences, $\mathbf{c}_j$ the collective-agreement provisions, and R_{dj} the remaining components of the fitted index. The baseline probability is $\Lambda^{-1}(\Lambda_{dj})$, where $\Lambda^{-1}(x) = \exp(x)/[1 + \exp(x)]$. Every exercise changes a set of preference parameters or job attributes, adds the implied shift to the index, and maps the result back into a probability through Λ^{-1}.

I.1 Design Features

Three features of the design guide interpretation. First, every exercise is conditional on a formalization opportunity: the counterfactuals change the probability of accepting a formal contract, not the probability of receiving one. Second, the exercises are partial equilibrium. They hold fixed informal wages (except where stated), other establishments' attributes, and workers' allocation across establishments, abstracting from re-sorting and employers' responses. Third, inference uses 1,000 establishment-level bootstrap replications to test whether the counterfactual gender gap differs from its baseline.

I.2 Equalizing Preferences

Holding each job fixed, the preference exercise replaces gender d's coefficients with those of the reference group:

$$s_{dj}^{\text{pref}} = (\delta_{\text{ref}} - \delta_d)\psi_{dj} + (\boldsymbol{\beta}_{\text{ref}} - \boldsymbol{\beta}_d)' \Delta \mathbf{a}_j + (\boldsymbol{\beta}_{\text{ref}}^c - \boldsymbol{\beta}_d^c)' \mathbf{c}_j. \tag{I.2}$$

The reference group is the majority gender in each workplace type: women in male-dominated industries receive men's coefficients, and men in female-dominated industries receive women's. The counterfactual probability is

$$p_{dj}^{\text{pref}} = \Lambda^{-1}\left(\Lambda_{dj} + s_{dj}^{\text{pref}}\right). \tag{I.3}$$

The exercise is implemented separately for the wage coefficient, establishment amenities, and collective-agreement provisions, and then jointly, holding the remaining components of the predicted log-odds fixed.

The exercise transfers the reference group's scale-normalized coefficients. It measures how take-up would change if the minority group's log-odds responded to job attributes as the reference group's do. Interpreting this as a change in preferences alone requires equal unobserved dispersion across genders. Transferring RMVs instead, while retaining the minority group's wage coefficient, would rescale the reference group's amenity coefficients by $\hat{\delta}_d/\hat{\delta}_{\text{ref}}$: 0.97 for women and 1.03 for men. The amenity-only exercises thus differ by a small rescaling of the transferred coefficients. The joint exercise additionally replaces the wage coefficient, changing both monetary valuations and wage responsiveness in log-odds.

Among individual amenities, the part-time share accounts for more than half of the change in male-dominated industries: transferring its coefficient alone raises women's predicted formalization to 56.4%, while transferring the female-manager coefficient raises it to 55.6%. In female-dominated industries, the largest individual contributions come from manual task content and the part-time share, which lower men's predicted formalization to 57.0 and 58.0%, respectively.

I.3 Equalizing Wage Premia

The wage-premium exercise of Panel C of Table 8 holds preferences and non-wage attributes fixed and assigns the minority group the reference group's formal wage premium:

$$s_{dj}^{\psi} = \delta_d \left(\psi_{\text{ref},j} - \psi_{dj} \right). \tag{I.4}$$

I.4 A Minimum-Wage Increase

Panel D of Table 8 considers a 10% increase in formal wages. Because minimum-wage increases can also raise informal wages, let ρ denote the pass-through of minimum-wage-induced formal wage gains to informal wages:

$$\Delta\psi_j^{MW} = \Delta w_{F,j}^{MW} - \Delta w_{I,s(j)}^{MW} = (1-\rho)\Delta w_{F,j}^{MW}. \tag{I.5}$$

The preferred calibration $\rho = 0.66$ follows Derenoncourt et al. (2025), who estimate that minimum-wage-induced wage gains among informal employees are 66% as large as those

among formal employees. The change in the wage premium shifts the fitted index by

$$s_{dj}^{MW} = \hat{\delta}_d \Delta\psi_j^{MW}, \tag{I.6}$$

and the counterfactual probability is

$$p_{dj}^{MW} = \Lambda^{-1}\left(\Lambda_{dj} + s_{dj}^{MW}\right). \tag{I.7}$$

I.5 Provision Floors

For each provision ℓ in bundle $\mathcal{B}$, the policy of Panel E of Table 8 raises its value to the sample mean at establishments initially below that mean:

$$c_{\ell j}^{CF} = \max\{c_{\ell j},\, \bar{c}_\ell\}, \tag{I.8}$$

which shifts the fitted index by

$$s_{dj}^{\mathcal{B}} = \sum_{\ell \in \mathcal{B}} \hat{\beta}_{d\ell}^{c} \left(c_{\ell j}^{CF} - c_{\ell j}\right). \tag{I.9}$$

J. Robustness of the Structural Estimates

This appendix assesses the robustness of the paper's main quantitative conclusions. I first examine the structural estimates—the wage response and relative valuations of non-wage attributes—under alternative outcomes, estimators, wage measures, amenity definitions, samples, and controls for revealed establishment value. I then report robustness for the survey–administrative comparison, the implied formal-opportunity rate in Section 9.1, and the mechanism decomposition in Section 8.4.

J.1 The Full Coefficient Battery

Tables J.1 and J.2 report the complete amenity-coefficient battery underlying the relative monetary values in Table 6. The four establishment-level amenity differences and fifteen collective-agreement groups enter jointly in the preferred establishment-premium specification; they are split across two tables only for presentation.

Table J.1: Formalization Log-Odds: Amenity Differences

	(1) Men	(2) Women
Δ Part-time share	−0.03	0.73
	(0.44)	(0.46)
Δ Female manager	−0.81***	0.29*
	(0.14)	(0.15)
Δ Manual task	0.21	−0.87***
	(0.13)	(0.16)
Δ Interpersonal task	−0.16*	−0.24**
	(0.10)	(0.10)
CBA clause groups (15)	✓	✓
Total clause intensity	✓	✓
Industry × state FE	✓	✓
Inspection-year FE	✓	✓
First-stage F	2529	1610
N (establishments)	13,059	9,683

Notes: The table reports the four amenity-difference coefficients from the establishment-premium specification in equation (14), estimated by 2SLS separately by gender using the cross-gender instrument. Each amenity difference is the formal-establishment value minus its industry × state informal benchmark. The part-time measure is the share of workers employed part-time, the female-manager measure indicates whether the establishment's highest-paid worker is a woman, and the task measures are mean O*NET z-scores. The same regressions include the fifteen collective-agreement clause groups reported in Table J.2. Standard errors clustered at the establishment level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table J.2: Formalization Log-Odds: Collective-Agreement Clause Groups

	(1) Men	(2) Women
Wage & compensation	−0.02	−0.66
	(0.78)	(0.87)
Bonuses & profit-sharing	−0.09	−0.71
	(0.49)	(0.59)
Social benefits	0.06	−0.82
	(0.44)	(0.55)
Contract & termination	0.32	0.66
	(0.50)	(0.60)
Maternity & childcare	0.12	0.47
	(0.30)	(0.35)
Leaves	0.13	0.65*
	(0.29)	(0.34)
Harassment & discrimination	0.01	0.63
	(0.48)	(0.59)
Flexibility & part-time	−0.25	0.60*
	(0.30)	(0.35)
Work organization	0.20	0.00
	(0.31)	(0.36)
Employment protection	−0.04	−0.44
	(0.35)	(0.42)
Work shifts	0.43	−0.54
	(0.55)	(0.63)
Health & safety	−0.01	−0.67
	(0.42)	(0.51)
Union provisions	−0.89	0.47
	(0.74)	(0.88)
CBA negotiation	0.40	1.79**
	(0.68)	(0.71)
Other clauses	−0.32	0.59
	(0.36)	(0.42)
Amenity differences (4)	✓	✓
Total clause intensity	✓	✓
Industry × state FE	✓	✓
Inspection-year FE	✓	✓
First-stage F	2529	1610
N (establishments)	13,059	9,683

Notes: The table reports the fifteen collective-agreement clause-group coefficients from the establishment-premium specification in equation (14), estimated by 2SLS separately by gender using the cross-gender instrument. Each regressor indicates whether the establishment's collective agreement contains at least one clause in the corresponding group in any of the three years preceding the inspection; Appendix Table C.1 reports the underlying classification. The same regressions include the four amenity differences reported in Table J.1. Standard errors clustered at the establishment level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J.2 Alternative Estimators and Take-Up Measures

Table J.3 asks whether the positive wage response depends on the take-up transformation, estimator, measure of establishment value, or earnings concept. Estimating take-up in levels, shrinking noisy establishment-level rates, and replacing the pay premium with the Sorkin (2018) revealed establishment value all preserve a positive relationship between establishment value and formalization. Columns (4)–(6) return to the 2SLS log-odds specification, without the amenity battery, and show that the estimated wage response is also stable across alternative earnings measures and employment weighting. I retain log-odds in the main specification because it follows directly from the choice model and maps predicted values into formalization probabilities.

J.3 Empirical-Bayes Shrinkage of the Outcome

Establishment-level take-up is measured with sampling error, particularly when inspections detect few informal workers. Following Walters (2024), I shrink each establishment's take-up rate toward the mean, with greater shrinkage for noisier estimates. Column (2) of Table J.3 shows that the wage coefficient falls from 1.11 to 0.65 for men and from 0.83 to 0.43 for women, but remains positive and statistically significant for both. The positive relationship between establishment pay and formalization is therefore robust to empirical-Bayes shrinkage.

J.4 Alternative Wage Concepts

The baseline uses net monthly earnings because formalization directly affects take-home pay and the informal benchmark is defined using the same earnings concept. Columns (4)–(6) of Table J.3 show that the 2SLS wage coefficients are stable across alternative earnings measures. Without the amenity battery, net monthly earnings yield coefficients of 3.17 for men and 3.04 for women; these become 3.03 and 2.98 using gross monthly earnings and 3.15 and 3.05 using net hourly earnings. The corresponding estimates in the preferred specification, which additionally includes the amenity battery, are 3.17 and 3.08. Employment weighting yields 3.11 for men and 2.42 for women. Overall, the strong wage response and the absence of a systematic gender difference are robust to the earnings concept and weighting scheme.

Table J.3: Estimator and Outcome Transformation

	(1)	(2)	(3)	(4)	(5)	(6)
	Take-up measure		*Firm value*	*Establishment pay premium*		
	Raw rate	EB-shrunk rate	Sorkin value	Gross monthly	Net hourly	Net monthly, emp.-weighted
Panel A. Men						
Coefficient	1.113***	0.654***	0.694***	3.030***	3.150***	3.108***
	(0.120)	(0.055)	(0.054)	(0.622)	(0.638)	(0.691)
Observations	20,248	20,248	20,231	13,288	13,288	13,288
Panel B. Women						
Coefficient	0.833***	0.432***	0.391***	2.977***	3.045***	2.420**
	(0.092)	(0.049)	(0.056)	(0.888)	(0.899)	(1.012)
Observations	15,596	15,596	14,922	9,847	9,847	9,847
Regressor	ψ_{dj}	ψ_{dj}	Sorkin value	ψ_{dj}	ψ_{dj}	ψ_{dj}
Outcome	Rate	Rate	Log-odds	Log-odds	Log-odds	Log-odds
Estimator	OLS	OLS	OLS	2SLS	2SLS	2SLS
Fixed effects	Industry × state, inspection year					

Notes: The table reports estimates separately for men (Panel A) and women (Panel B); the regressor in each specification is listed at the bottom of the corresponding column. Column (1) estimates by OLS a linear probability model of the raw single-difference take-up rate on the establishment pay premium ψ_{dj}, while column (2) replaces the outcome with the empirical-Bayes shrunk take-up rate. Column (3) estimates the log-odds of take-up by OLS using the Sorkin (2018) revealed-preference firm value in place of ψ_{dj}. Columns (4)–(6) estimate the log-odds by 2SLS, instrumenting each gender's establishment pay premium with the other gender's premium at the same establishment. These columns differ in the wage measure used to estimate ψ_{dj}: gross monthly earnings in column (4), net hourly earnings in column (5), and net monthly earnings in column (6). Column (6) additionally weights observations by average monthly formal employment at the establishment before inspection. All specifications include industry × state and inspection-year fixed effects, with standard errors clustered at the establishment level reported in parentheses. Amenity differences and collective-agreement provisions are excluded from all specifications. Because columns (1)–(2) model take-up rates rather than log-odds, their coefficients are not directly comparable with those in columns (3)–(6). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J.5 Hire Windows

The baseline measures new formal hires within six months of inspection, allowing for delayed registration while limiting unrelated subsequent hiring. Table J.4 compares this outcome with a three-month window and a window extending through year-end. The wage coefficient remains positive for both genders under all three definitions, ranging from 2.59 to 3.42 for men and from 2.70 to 3.28 for women.

Table J.4: Alternative Hire Windows

	(1)	(2)	(3)	(4)	(5)	(6)
		Men			Women	
Hire window	3 months	**6 months** (baseline)	Through year-end	3 months	**6 months** (baseline)	Through year-end
Establishment pay premium $\hat{\psi}_{dj}$	2.585***	3.167***	3.424***	2.702***	3.083***	3.276***
	(0.648)	(0.649)	(0.647)	(0.925)	(0.918)	(0.912)
First-stage F	2,529.2	2,529.2	2,529.2	1,609.8	1,609.8	1,609.8
Observations	13,059	13,059	13,059	9,683	9,683	9,683

Notes: The table reports 2SLS estimates of the preferred log-odds specification, varying the window used to measure new formal hires. Columns (1)–(3) report estimates for men and columns (4)–(6) for women; the six-month specification is the baseline. Each gender's establishment pay premium is instrumented with the other gender's premium at the same establishment. All specifications include the amenity differences, collective-agreement clause-group indicators, total clause intensity, the CBA coverage indicator, and industry × state and inspection-year fixed effects. Standard errors clustered at the establishment level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J.6 Clause Counts Instead of Indicators

The baseline measures each collective-agreement group with an indicator for whether the agreement contains at least one provision in that category in any of the three years preceding the inspection. Table J.5 instead uses the average number of provisions over those years, allowing contractual coverage to vary along the intensive margin. The wage response is essentially unchanged: the establishment-premium coefficients are 3.23 for men and 3.04 for women, compared with 3.17 and 3.08 in the baseline; the corresponding wage-premium estimates are 2.76 and 2.73. The clause-group coefficients are not directly comparable in magnitude with those in Table J.2, since they now measure the effect of an additional provision rather than the presence of any provision in the group. The qualitative pattern is broadly similar for women, with 12 of the 15 coefficients retaining the same sign and flexibility and part-time provisions remaining positive and statistically significant. The estimates for men are less stable across measures, but are also generally imprecise under both specifications. Overall, the main wage estimates and the evidence of greater responsiveness among women to flexibility and part-time provisions do not depend on measuring collective-agreement provisions at the extensive or intensive margin. Total clause intensity is omitted because it is the sum of the category-specific counts.

Table J.5: Clause Counts Instead of Indicators

	(1)	(2)	(3)	(4)
	Establishment-premium spec.		Wage-premium spec.	
	Men	Women	Men	Women
Establishment pay premium ψ_{dj}	3.232***	3.036***		
	(0.647)	(0.919)		
Wage premium Δw_{dj}			2.757***	2.734**
			(0.854)	(1.191)
Wage & compensation	0.016	0.046	0.064*	0.037
	(0.035)	(0.040)	(0.036)	(0.051)
Bonuses & profit-sharing	0.111	−0.055	0.037	0.044
	(0.068)	(0.084)	(0.100)	(0.106)
Social benefits	−0.208***	−0.091	−0.219**	−0.200
	(0.078)	(0.100)	(0.102)	(0.126)
Contract & termination	0.033	−0.087	0.051	0.129
	(0.078)	(0.101)	(0.112)	(0.136)
Maternity & childcare	0.201	0.216	0.239	0.242
	(0.208)	(0.238)	(0.270)	(0.301)
Leaves	0.137	0.034	0.440**	0.333
	(0.128)	(0.167)	(0.183)	(0.216)
Harassment & discrimination	0.076	0.007	0.071	−0.152
	(0.095)	(0.251)	(0.105)	(0.261)
Flexibility & part-time	0.092	0.149**	0.151*	0.135
	(0.073)	(0.075)	(0.088)	(0.090)
Work organization	0.063	0.005	0.090*	0.051
	(0.047)	(0.054)	(0.055)	(0.062)
Employment protection	−0.009	−0.048	−0.158	−0.284
	(0.120)	(0.145)	(0.151)	(0.181)
Work shifts	−0.152***	−0.129*	−0.205***	−0.206**
	(0.056)	(0.068)	(0.077)	(0.092)
Health & safety	0.016	−0.007	−0.029	−0.042
	(0.073)	(0.090)	(0.101)	(0.120)
Union provisions	0.060	0.129**	0.151**	0.176**
	(0.056)	(0.064)	(0.072)	(0.079)
CBA negotiation	−0.006	−0.007	−0.025	0.006
	(0.018)	(0.024)	(0.025)	(0.033)
Other clauses	0.090	0.051	0.025	0.106
	(0.083)	(0.101)	(0.110)	(0.126)
Estimator	2SLS	2SLS	2SLS	2SLS
Amenity differences	✓	✓	✓	✓
Industry × state FE	✓	✓		
Industry FE, state FE			✓	✓
Inspection-year FE	✓	✓	✓	✓
First-stage F	2,528	1,602	1,651	1,261
Observations	13,059	9,683	7,955	6,093

Notes: The table re-estimates the specifications in Table 5, replacing each collective-agreement clause-group indicator with the average number of clauses in that category over the three years preceding the inspection. Total clause intensity is omitted because it equals the sum of the category-specific counts. The dependent variable is the log-odds of the six-month formalization rate, $\tilde{\lambda}_{dj}$. Columns (1)–(2) instrument each gender's establishment pay premium with the other gender's premium at the same establishment; columns (3)–(4) analogously instrument the formal–informal wage premium, $\Delta w_{dj} = \psi_{dj} - \pi_{ds(j)}$. All specifications include the four amenity differences and the collective-agreement-coverage indicator. Standard errors clustered at the establishment level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J.7 Robustness of the Valuations: Outcome, Sample, and Unobserved Firm Quality

Table J.6 examines whether the estimated relative monetary values are sensitive to the definition of take-up, inspections with small denominators, or broader differences in establishment desirability. Across these alternatives, the main pattern is stable: women's formalization decisions remain responsive to several non-wage job attributes, and the amenities that matter in the baseline generally retain the same sign and similar magnitudes.

Columns (1)–(8) vary the take-up outcome. The baseline uses the establishment-level single difference in equation (12); the alternatives apply empirical-Bayes shrinkage, replace the outcome with the matched-pair difference-in-differences, or combine the two. The empirical-Bayes single difference produces valuations close to the baseline, with rank correlations of 0.95 for men and 0.96 for women. For women, the relative monetary value of flexibility and part-time provisions remains positive, while that of manual task content remains negative. The double-difference specifications preserve these patterns but are considerably less precise. Their wage coefficients fall from 3.17 and 3.08 in the baseline to 1.46 and 1.79 under the raw double difference and to 0.78 and 1.16 after shrinkage. Because relative monetary values scale amenity coefficients by the estimated wage coefficient, these smaller denominators also magnify the RMVs and their uncertainty. I therefore retain the single-difference outcome as the preferred measure.

Columns (9)–(10) restrict the sample to inspections detecting more than one informal worker, reducing the influence of take-up rates constructed from very small denominators. The main amenity patterns again remain similar. For women, the relative monetary value of flexibility and part-time provisions increases from 0.18 to 0.43, while that of manual task content changes from -0.33 to -0.43; the positive valuations of leaves and CBA negotiation are also preserved. The wage coefficients become 2.69 for men and 1.88 for women, and the hypothesis that they are equal is not rejected; as in the baseline, the gender comparison of coefficient levels is not informative about relative wage preferences (Section 5). Thus, restricting attention to inspections with more informative take-up rates does not alter the main pattern of non-wage valuations.

Finally, columns (11)–(12) add the component of the Sorkin (2018) revealed establishment value that is orthogonal to the establishment pay premium, capturing broader establishment desirability not reflected in pay. The estimated valuations move little: their rank correlation with the baseline is 0.88 for men and 0.96 for women, while the wage coefficients decline modestly to 2.90 and 2.78. For women, flexibility and part-time provisions remain positively valued at 0.20, leaves and CBA negotiation remain positive at 0.27 and 0.48, and manual

Table J.6: Relative Monetary Values Under Alternative Take-Up Outcomes, Samples, and Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Outcome definition, full sample								*Sample*		*Specification*	
	Single difference		EB single diff.		Diff-in-diff.		EB diff-in-diff.		> 1 detected		Sorkin control	
	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women
CBA clause groups												
Wage & compensation	−0.006	−0.239	−0.058	−0.338	−0.214	−0.293	−0.237	−0.544	0.105	0.234	−0.011	−0.037
Bonuses & profit-sharing	−0.030	−0.260	−0.015	−0.286	0.146	−0.572	0.305	−0.365	−0.262	−0.631	0.023	−0.296
Social benefits	0.017	−0.303	0.082	−0.289	0.210	−0.512	0.325	−0.569	0.154	−0.579	−0.026	−0.373
Contract & termination	0.096	0.193	0.140	0.226	0.102	0.059	0.168	−0.082	−0.010	0.178	0.073	0.168
Maternity & childcare	0.038	0.142	0.066	0.142	−0.008	0.167	0.116	0.159	−0.060	0.261	0.075	0.146
Leaves	0.040	0.189**	0.084	0.216**	0.106	0.241	0.213	0.224	0.183	0.247	0.065	0.271***
Harassment & discrimination	0.003	0.184	0.026	0.270	0.107	−0.192	0.160	−0.241	0.015	0.099	0.009	0.298
Flexibility & part-time	−0.081	0.178*	−0.099	0.208*	−0.019	0.444**	−0.067	0.477*	0.066	0.429*	−0.075	0.199*
Work organization	0.062	0.000	0.108	0.032	0.121	0.051	0.227	0.067	0.198*	0.374*	0.088	0.016
Employment protection	−0.014	−0.154	−0.078	−0.205	−0.037	0.094	−0.200	0.127	−0.136	−0.329	−0.054	−0.107
Work shifts	0.128	−0.191	0.108	−0.114	0.086	−0.227	−0.063	−0.161	0.032	−0.002	0.143	−0.139
Health & safety	−0.003	−0.242	0.004	−0.311	0.158	−0.675	0.320	−0.673	0.126	−0.923*	−0.026	−0.376*
Union provisions	−0.326	0.142	−0.487	0.002	−1.564	0.411	−6.480	0.351	−0.666	−0.278	−0.229	0.067
CBA negotiation	0.120	0.440**	0.216	0.472**	−0.010	0.666*	0.432	0.651	0.319	0.612	0.065	0.481**
Other clauses	−0.107	0.174	−0.169	0.181	−0.136	0.127	−0.150	0.064	−0.038	0.299	−0.110	0.165
Total clause intensity	0.004	−0.003	0.004	−0.003	0.005	−0.007	0.005	−0.006	0.004	−0.003	0.003	−0.005
Amenity differences (formal − informal)												
Δ Part-time	−0.010	0.210	0.013	0.194	0.015	0.249	−0.006	0.138	0.096	0.341	0.038	0.243
Δ Female manager	−0.290***	0.089**	−0.339***	0.055	−0.939**	0.161*	−1.673	0.192*	−0.249***	0.071	−0.283***	0.086
Δ Manual task	0.064	−0.327***	0.058	−0.360***	0.210**	−0.435**	0.261	−0.528*	0.076	−0.430*	0.063	−0.414***
Δ Interpersonal task	−0.052*	−0.082**	−0.026	−0.078*	−0.033	−0.093	−0.050	−0.145*	−0.067	−0.192*	−0.051	−0.104***
Firm quality net of pay											0.169***	0.093***
$\hat{\delta}_d$	3.167	3.083	1.759	1.923	1.461	1.793	0.783	1.156	2.686	1.876	2.901	2.775
N (establishments)	13,059	9,683	13,059	9,683	13,059	9,683	13,059	9,683	8,723	6,440	12,358	8,901

Notes: The table reports relative monetary values under alternative definitions of take-up, samples, and controls. Relative monetary values are computed as $\text{RMV}_{d\ell} = 1 - \exp(-\hat{\beta}_{d\ell}/\hat{\delta}_d)$ using the coefficients of equation (14). Columns (1)–(2) report the baseline single-difference specification of equation (12). Columns (3)–(4) apply empirical-Bayes shrinkage following Walters (2024). Columns (5)–(6) replace the outcome with the matched-pair difference-in-differences, and columns (7)–(8) apply empirical-Bayes shrinkage to this outcome. Columns (9)–(10) return to the baseline outcome and restrict the sample to inspections detecting more than one informal worker. Columns (11)–(12) instead add the Sorkin (2018) revealed establishment value residualized on the establishment pay premium and standardized to unit variance. All specifications include the four amenity differences, the fifteen collective-agreement clause groups, total clause intensity, industry × state and inspection-year fixed effects, and are estimated separately by gender using 2SLS, instrumenting each gender's establishment pay premium with the other gender's premium at the same establishment. Statistical significance is based on percentile intervals from 1,000 establishment-level bootstrap replications: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

task content remains negatively valued at −0.41. The stability of these estimates indicates that the non-wage valuation patterns are not driven by sorting toward establishments that are more desirable along dimensions not captured by pay.

Overall, the robustness exercises preserve the central pattern in the administrative estimates: men and women respond positively to the formal wage premium, while the larger differences across genders arise in their valuations of non-wage job attributes.

J.8 Survey–Administrative Amenity Crosswalk

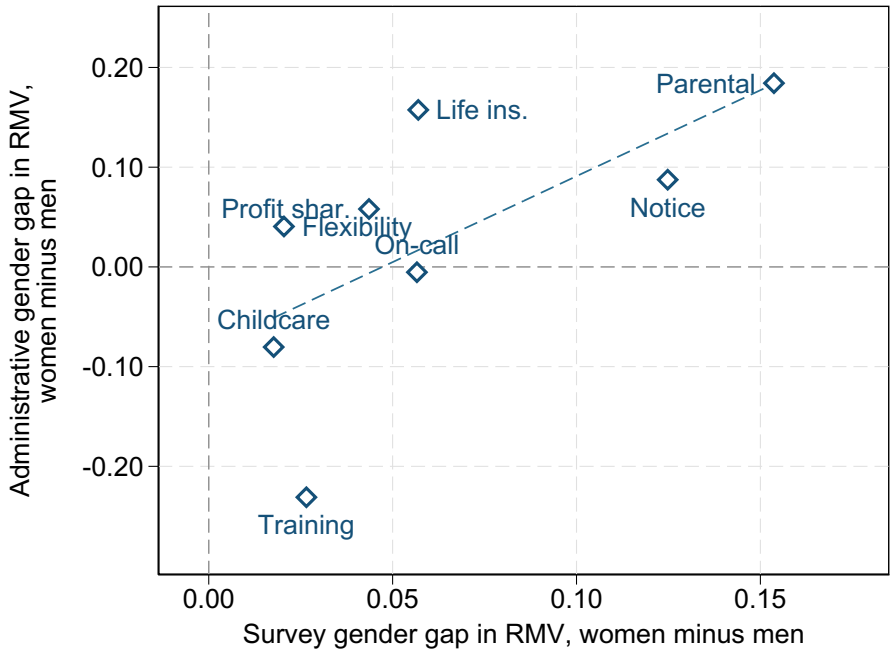
The main text compares women–men differences in survey and administrative valuations across thirteen amenities with an administrative counterpart. Figure J.1 restricts the comparison to the eight amenities with a direct one-to-one mapping across the two data sources. The Pearson correlation increases from 0.42 in the full crosswalk to 0.65 under the direct mapping. Although the comparison contains only eight amenities, the substantially stronger correlation shows that the correspondence between stated and revealed gender differences is sharper when the two measures capture the same underlying job attribute.

J.9 The Formal-Opportunity Slope: Robustness and Binned Scatter Plots

Section 9.1 combines inspection-based take-up with PNAD-C informal-to-formal transitions to recover the implied formal-job opportunity rate, $\hat{q}_d$. Table J.7 shows how this accounting exercise changes with the transition definition, horizon, and level of aggregation.

The baseline uses six-month, same-employer transitions because they most closely match the opportunity created by an inspection. Counting transitions to any employer raises the implied rates for both genders but leaves their relative magnitude similar: the women–men ratio moves from 0.77 to 0.74. Extending both measures through year-end narrows the gender difference, while aggregating to state × year cells yields a ratio of 0.92 that is not statistically distinguishable from one. The baseline gender difference is therefore not specific to the same-employer restriction, but it is concentrated in variation across industries.

Figure J.1: Survey and Administrative Gender Differences: Directly Mapped Amenities



Notes: Each point represents one amenity. The horizontal axis reports the women–men difference in survey relative monetary values (RMVs), and the vertical axis reports the corresponding difference in administrative RMVs. The administrative estimates come from a variant of the preferred establishment-premium specification that replaces the fifteen CBA clause-group indicators with indicators for whether the establishment's collective agreement contains a provision corresponding to each survey attribute; Appendix Table K.9 reports the crosswalk. The specification is estimated separately by gender using 2SLS, instrumenting each gender's establishment pay premium with the other gender's premium at the same establishment. It includes total clause intensity, the CBA coverage indicator, and industry × state and inspection-year fixed effects; the four amenity differences are omitted. Administrative RMVs are computed as in equation (10), $1 - \exp(-\hat{\beta}_{d\ell}/\hat{\delta}_d)$, using the wage coefficient from the same specification ($\hat{\delta}_m = 3.18$ and $\hat{\delta}_w = 3.03$). Survey RMVs are constructed analogously from the choice-experiment logit estimates. The figure restricts the comparison to the eight attributes with a direct one-to-one crosswalk between the survey and collective-agreement measures: schedule flexibility, parental leave, training, job stability and advance notice, childcare assistance, profit sharing, on-call pay, and group life insurance. Five additional attributes with administrative counterparts—health insurance, unemployment insurance, commuting support, paid caregiving days, and part-time hours—are excluded because their mappings to collective-agreement provisions are less direct; the *carteira* and home office attributes are excluded because they have no collective-agreement counterpart. The dashed line reports the OLS fit. The Pearson correlation is 0.65 ($p = 0.08$), and the Spearman correlation is 0.81. Estimates use the ever-informal survey subsample.

Table J.7: Formal-Opportunity Slope under Alternative Definitions

	(1) Men	(2) Women	(3) W/M	(4) p-value	(5) Cells (M/W)
Industry × state × year cells					
(a) 6 months, same employer	0.383	0.294	0.77	0.001	281/199
	(0.014)	(0.023)			
(b) 6 months, all employers	0.527	0.389	0.74	<0.001	281/199
	(0.018)	(0.035)			
(c) Through year-end, same employer	0.412	0.367	0.89	0.129	188/133
	(0.017)	(0.024)			
(d) Through year-end, all employers	0.607	0.525	0.87	0.025	188/133
	(0.023)	(0.028)			
State × year cells					
(e) 6 months, same employer	0.230	0.212	0.92	0.251	144/153
	(0.013)	(0.008)			

Notes: Each row estimates $p_{dc} = q_d \lambda_{dc}$ by weighted least squares without a constant, separately by gender, as in Table 10. p_{dc} is the PNAD-C share of informal workers who formalize at some point within the horizon, cumulated from quarterly hazards, and λ_{dc} is inspection-based take-up over the same horizon. "Same employer" counts only formalization with the worker's existing employer. W/M is the women/men ratio of the slopes. Robust standard errors in parentheses. The p-value tests equal slopes in a pooled regression with a female × λ_{dc} interaction and no constant. Weights are the PNAD-C informal worker-quarter observations entering the hazard in each cell.

J.10 Mechanism Decomposition: Alternative Form and Sample

Tables J.8 and J.9 show that the mechanism decomposition in Section 8.4 is robust to the decomposition form and sample. Using a pooled two-fold decomposition, differences in observed characteristics account for 53% of the women–men take-up gap (Table J.8). Extending the baseline three-fold decomposition to all 1,298 survey respondents yields a similar pattern: observed characteristics account for 48% of the gap, with the remainder attributable to coefficient differences and their interaction (Table J.9). The main conclusions are therefore similar across decomposition methods and sample definitions.

Table J.8: Mechanism Decomposition, Two-Fold Form with the Pooled Reference

Component	(1) Contribution	(2) % of raw gap
Raw gender gap (W−M)	−0.089	
	(0.032)	
Explained (different characteristics)		
Household and care	−0.016	17.9
Children	0.002	−2.6
Knowledge and trust	−0.001	1.6
Formal experience	−0.006	6.4
Demographics	−0.005	6.0
Sorting: state and industry	−0.021	23.9
Total explained	−0.048	53.1
	(0.020)	
Unexplained (different returns)		
Household and care	−0.030	33.7
Children	−0.005	6.0
Knowledge and trust	−0.153	171.4
Formal experience	0.064	−71.9
Demographics	0.057	−63.8
Sorting: state and industry	0.143	−160.4
Constant	−0.118	131.9
Total unexplained	−0.042	46.9
	(0.034)	

Notes: The table reports the two-fold decomposition of the women–men take-up gap for the ever-informal respondents used in the main analysis. Reference coefficients are estimated from a pooled model including a female indicator. The explained component captures differences in observed characteristics; the unexplained component captures differences in coefficients. Covariates are grouped as follows. *Household and care* includes inability to accept a better-paying job with fixed hours from 8 a.m. to 6 p.m. because of childcare, other family responsibilities, or a second job; weekly hours of care work and housework; a household financial buffer of at most one month; being the only household earner; receipt of government benefits; and spouse formal-employment status. Indicators for unknown financial buffer and missing partner or spouse status are included separately. *Children* includes having children, having a child aged 0–5, being the primary caregiver for one's children, and number of children. *Knowledge and trust* includes a twelve-item score measuring knowledge of *carteira* entitlements, knowledge of dismissal payments and parental leave, and trust that employers provide these benefits. *Formal experience* includes ever having held a *carteira*, currently holding one, and having worked at an inspected establishment, with a separate indicator for not remembering inspection status. *Demographics* includes age and tertiary education. *Sorting* includes state and two-digit industry indicators, normalized to sum to zero, with separate categories for unreported state and industry. Robust standard errors are reported in parentheses.

Table J.9: Mechanism Decomposition, Full Survey Sample

	(1) Endowments	(2) Coefficients	(3) Interaction	(4) Total
Household and care	−0.028	−0.009	0.014	−0.023
	(0.013)	(0.048)	(0.017)	(0.054)
Children	−0.003	0.000	0.003	0.000
	(0.006)	(0.036)	(0.009)	(0.037)
Knowledge and trust	−0.001	0.027	−0.002	0.023
	(0.005)	(0.118)	(0.003)	(0.117)
Formal experience	0.004	0.055	−0.007	0.052
	(0.009)	(0.092)	(0.012)	(0.088)
Demographics	0.008	0.136	−0.018	0.126
	(0.007)	(0.120)	(0.010)	(0.117)
Sorting: state and industry	−0.017	0.080	0.002	0.065
	(0.017)	(0.084)	(0.025)	(0.087)
Constant		−0.323		−0.323
		(0.208)		(0.208)
Total (raw gap W−M)	−0.037	−0.033	−0.008	−0.079
	(0.025)	(0.035)	(0.034)	(0.029)
% of raw gap	47.5	42.1	10.4	100.0

Notes: The table reports the three-fold decomposition in Table 9 using all survey respondents rather than only the ever-informal respondents used in the main analysis. The decomposition separates the women–men take-up gap into differences in observed characteristics (endowments), differences in coefficients, and their interaction. Covariates are grouped as follows. *Household and care* includes inability to accept a better-paying job with fixed hours from 8 a.m. to 6 p.m. because of childcare, other family responsibilities, or a second job; weekly hours of care work and housework; a household financial buffer of at most one month; being the only household earner; receipt of government benefits; and spouse formal-employment status. Indicators for unknown financial buffer and missing partner or spouse status are included separately. *Children* includes having children, having a child aged 0–5, being the primary caregiver for one's children, and number of children. *Knowledge and trust* includes a twelve-item score measuring knowledge of *carteira* entitlements, knowledge of dismissal payments and parental leave, and trust that employers provide these benefits. *Formal experience* includes ever having held a *carteira*, currently holding one, and having worked at an inspected establishment, with a separate indicator for not remembering inspection status. *Demographics* includes age and tertiary education. *Sorting* includes state and two-digit industry indicators, normalized to sum to zero, with separate categories for unreported state and industry. Robust standard errors are reported in parentheses.

K. Survey Experiment: Design and Additional Results

This appendix provides details on the survey experiment used to measure workers' valuations of formal employment and its attributes. The survey serves three purposes in the paper. First, randomized wage–attribute trade-offs provide an independent measure of workers' preferences over wages and non-wage job attributes. Second, an inspection-style vignette measures stated take-up of formalization and isolates the role of the associated reduction in take-home pay. Third, combining the two exercises identifies the residual value of the informal outside option required to reconcile workers' stated valuations with their formalization decisions.

K.1 Fielding and Sample

The survey was administered online in Brazil through the Netquest panel.[47] A pilot of 163 respondents and the main wave, which yielded 1,501 completed responses, were fielded between June and August 2026. Eligibility required respondents to be aged 18–65 and either to have worked for pay during the previous twelve months, including in self-employment, or to be currently available for work. Recruitment used gender and age quotas, and randomization was stratified by gender and current employment status. The survey experiment was approved by the UBC Behavioural Research Ethics Board under certificate H26-01025. Respondents reviewed an information letter before beginning the survey and consented by proceeding.

I exclude respondents who completed the survey in under ten minutes, reported not paying careful attention, or stated that they ignored differences between the jobs presented. Accounting for overlap across these criteria leaves 1,298 respondents. The analysis in the paper focuses on the 942 respondents who currently hold or previously held a job without a signed *carteira*. This ever-informal sample most closely corresponds to the population studied in the administrative analysis. The full analysis sample is used only as a robustness check where indicated.

Table K.1 describes the ever-informal sample. Respondents are approximately evenly divided by gender, have a mean age of 40, and 29% have completed tertiary education. Their current employment arrangements are heterogeneous: 35% currently hold a signed

[47] Netquest is a commercial online research panel operated by Bilendi. Netquest recruited eligible panel members, programmed and hosted the questionnaire, implemented the randomization and sampling quotas, and managed fieldwork. Respondents received 46 Korus points for completing the survey. Netquest provided only de-identified responses and did not share respondents' contact information with the research team.

carteira, 8% are self-employed, and 16% are not working.

Table K.1: Survey Sample Composition

	Mean	SD	*N*
Female	0.524	0.500	942
Age (years)	40.103	12.698	942
Completed tertiary education	0.290	0.454	942
Currently holds a signed *carteira*	0.352	0.478	942
Self-employed (registered or employer)	0.081	0.272	942
Not currently working	0.158	0.365	942
Monthly reference salary (R$)	3,538	4,002	623
Salary between 1 and 1.5 minimum wages	0.286	0.452	623
Ever had a signed *carteira*	0.829	0.377	942
Has children	0.538	0.499	942
Has a child aged 0–5	0.111	0.315	942
Has children and is their primary caregiver	0.139	0.346	942
Cannot accept a fixed 8 a.m.–6 p.m. schedule	0.132	0.338	942
Household could last ≤1 month without income	0.516	0.500	699
Expects a new job would pay a higher salary	0.364	0.482	815
A workplace was ever inspected	0.282	0.450	766

Notes: The table describes respondents in the ever-informal analysis sample, fielded through the Netquest panel in Brazil between June and August 2026. A respondent is classified as ever informal if they report currently holding or previously having held a job without a signed *carteira*. The analysis excludes respondents who completed the survey in under ten minutes, reported not paying careful attention, or stated that they ignored differences between the jobs presented. Monthly reference salary is the salary used to anchor the choice tasks: current salary for employed respondents and reservation wage otherwise. Salary statistics exclude respondents who did not report a salary or who reported salaries below R$500 or above R$50,000. The minimum-wage band is defined relative to the 2026 federal monthly minimum wage of R$1,621. The median reported reference salary is R$2,200, or 1.36 minimum wages. Respondents who did not report a salary were shown a default reference salary of R$2,300 in the choice tasks. Cannot accept a fixed 8 a.m.–6 p.m. schedule equals one if the respondent reports being unable to accept a better-paying job with fixed hours from 8 a.m. to 6 p.m., Monday to Friday, because of childcare, other family responsibilities, or a second job.

K.2 Experimental Design

Each respondent completed fifteen binary choice tasks designed to recover the value of wages and non-wage job attributes. Job A paid the respondent's current salary, or reservation wage if not employed, and did not include the attribute being evaluated. Respondents who did not report a salary, 32% of the ever-informal sample, were shown a default reference salary of R$2,300, close to the median of R$2,200 among respondents who reported a plausible salary. Job B offered the attribute at a lower salary, holding other

characteristics fixed. The salary reduction was independently randomized in each task over $\{5, 10, 20, 30, 40\}$%. Task order and the left–right position of Job A were also randomized.

Table K.2 lists the fifteen attributes and the information presented to respondents. The attributes include statutory protections associated with formal employment, such as a signed *carteira*, unemployment insurance, and advance notice, as well as flexibility, family-related benefits, training, and other workplace amenities. The part-time task differs slightly from the others: it reduces weekly hours from forty to thirty-two and additionally reduces earnings by 5, 10, 20, or 30%. Because this treatment changes both hours and the implied hourly wage, its estimated valuation should be interpreted as a lower bound on the value of shorter hours.

K.3 Model

Respondent i of gender d chooses between Job A, which pays the reference wage w_{iA} and does not include attribute k, and Job B, which pays $w_{iB} < w_{iA}$ and includes the attribute. Utility from option $j \in \{A, B\}$ is

$$U_{ijk} = \delta_d \ln w_{ij} + \beta_{kd} \mathbf{1}\{j = B\} + \varepsilon_{ijk}, \tag{K.1}$$

where δ_d measures responsiveness to log wages and β_{kd} captures the value of attribute k, both allowed to vary by gender. Assuming independent and identically distributed type I extreme value utility shocks gives

$$\Pr(\text{choose Job B in task } k \mid d) = \frac{\exp\left[\beta_{kd} + \delta_d \ln(w_{iB}/w_{iA})\right]}{1 + \exp\left[\beta_{kd} + \delta_d \ln(w_{iB}/w_{iA})\right]}. \tag{K.2}$$

Random assignment of the wage reduction identifies δ_d separately from β_{kd}. I express the valuation of each attribute as the proportional wage reduction that makes the respondent indifferent between the two jobs:

$$\mathrm{RMV}_{kd} = 1 - \exp\left(-\frac{\beta_{kd}}{\delta_d}\right). \tag{K.3}$$

This relative monetary value is defined analogously to equation (10) in the administrative analysis and expresses the value of attribute k as a proportion of the reference wage.

Table K.2: Attributes in the Discrete-Choice Experiment

Task	Attribute	Content of Job B
1	*Carteira* signed	Paid vacation, thirteenth salary, FGTS, INSS, and unemployment insurance
2	Schedule flexibility	Choice of start and end times, holding daily hours fixed
3	Unemployment insurance	Three to five monthly payments following dismissal without cause
4	Paid parental leave	120 paid days for mothers and 5 for fathers, with job protection
5	Training and growth	Employer-funded training and a defined promotion path
6	Shorter commute	Fifteen minutes each way
7	Advance notice	At least thirty days' notice before dismissal, plus severance
8	Health insurance	Employer-paid private coverage for the worker and dependents
9	Childcare assistance	R$400 per month for each child under six
10	Profit sharing	Annual bonus of approximately one month's salary if targets are met
11	On-call pay	One-third of the hourly wage for each standby hour
12	Group life insurance	Twenty-four months of salary upon death or permanent disability
13	Paid family-care days	Three paid days per year to care for a sick relative
14	Part-time hours	Thirty-two rather than forty hours per week
15	Home office	Work from home up to three days per week

Notes: The table lists the attribute presented in Job B in each of the fifteen binary choice tasks. Job A pays the respondent's current salary, reservation wage if not employed, or the default reference salary when no salary is reported. Job B offers the listed attribute at a salary reduced by a randomly assigned 5, 10, 20, 30, or 40%, with all other job characteristics held fixed. The part-time task instead reduces weekly hours from forty to thirty-two and applies an additional wage reduction of 5, 10, 20, or 30%.

K.4 Estimated Valuations

I estimate equation (K.2) jointly across the fifteen tasks, allowing both the attribute intercepts and the wage coefficient to vary by gender. Table K.3 reports the resulting relative monetary values.

The first result is that both genders respond strongly to wages. The estimated wage coefficient is 3.27 for men and 2.99 for women, and the difference is not statistically significant. Because the logit identifies coefficients only relative to the dispersion of unobserved utility, this comparison relies on the pooled specification's restriction that

Table K.3: Willingness to Pay for Job Attributes

		All	Men		Women		
Rank	Attribute	RMV	RMV	Rank	RMV	Rank	Δ (W−M)
1	Unemployment insurance	0.45	0.40	2	0.50	1	0.10***
			(0.02)		(0.02)		(0.03)
2	Training and growth	0.43	0.42	1	0.44	3	0.03
			(0.02)		(0.02)		(0.03)
3	Advance notice	0.40	0.33	6	0.45	2	0.12***
			(0.02)		(0.02)		(0.03)
4	On-call pay	0.38	0.35	4	0.40	5	0.06*
			(0.02)		(0.02)		(0.03)
5	*Carteira* signed	0.38	0.35	5	0.40	6	0.06*
			(0.02)		(0.02)		(0.03)
6	Health insurance	0.37	0.36	3	0.38	7	0.02
			(0.02)		(0.02)		(0.03)
7	Profit sharing	0.35	0.33	7	0.37	8	0.04
			(0.02)		(0.02)		(0.03)
8	Paid parental leave	0.35	0.27	9	0.42	4	0.15***
			(0.02)		(0.02)		(0.03)
9	Group life insurance	0.31	0.28	8	0.34	9	0.06*
			(0.02)		(0.02)		(0.03)
10	Paid family-care days	0.28	0.25	10	0.32	11	0.07**
			(0.02)		(0.02)		(0.03)
11	Home office	0.26	0.19	12	0.33	10	0.14***
			(0.02)		(0.02)		(0.03)
12	Childcare assistance	0.23	0.23	11	0.24	13	0.02
			(0.02)		(0.02)		(0.03)
13	Shorter commute	0.21	0.17	13	0.24	12	0.07**
			(0.03)		(0.02)		(0.03)
14	Schedule flexibility	0.14	0.13	14	0.15	14	0.02
			(0.03)		(0.03)		(0.04)
15	Part-time hours	0.04	0.01	15	0.07	15	0.06
			(0.04)		(0.04)		(0.05)

Notes: The table reports relative monetary values for ever-informal respondents. Estimates come from the saturated logit in equation (K.2), which includes task-by-gender intercepts and gender-specific coefficients on the log wage ratio between Jobs B and A. Relative monetary values are computed as $\text{RMV}_{kd} = 1-\exp(-\hat{\beta}_{kd}/\hat{\delta}_d)$ and express the value of attribute k as a proportion of the reference wage. The estimated wage coefficients are $\hat{\delta}_m = 3.27$ and $\hat{\delta}_w = 2.99$. Standard errors in parentheses are obtained from 1,000 bootstrap replications resampling respondents with replacement. Statistical significance is based on the corresponding bootstrap percentile intervals: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

this dispersion is the same for men and women. Under that restriction, the experiment indicates no systematic gender difference in wage responsiveness. The female-to-male ratio of the coefficients, 0.91, is close to the ratio of 0.97 in the administrative data, where the

model is estimated separately by gender and the restriction is not imposed; the similarity of the two ratios is consistent with, but does not by itself establish, equal wage responsiveness. For this reason, I compare the two designs using valuations expressed in wage units.

Workers also place substantial value on non-wage attributes. Unemployment insurance, training and growth, and advance notice receive the highest valuations overall. Women's point estimates exceed men's for all fifteen attributes, although many of the individual gender differences are imprecisely estimated. The largest differences are for paid parental leave, home office, advance notice, and unemployment insurance. These results reinforce the administrative evidence that gender differences are more pronounced along some non-wage dimensions than in responsiveness to the formal wage premium.

K.5 Formalization Vignette

The choice experiment measures workers' valuations of formal employment and its attributes. I separately elicit whether respondents would accept formalization in a setting designed to mirror the labor-inspection variation used in the administrative analysis. Respondents are asked to imagine that an unannounced inspection discovers that they are working without a signed *carteira* and that their employer must offer them a formal contract.

Table K.4 reports responses under two pricing scenarios. In the first, gross salary, hours, and duties remain unchanged, so the employee social-security contribution reduces take-home pay. In the second, gross salary increases enough to preserve take-home pay. Acceptance includes respondents answering *definitely yes* or *probably yes*; the stricter *definitely yes* response is reported separately.

A majority of respondents accept formalization in each scenario, but the immediate monetary cost matters substantially. When take-home pay falls, 60.9% accept the formal contract. When take-home pay is protected, acceptance rises to 78.9%. The increase is particularly large among women: acceptance rises from 56.7 to 77.3%, compared with a rise from 65.6 to 80.6% among men. As a result, the 8.9-percentage-point gender gap observed when take-home pay falls narrows to 3.3 points and is no longer statistically significant when take-home pay is preserved.

Responses across the two scenarios sharpen this interpretation. Among women, 30.0% accept formalization only when take-home pay is protected, compared with 20.5% of men. By contrast, the share declining formalization under both scenarios is nearly identical across genders, at 13.4% for women and 13.8% for men. The lower stated take-up among women therefore does not reflect a uniformly greater propensity to reject formal employment.

Table K.4: Stated Take-Up Under Alternative Pricing Scenarios

	Men	Women	Gap (W − M)
Panel A. Take-home pay falls			
Accept	0.656	0.567	−0.089***
Definitely accept	0.366	0.308	−0.058*
Panel B. Take-home pay protected			
Accept	0.806	0.773	−0.033
Definitely accept	0.549	0.458	−0.092***
Panel C. Responses across scenarios			
Accepts under both	0.600	0.474	−0.127***
Accepts only when pay is protected	0.205	0.300	0.094***
Declines under both	0.138	0.134	−0.005
Accepts only when pay falls	0.056	0.093	0.037**

Notes: The table reports stated acceptance of a formal contract following an unannounced labor inspection for ever-informal respondents. In Panel A, gross salary, hours, and duties remain unchanged, so take-home pay falls by the employee's social-security contribution. In Panel B, gross salary rises enough to hold take-home pay fixed. Acceptance denotes answering *definitely yes* or *probably yes*; the "definitely accept" rows use only the former response. Panel C classifies respondents by their answers across the two scenarios. The gap is the female share minus the male share, and equality across genders is tested using a saturated linear probability model. Overall acceptance is 0.609 when take-home pay falls and 0.789 when it is protected. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Instead, women are more likely to change their decision when formalization reduces current disposable income.

Table K.5 turns to respondents with direct experience of an inspection. Of the 1,501 main-wave respondents, 342 report that a labor inspector visited one of their workplaces, and 57 of these report having been informal at the time and having received an offer to sign a *carteira*. Among these 57, 78.9% accepted the offer: 81.5% of men and 76.7% of women. The gender difference is not statistically significant, and with 57 respondents it is imprecisely estimated. Among the 342 respondents whose workplace was ever inspected, stated take-up follows the same pattern as in the analysis sample. Women are 9.6 percentage points less likely than men to accept when take-home pay falls, and the gap shrinks to 1.9 points when take-home pay is protected.

Table K.6 examines whether stated acceptance reflects attachment to the current employer rather than the value of formalization. Because the vignette holds the worker–employer match fixed, workers may accept a *carteira* to retain their jobs. I classify 471 employees by their job-search intentions over the previous three months. Contrary to this explanation, workers not interested in changing jobs have the lowest acceptance when take-home pay falls (55.0%), compared with 77.6% among active job seekers ($p = 0.001$).

Table K.5: Take-Up Among Respondents Who Experienced an Inspection

	Men	Women	Gap (W − M)
Panel A. Revealed take-up at a real inspection			
Accepted the *carteira* offered	0.815	0.767	−0.048
Respondents	27	30	
Panel B. Stated take-up, respondents ever inspected			
Accept, take-home pay falls	0.709	0.613	−0.096*
Accept, take-home pay protected	0.863	0.844	−0.019
Respondents	182	160	

Notes: The table uses all 1,501 main-wave respondents, without the attention and completion-time exclusions applied to the analysis sample. Panel A is restricted to respondents who report that a labor inspector visited a workplace where they were working without a signed *carteira* and that the employer offered to sign one. It reports the share who accepted. Panel B is restricted to respondents who report that a labor inspector ever visited one of their workplaces. It reports stated acceptance in the formalization vignette under the two pricing scenarios of Table K.4. Acceptance denotes answering *definitely* or *probably yes*. The gap is the female share minus the male share, and equality across genders is tested using a linear probability model with robust standard errors. Overall revealed take-up in Panel A is 0.789. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

When pay is protected, acceptance ranges from 79.1 to 85.6% across groups, with no significant differences ($p = 0.494$). The only significant gender gap under protected pay is among workers not interested in changing jobs: 70.8% of women accept, compared with 87.5% of men.

Among respondents who decline formalization when take-home pay falls, the most common reason for refusal is the reduction in net pay (Figure K.1), cited by 36% of men and 30% of women. Other reasons are more dispersed and point to differences in the perceived value of formalization and the informal outside option. Women are more likely to cite uncertainty that formal benefits will be delivered or the potential loss of government benefits, whereas men are more likely to cite schedule flexibility or a preference for working on their own terms. These responses reinforce the vignette results: the immediate monetary cost is the primary stated reason for refusing formalization, while non-wage considerations account for additional heterogeneity in take-up.

K.6 Reconciling Valuations and Formalization Take-Up

I combine the choice experiment and formalization vignette to recover the residual value of remaining informal. Let β_{1d} denote the estimated value of a signed *carteira* and δ_d the wage coefficient. In the vignette, formalization leaves gross pay unchanged but reduces take-home pay by the employee contribution p. Predicted acceptance absent an additional

Table K.6: Stated Take-Up by On-the-Job Search

	All	Men	Women	Gap (W − M)
Panel A. Take-home pay falls				
Actively looking	0.776	0.857	0.676	−0.181*
Keeping an eye on opportunities	0.585	0.702	0.492	−0.211**
Would switch for the right job	0.613	0.680	0.553	−0.127*
Not interested in changing jobs	0.550	0.641	0.462	−0.179**
All employees	0.616	0.706	0.531	−0.175***
Panel B. Take-home pay protected				
Actively looking	0.842	0.833	0.853	0.020
Keeping an eye on opportunities	0.811	0.809	0.814	0.005
Would switch for the right job	0.856	0.867	0.847	−0.020
Not interested in changing jobs	0.791	0.875	0.708	−0.167**
All employees	0.826	0.851	0.802	−0.048

Notes: The table reports stated acceptance of formalization among 471 currently employed respondents with prior informal work experience, of whom 332 have a signed *carteira* and 139 do not. Respondents are grouped by job-search intentions over the previous three months: actively looking (76), keeping an eye on opportunities (106), willing to switch for the right job (160), and not interested in changing jobs (129). Acceptance is defined as answering *definitely* or *probably yes*. Gender gaps are women's minus men's acceptance rates, with significance based on linear probability models with robust standard errors. Differences across search groups are jointly significant when take-home pay falls ($p = 0.004$), but not when it is protected ($p = 0.494$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

value of remaining informal is

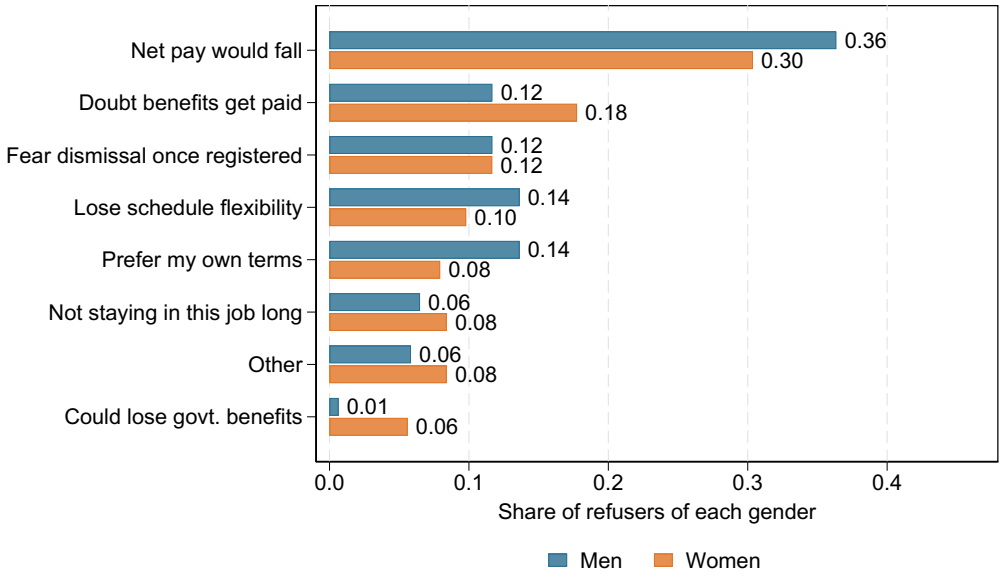
$$\Pr(\text{accept} \mid \text{net pay falls by } p) = \Lambda^{-1}\left[\beta_{1d} + \delta_d \ln(1-p)\right]. \tag{K.4}$$

Setting $p = 0.09$, the midpoint of the 7–11% contribution stated in the vignette, yields predicted acceptance of 74.6% for men and 78.0% for women. Observed acceptance instead equals 65.6 and 56.7%, respectively.

Equation (15) reconciles the two by allowing the informal option to carry residual value γ_d. The implied values are $\gamma_m = 0.43$ and $\gamma_w = 1.00$, and equality between predicted and observed acceptance is rejected ($p = 0.004$). Thus, women's lower take-up cannot be explained by lower measured valuations of formal employment; reconciling the two exercises requires a higher residual value of their informal alternative. I interpret γ_d as an outside-option residual capturing unmeasured job attributes and household circumstances, rather than as a primitive preference parameter.

As complementary evidence, I construct a price-adjusted index comparing preferences for gig and on-demand attributes—schedule flexibility, on-call pay, and home office—with preferences for formal protections—unemployment insurance, advance notice, and health

Figure K.1: Main Reason for Refusing Formalization When Take-Home Pay Falls



Notes: The figure reports the main reason for refusing formalization among ever-informal respondents who decline the formal contract when take-home pay falls. Respondents choose one main reason for refusal. Shares are computed separately by gender and sum to one within gender.

and life insurance. Table K.7 shows that greater relative preference for gig-style attributes is associated with lower formalization take-up. A one-standard-deviation increase in the relative index is associated with a 2.3-percentage-point lower probability of acceptance, or 1.9 percentage points with controls, relative to a baseline acceptance rate of 60.9%. Neither estimate is statistically significant, so the evidence on what γ_d captures is suggestive.

K.7 Parenthood and Stated Valuations

One interpretation of the gender differences in stated valuations is that they reflect current childcare responsibilities rather than broader differences in preferences over job attributes. Table K.8 examines this possibility by separating respondents with a child under six from those without one.

Current parenthood affects the valuation of attributes directly related to young children. Among women, having a child under six raises the estimated relative monetary value of childcare assistance by 0.19 and that of paid parental leave by 0.18. Most other valuations change little. Importantly, however, the main gender differences are not confined to parents of young children. Among respondents without a child under six, women continue to value parental leave, advance notice, home office, and unemployment insurance more than

Table K.7: Taste for Gig and On-Demand Arrangements and Acceptance of a Formal Contract

	(1)	(2)	(3)	(4)
Taste for gig and on-demand work	−0.010	−0.046		
	(0.050)	(0.054)		
Taste for formal protections		0.090		
		(0.056)		
Relative gig and on-demand taste			−0.067	−0.054
			(0.046)	(0.046)
Controls				✓
Respondents	942	942	942	942

Notes: The table reports average marginal effects from logit regressions of accepting a formal contract when formalization lowers take-home pay among ever-informal respondents. Taste indices are respondent-level mean choices residualized on task × assigned wage penalty. The gig and on-demand index includes schedule flexibility, on-call pay, and home office; formal protections include unemployment insurance, advance notice, and health and life insurance. Relative taste is the former minus the latter; the *carteira* task is excluded. Coefficients are effects of a one-unit change in each index; the relative index has a standard deviation of 0.345, so the estimates in columns (3) and (4) correspond to 2.3 and 1.9 percentage points per standard deviation, respectively. Controls are age, education, socioeconomic class, region, and current formality. Standard errors are in parentheses.

men. The corresponding women–men differences are 0.14, 0.12, 0.12, and 0.10, respectively.

These patterns indicate that current childcare responsibilities amplify valuations of amenities directly related to children but do not account for the broader gender differences in stated job preferences.

K.8 Internal Validation

The choice data display three patterns consistent with respondents trading off wages against job attributes. First, the probability of choosing Job B declines monotonically with the randomly assigned wage penalty, and the estimated wage coefficient is positive and statistically significant across subgroup specifications. Second, choices exhibit substantial interior variation: only 3.6% of respondents choose the attribute job in every task and 2.4% never choose it. Excluding these respondents raises the estimated wage coefficients to 3.54 for men and 3.20 for women but leaves their ratio close to its baseline value of 0.91. Third, stated attitudes covary with the corresponding experimental choices: respondents who report placing greater value on a signed *carteira* select the formal-contract job more often, whereas respondents who prioritize autonomy select flexibility and home-office jobs more

Table K.8: Willingness to Pay by Gender and Parenthood

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Levels				Differences			
	Mothers	Fathers	Women, no child	Men, no child	Mothers − fathers	W − M, no child	Mothers − women, no child	Fathers − men, no child
Unemployment insurance	0.49***	0.40***	0.50***	0.40***	0.09	0.10***	−0.01	−0.00
	(0.08)	(0.08)	(0.02)	(0.02)	(0.11)	(0.03)	(0.09)	(0.08)
Training and growth	0.44***	0.47***	0.44***	0.41***	−0.03	0.03	−0.00	0.06
	(0.08)	(0.07)	(0.02)	(0.02)	(0.10)	(0.03)	(0.08)	(0.07)
Advance notice	0.41***	0.24***	0.46***	0.34***	0.17*	0.12***	−0.05	−0.10
	(0.07)	(0.08)	(0.02)	(0.02)	(0.11)	(0.03)	(0.08)	(0.08)
On-call pay	0.44***	0.31***	0.40***	0.35***	0.12	0.05	0.04	−0.04
	(0.07)	(0.07)	(0.02)	(0.02)	(0.10)	(0.03)	(0.08)	(0.08)
Carteira signed	0.47***	0.24***	0.39***	0.36***	0.23**	0.04	0.08	−0.11
	(0.07)	(0.08)	(0.02)	(0.02)	(0.11)	(0.03)	(0.07)	(0.08)
Health insurance	0.31***	0.39***	0.39***	0.36***	−0.07	0.03	−0.08	0.03
	(0.08)	(0.08)	(0.02)	(0.02)	(0.11)	(0.03)	(0.08)	(0.08)
Profit sharing	0.37***	0.37***	0.37***	0.32***	0.00	0.05	0.00	0.05
	(0.07)	(0.07)	(0.02)	(0.02)	(0.10)	(0.03)	(0.08)	(0.08)
Paid parental leave	0.58***	0.35***	0.40***	0.26***	0.23*	0.14***	0.18**	0.09
	(0.08)	(0.08)	(0.02)	(0.02)	(0.12)	(0.03)	(0.08)	(0.09)
Group life insurance	0.37***	0.23**	0.33***	0.29***	0.14	0.05	0.04	−0.06
	(0.08)	(0.08)	(0.02)	(0.02)	(0.11)	(0.03)	(0.08)	(0.08)
Paid family-care days	0.33***	0.26***	0.31***	0.24***	0.08	0.07**	0.02	0.01
	(0.08)	(0.08)	(0.02)	(0.02)	(0.11)	(0.03)	(0.08)	(0.08)
Home office	0.35***	0.01	0.33***	0.21***	0.33***	0.12***	0.02	−0.20**
	(0.07)	(0.13)	(0.02)	(0.02)	(0.15)	(0.03)	(0.08)	(0.13)
Childcare assistance	0.41***	0.30***	0.22***	0.22***	0.11	0.00	0.19***	0.08
	(0.07)	(0.08)	(0.03)	(0.03)	(0.10)	(0.04)	(0.07)	(0.08)
Shorter commute	0.24**	0.20**	0.25***	0.17***	0.04	0.08**	−0.01	0.03
	(0.08)	(0.08)	(0.02)	(0.03)	(0.12)	(0.04)	(0.09)	(0.08)
Schedule flexibility	0.20**	0.03	0.14***	0.14***	0.17	0.00	0.05	−0.11
	(0.08)	(0.11)	(0.03)	(0.03)	(0.13)	(0.04)	(0.09)	(0.11)
Part-time hours	0.17	0.09	0.06	0.00	0.08	0.06	0.11	0.09
	(0.10)	(0.10)	(0.04)	(0.05)	(0.14)	(0.06)	(0.11)	(0.11)
$\hat{\delta}$	2.72	3.28	3.05	3.27				

Notes: The table reports estimates for ever-informal respondents. A parent is a respondent with a child under six; "no child" refers to respondents without a child under six. Relative monetary values are computed as $\text{RMV}_{kd} = 1 - \exp(-\hat{\beta}_{kd}/\hat{\delta}_d)$ from a saturated logit with group-specific wage slopes and task intercepts, which is numerically equivalent to estimating separate logits for the four groups. Columns (5)–(8) report differences between the indicated estimates in columns (1)–(4), constructed replication by replication from the same bootstrap draws. Standard errors in parentheses are obtained from 1,000 respondent-level bootstrap replications. Statistical significance is based on the corresponding bootstrap percentile intervals: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

often and the formal-contract job less often. Together, these patterns support interpreting the experimental choices as informative about respondents' job preferences.

K.9 External Comparison with the Administrative Estimates

The survey provides an independent comparison with the preference estimates obtained from the administrative data. The two designs use different populations and sources of

variation: survey wages and attributes are randomly assigned, whereas the administrative estimates use observed formalization following inspections and establishment-level job characteristics. Agreement between them is therefore informative about whether the main qualitative patterns depend on the administrative design.

The amenity coefficients cannot be compared directly in levels. The survey estimates the value of receiving a job attribute, whereas the administrative specification relates formalization to collective-agreement provisions corresponding to that attribute. I therefore focus on the women–men difference in relative monetary values within each design, which is less sensitive to differences in scale across the two measurement approaches.

Figure 9 compares the women–men differences in stated and administrative valuations across the thirteen attributes with an administrative counterpart; the Pearson correlation is 0.42. Restricting the comparison to the eight attributes with a direct mapping raises the correlation to 0.65 (Appendix Figure J.1). Given the small number of attributes, these estimates are imprecise and provide suggestive evidence of correspondence across the two designs.

K.10 Mapping Survey Attributes to Contract Clauses

Table K.9 reports the crosswalk used to compare the survey and administrative amenity estimates. Thirteen of the fifteen experimental attributes have a counterpart in the collective-agreement taxonomy. The signed *carteira* is the formal contract itself and therefore has no clause-level analogue, while home office does not appear in the administrative clause taxonomy.

Table K.9: Crosswalk from Experimental Attributes to CBA Clauses

Attribute	Task	Corresponding contract clauses
Parental leave	4	Maternity leave; adoption leave
Childcare	9	Daycare allowance
Family-care days	13	Leave for family reasons
Part-time	14	Fixed-term contracts; special or reduced schedules
Schedule flexibility	2	Working-time duration; reduced hours; schedule autonomy
Advance notice	7	Advance notice
Unemployment insurance	3	Job-stability and related provisions
Life insurance	12	Group life insurance
Health insurance	8	Health allowances, programs, and coverage
On-call pay	11	Standby supplement and scheduling
Profit sharing	10	Profit and results sharing
Training	5	Training, qualification, and education allowances
Commute	6	Transport allowance
Carteira	1	*No contractual analogue*
Home office	15	*No analogue in the clause taxonomy*

Notes: The table maps attributes in the discrete-choice experiment to the corresponding provisions in the collective-agreement taxonomy used in the administrative analysis. The crosswalk covers 27 of the 137 clauses. The signed *carteira* has no clause-level analogue because it is the formal contract itself, and home office has no analogue in the clause taxonomy.

K.11 Questionnaire

The following pages reproduce the information letter and the English master questionnaire. The fielded survey used its Portuguese translation.

Information Letter for Participants

UBC Research Study

Study Title: *Sorting into (In)Formality: The Role of Gender and Amenities*

This survey is being conducted by researchers at the University of British Columbia (UBC) for research purposes.

Principal Investigator:
Professor Claudio Ferraz
Department of Economics
University of British Columbia

Student Researcher:
Carla Srebot, PhD Candidate
Department of Economics
University of British Columbia

This research contributes to a doctoral dissertation (a public document). The information collected will be used for academic research purposes; findings may be published in aggregate form and de-identified data may be shared in accordance with journal and funding-agency requirements. Only authorized members of the research team will have access to the de-identified data.

Participation is entirely voluntary and takes approximately **25–35 minutes**. You may skip any question or stop before submitting, but because the survey is anonymous, responses cannot be withdrawn once submitted. Upon completion, you will receive **46 Korus** through the Netquest panel.

The survey is administered through the Netquest online panel, and data are stored on secure Amazon Web Services (AWS) servers in North Virginia, United States. De-identified data are retained in accordance with UBC policies, the Principal Investigator remains responsible for the study data, and access is limited to the study team.

Risks are minimal and no greater than those of daily life; your participation will help improve understanding of labor market choices and may inform future policy and research. Every reasonable effort will be made to maintain confidentiality, and because the survey is anonymous there are no anticipated limits to confidentiality beyond those required by law.

If you have any questions about the study, please contact Carla Srebot at `csrebot@student.ubc.ca`.

If you have any concerns or complaints about your rights as a research participant and/or your experiences while participating in this study, contact the Research Participant Complaint Line in the UBC Office of Research Ethics at 604-822-8598, or if long distance e-mail `RSIL@ors.ubc.ca` or call toll free 1-877-822-8598.

By completing this questionnaire, you are consenting to participate in this study.

UBC Behavioural Research Ethics Board ID Number: H26-01025

1 Module S: Screening

S1. How old are you? _____ **years**

S2. Have you worked for pay (including self-employment) at any point in the past 12 months?

(a) Yes
(b) No

S3. Are you currently looking for work, or would you be available to work if offered a job?

(a) Yes
(b) No

2 Module A: Demographics

A1. What is your gender?

(a) Male
(b) Female
(c) Non-binary
(d) You don't have an option that applies to me. I identify as: _____
(e) Prefer not to say

A2. The following categories follow the official IBGE classification and help us understand differences across groups. How would you describe your ethnicity?

(a) Branca (White)
(b) Preta (Black)
(c) Parda (Mixed)
(d) Amarela (Asian)
(e) Indígena (Indigenous)
(f) Prefer not to say

A3. What is your marital status?

(a) Single (never married)
(b) Married or in a civil union
(c) Living with a partner (not married)
(d) Divorced or separated
(e) Widowed

A4. What is the highest level of education you have **completed**?

(a) No schooling
(b) Ensino Fundamental incompleto (incomplete primary)
(c) Ensino Fundamental completo (complete primary)

(d) Ensino Médio incompleto (incomplete secondary)

(e) Ensino Médio completo (complete secondary)

(f) Ensino Superior incompleto (incomplete tertiary / ongoing)

(g) Ensino Superior completo (complete tertiary — bachelor's)

(h) Pós-graduação (postgraduate)

A5. In which state (UF) and municipality do you currently **live**?

UF: _____ Município: _____

A6. Where do you currently **work** (or most recently worked)?

(a) Same municipality as where I live

(b) Different municipality — UF: _____ Município: _____

(c) I work from home / fully remote

(d) Not applicable (not currently working)

A7. How long does it usually take you to get from home to work **one way**?

_____ **minutes** ☐ Work from home / not applicable

A8. Do you have any children?

(a) No

(b) Yes — How many? _____

A9. **[If A8 = Yes]** What is the age of your youngest child?

(a) 0–1 year

(b) 2–3 years

(c) 4–5 years

(d) 6–12 years

(e) 13–17 years

(f) 18 years or older

A10. **[If A8 = Yes]** Who is the primary caregiver for your children during your working hours?

(a) Myself

(b) My spouse / partner

(c) Grandparent or other family member

(d) Paid nanny / babysitter

(e) Public daycare / creche

(f) Private daycare / school

(g) Children are old enough to stay alone

(h) Other: _____

3 Module B: Work History and Current Employment

B1. What is your **current employment status**? (Select all that apply if you have multiple jobs)

(a) Working as a salaried employee **with** *carteira de trabalho*

(b) Working as a salaried employee **without** *carteira de trabalho*

(c) Working as an MEI / autônomo (self-employed, registered)

(d) Working as self-employed (not registered)

(e) Working as an employer (business owner with employees)

(f) Working as unpaid family helper

(g) Unemployed, actively looking for work

(h) Unemployed, not looking for work but available

B2. **[If B1 = (a) or (b)]** What is your job title or main activity? Please be as specific as possible (e.g., "cashier at supermarket", "construction helper", "domestic worker", "store manager").

_____

B3. **[If B1 = (a) or (b)]** What sector or type of business do you work in?
(e.g., "restaurant", "clothing factory", "hair salon", "construction firm", "hospital").

_____

B4. How many employees work at the establishment where you work (or most recently worked)?

(a) Just me (self-employed, no employees)

(b) 2–5 employees

(c) 6–10 employees

(d) 11–50 employees

(e) 51–100 employees

(f) 101–500 employees

(g) More than 500 employees

(h) Don't know

B5. **[If B1 = (a)–(e)]** What is your **total monthly earnings** from your main job, before any deductions (*rendimento mensal bruto*)? If paid by the day or week, please convert to a monthly amount.

R$ _____ per month ☐ Prefer not to say

B6. Besides your main job, do you receive any money from additional work, odd jobs ("bicos"), or side activities?

(a) No

(b) Yes — approximately R$ _____ per month

B7. How many hours per week do you usually work in your main job? _____ hours

B8. What is your typical work schedule?

(a) Fixed schedule (same hours every day, set by employer)

(b) Rotating shifts (varying hours, set by employer)

(c) Flexible schedule (I choose when to start and end)

(d) On-demand / as needed (work when there is demand)

(e) Other: _____

B9. How long have you been in your current job (or most recent job, if not currently employed)?

(a) Less than 3 months

(b) 3–6 months

(c) 7–12 months

(d) 1–2 years

(e) 3–5 years

(f) More than 5 years

B10. In the past 5 years, how many different jobs have you held (including the current one)?

(a) 1 (only current job)

(b) 2

(c) 3

(d) 4

(e) 5 or more

B11. Have you ever in your life had a job with *carteira de trabalho*?

(a) Yes, I currently have one

(b) Yes, but not currently

(c) No, never

B12. In total, counting all the jobs you have ever held, how many did **not** have a signed *carteira de trabalho*?

(a) None

(b) 1

(c) 2

(d) 3–4

(e) 5 or more

(f) Don't know

B13. **[If B12 = (b)–(e)]** When did your **first** job without a signed *carteira de trabalho* begin?

(a) Less than 1 year ago

(b) 1–2 years ago

(c) 3–5 years ago

(d) 6–10 years ago

(e) More than 10 years ago

B14. [If B11 = (b)] What was the main reason you left (or lost) your most recent job with *carteira de trabalho*?

(a) I was dismissed / laid off
(b) I resigned because I found a better-paying job
(c) I resigned because I wanted more flexibility in my schedule
(d) I resigned for family or personal reasons (e.g., childcare, move)
(e) My contract ended
(f) The firm closed / went out of business
(g) I resigned to start my own business
(h) I resigned because I did not like the work environment
(i) Other: _____

B15. Does your current job (or most recent job) provide you with any of the following? (Select all that apply)

☐ Signed *carteira de trabalho*
☐ FGTS contributions (severance fund)
☐ INSS contributions (social security / retirement)
☐ Paid vacation (30 days per year)
☐ 13th salary (*décimo terceiro*)
☐ Vale transporte (transportation voucher)
☐ Vale refeição / vale alimentação (meal / food voucher)
☐ Health insurance (*plano de saúde*)
☐ Dental insurance
☐ Maternity or paternity leave
☐ Overtime pay
☐ Learning / training opportunities
☐ Childcare assistance
☐ None of the above
☐ Don't know

B16. [If B1 = (f)–(h): not currently working for pay] If you were offered a job today, what monthly salary (gross, before deductions) would you realistically expect to earn?

R$ _____ per month ☐ Don't know

4 Module C: Knowledge and Attitudes about the Carteira de Trabalho

C1. In your own words, what rights or benefits does a worker receive when their employer signs their *carteira de trabalho*?

_____

_____

C2. For each of the following, please indicate whether you believe it is guaranteed to workers with a signed *carteira de trabalho*:

	Yes	**No**	**Don't know**
Paid vacation once a year	☐	☐	☐
A year-end bonus (equivalent to one extra month of pay)	☐	☐	☐
A fund that grows while you work and is released when you leave (FGTS)	☐	☐	☐
Contributions to your retirement pension (INSS)	☐	☐	☐
Monthly payments if you are dismissed without cause	☐	☐	☐
Paid time off after having or adopting a child	☐	☐	☐
Extra pay for working more hours	☐	☐	☐
Advance notice of at least 30 days before being dismissed	☐	☐	☐
The right to take a work dispute to the labor court	☐	☐	☐
A private health insurance plan paid by the employer	☐	☐	☐
A monthly voucher for meals or groceries	☐	☐	☐
An annual profit-sharing bonus when the company does well	☐	☐	☐

C3. In general, do you think that having a signed *carteira de trabalho* is:

(a) Very important for a worker

(b) Somewhat important

(c) Neither important nor unimportant

(d) Not very important

(e) Not important at all

C4. Please read each statement below and indicate how much you agree or disagree.

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
For me, having a signed *carteira de trabalho* is the most important thing in a job.	☐	☐	☐	☐	☐
I would accept a lower take-home pay if the job comes with a signed *carteira de trabalho*.	☐	☐	☐	☐	☐
Having autonomy and flexibility in my work hours matters more to me than having the protections of a *carteira de trabalho*.	☐	☐	☐	☐	☐
I trust that employers who offer a signed *carteira de trabalho* will actually deliver all the associated benefits on time.	☐	☐	☐	☐	☐

5 Module D: Discrete Choice Experiment

5.1 Instructions

How this section works

You will now see **15 job comparisons**. In each one, you will see two hypothetical job offers—**Job A** and **Job B**. The jobs are similar in most ways (same employer, same tasks, same location), but differ in **a few specific characteristics**.

For each comparison:

- Read both job descriptions carefully.
- Select the job you would personally prefer.
- There are no right or wrong answers—we want your honest opinion.

5.2 Choice Tasks

Task D1: Contract Type (*Carteira de Trabalho*)

What is the difference between the two jobs?

- **Job A:** No signed *carteira de trabalho* — the job does not include the legal benefit package (no paid vacation, 13th salary, FGTS, INSS contributions, or unemployment insurance).
- **Job B:** Signed *carteira de trabalho* — includes the full legal benefit package: paid vacation, 13th salary, FGTS, INSS contributions, and unemployment insurance if dismissed without cause.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Carteira de trabalho	**No**	**Yes**
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D2: Flexible Working Hours

What is the difference between the two jobs?

- **Job A:** Fixed schedule set by employer (e.g., 8h to 17h, Monday–Friday).
- **Job B:** Flexible hours—you choose when to start and end your workday (e.g., 7h–16h or 10h–19h instead of a fixed 8h–17h). Total daily hours (8h, Mon–Fri) are the same in both jobs.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Work schedule	**Fixed** (set by employer)	**Flexible** (you choose start/end)
Tasks and responsibilities	Same	Same
Benefits	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D3: Unemployment Insurance (*Seguro Desemprego*)

What is the difference between the two jobs?

- **Job A:** No unemployment insurance — if you are dismissed, you receive nothing.
- **Job B:** Unemployment insurance (*seguro desemprego*) — if dismissed without cause, you receive monthly payments for 3 to 5 months (equivalent to 1 to 1.5 minimum wages per month).

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
If dismissed	**No insurance**	**Unemployment insurance** (3–5 monthly payments)
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D4: Paid Parental Leave

What is the difference between the two jobs?

- **Job A:** No paid parental leave — if you have a child, you do not receive paid time off.
- **Job B:** Paid parental leave — 120 paid days for mothers and 5 paid days for fathers, with a guaranteed right to return to the same position afterward.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Parental leave	**None**	**Paid leave** (120 days mothers / 5 days fathers) + guaranteed return
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	☐ **Job A**	☐ **Job B**

Task D5: Training and Professional Growth

What is the difference between the two jobs?

- **Job A:** No training or courses provided; no clear path for promotion.
- **Job B:** The employer provides training, courses, or certifications at no cost to you, and there are clear opportunities for promotion to better-paying positions.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Training / growth	**None**	**Training + clear promotion path**
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	☐ **Job A**	☐ **Job B**

Task D6: Commute Time

What is the difference between the two jobs?

- **Job A:** Located farther from your home — one-way commute: **[commute from A7, or 60 min if not available]**.
- **Job B:** Located closer to your home — one-way commute: **15 minutes**.
- Both are the same type of work and the same employer, just at different locations.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Commute (one way)	**[A7 value] min**	**15 min**
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same type of work	Same type of work
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D7: Job Stability (*Aviso Prévio*)

What is the difference between the two jobs?

- **Job A:** No job protection — you can be dismissed at any time, without advance notice and without severance pay.
- **Job B:** Job protection (*aviso prévio*) — the employer must give at least 30 days' advance notice before dismissing you, plus pay a severance amount.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Job protection	**None** (dismissed at any time)	**30-day notice** + severance pay
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D8: Health Insurance (*Plano de Saúde*)

What is the difference between the two jobs?

- **Job A:** No health insurance provided — you rely on the public health system (SUS) or pay out of pocket.
- **Job B:** The employer covers a private health plan (*plano de saúde*) for you and your dependents, at no cost to you.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Health coverage	**None** (SUS or out of pocket)	**Private health plan** for you + dependents (paid by employer)
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D9: Childcare Assistance (*Auxílio Creche*)

What is the difference between the two jobs?

- **Job A:** No childcare assistance from the employer.
- **Job B:** Monthly allowance of **R$400 per child under 6 years old** toward daycare or preschool, paid by the employer in addition to your salary.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Childcare support	**None**	**R$400/month per child** under 6 (extra, on top of salary)
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	☐ **Job A**	☐ **Job B**

Task D10: Annual Profit Sharing (*Participação nos Lucros — PLR*)

What is the difference between the two jobs?

- **Job A:** Only regular monthly salary; no annual bonus.
- **Job B:** Annual profit sharing (PLR) — once a year, if the company meets its performance targets, you receive a bonus equivalent to **approximately one extra month's salary**. This is paid on top of your regular salary and 13th salary. The bonus is not guaranteed every year.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Annual bonus	**None**	**PLR bonus** ≈ 1 month's salary/year (when firm meets targets)
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	☐ **Job A**	☐ **Job B**

Task D11: On-Call Pay (*Sobreaviso*)

What is the difference between the two jobs?

- **Job A:** No on-call pay — time spent waiting to be called in outside regular hours is unpaid.
- **Job B:** On-call pay (*sobreaviso*) — you receive extra pay for being available to work outside your regular schedule, even if you are not actually called in. The rate is **one-third (1/3) of your regular hourly rate** for each standby hour.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
On-call availability	**No on-call pay** (standby is unpaid)	**On-call pay** at 1/3 hourly rate per standby hour
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D12: Group Life Insurance (*Seguro de Vida em Grupo*)

What is the difference between the two jobs?

- **Job A:** No life insurance from the employer.
- **Job B:** The employer provides group life insurance at no cost to you — a lump-sum payment of **24 months of your salary** to your family in case of your death or permanent disability while employed.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Life insurance	**None**	**Group life insurance**: 24 months' salary to family upon death or permanent disability
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D13: Paid Days Off for Family Care

What is the difference between the two jobs?

- **Job A:** Absences for family emergencies are unpaid, or you must use your vacation days.
- **Job B:** You may miss up to **3 workdays per year** to care for a sick child, spouse, or parent, at **full salary** — without using vacation days or losing any pay.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Family emergencies	**Unpaid absences** (or use vacation days)	**3 paid days/year** to care for sick child, spouse, or parent
Tasks and responsibilities	Same	Same
Work schedule	Same	Same
Employer	Same	Same
Which job would you prefer?	**☐ Job A**	**☐ Job B**

Task D14: Part-Time Work (32 Hours per Week)

What is the difference between the two jobs?

- **Job A:** Full-time work, 40 hours per week (Monday–Friday), at your regular monthly salary and hourly rate.
- **Job B:** Part-time work, 32 hours per week (Monday–Friday) — a 20% reduction in weekly hours relative to Job A.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary × 0.80 × (1−X%)]
Weekly hours	**Full-time** (40 hours/week)	**Part-time** (32 hours/week)
Earnings per hour	**Regular rate**	**X% below Job A's rate**
Tasks and responsibilities	Same	Same
Benefits	Same	Same
Employer	Same	Same
Which job would you prefer?	☐ **Job A**	☐ **Job B**

Task D15: Home Office Option (*Trabalho Remoto / Home Office*)

What is the difference between the two jobs?

- **Job A:** In-person work only — you must be at the workplace every day.
- **Job B:** Hybrid work (*home office*) — you can work from home **up to 3 days per week** (you choose which days). On the remaining days, you go to the workplace. Total hours per week are the same in both jobs.

Attribute	Job A	Job B
Monthly salary	R$ [ref. salary]	R$ [ref. salary − X%]
Work location	**In-person only** (at workplace every day)	**Hybrid** (up to 3 days/week from home)
Total weekly hours	Same	Same
Tasks and responsibilities	Same	Same
Benefits	Same	Same
Employer	Same	Same
Which job would you prefer?	☐ **Job A**	☐ **Job B**

5.3 Post-Experiment Questions

D16. Thinking about the job choices you just made: which characteristic influenced your decisions the most?

(a) The salary difference

(b) The specific benefit or feature offered by Job B

(c) Both equally

(d) I was not paying much attention to the differences

D17. Were any of the job comparisons confusing or hard to understand? If so, which ones?

_____

_____

6 Module E: Household Context and Constraints

E1. Does anyone else in your household contribute to the household income?

(a) No, I am the only income earner

(b) Yes, my spouse / partner also works

(c) Yes, other household members also work

(d) I am not the main income earner

E2. [If E1 = (b)] Does your spouse / partner have a signed *carteira de trabalho*?

(a) Yes

(b) No

(c) Self-employed

(d) Not currently working

(e) Don't know

E3. What is your **total household monthly income** (from all sources and all members)?

(a) Less than R$1,500

(b) R$1,500 – R$2,500

(c) R$2,501 – R$4,000

(d) R$4,001 – R$6,000

(e) R$6,001 – R$10,000

(f) More than R$10,000

(g) Prefer not to say

E4. Does your household currently receive any of the following government benefits? (Select all that apply)

☐ Bolsa Família / Auxílio Brasil

☐ BPC (Benefício de Prestação Continuada)

☐ Seguro desemprego (currently receiving)

☐ Aposentadoria / pension

☐ None of the above

E5. How many hours per week do you spend on **unpaid household tasks** (cooking, cleaning, laundry, etc.)?
_____ hours per week

E6. How many hours per week do you spend on **childcare or care for elderly / disabled family members**?
_____ hours per week

E7. If you lost your job today, how long could your household manage financially without your income?

(a) Less than 1 week
(b) 1–2 weeks
(c) About 1 month
(d) 2–3 months
(e) More than 3 months
(f) Don't know

E8. If you were offered a better-paying job but it required you to work **fixed hours from 8am to 6pm, Monday to Friday**, would you be able to accept it?

(a) Yes, no problem
(b) Probably, but it would be difficult
(c) No, because of childcare responsibilities
(d) No, because of other family responsibilities
(e) No, because of a second job or other commitments
(f) No, for other reasons: _____

7 Module F: Outside Options and Job Search

F1. If you had to leave your current job today (or, if not currently employed, if you started looking for work today), how long do you think it would take to find a new job?

(a) Less than 1 month
(b) 1–3 months
(c) 4–6 months
(d) 7–12 months
(e) More than 1 year
(f) Don't know

F2. If you found a new job, do you think your **salary** would be:

(a) Much higher than my current salary (more than 20% higher)
(b) Somewhat higher (5–20% higher)
(c) About the same

(d) Somewhat lower (5–20% lower)

(e) Much lower than my current salary (more than 20% lower)

(f) Don't know

F3. Do you think the new job you would find would have a signed *carteira de trabalho*?

(a) Very likely yes

(b) Probably yes

(c) Not sure

(d) Probably not

(e) Very likely not

F4. In the past 3 months, have you looked for or considered looking for a different job?

(a) Yes, I am actively looking

(b) Yes, I have been keeping an eye on opportunities

(c) No, but I would consider switching if the right opportunity came

(d) No, I am not interested in changing jobs

F5. If you were to negotiate with your current employer for better conditions, what would you ask for first?

(a) Higher salary

(b) Better work schedule / more flexibility

(c) Signing my *carteira de trabalho*

(d) More benefits (vale, health insurance, etc.)

(e) Better treatment / work environment

(f) I would not feel comfortable negotiating

(g) Other: _____

8 Module G: Labor Inspections

Imagine the following situation: **you are working without** a signed *carteira de trabalho*. A labor inspector (*fiscal do trabalho*) visits your workplace **unannounced**. After the inspection, your employer offers to sign your *carteira de trabalho*—keeping your **gross salary** (*salário bruto*), hours, and job tasks **exactly the same**.

Note: because INSS contributions (around 7–11% of gross wages) would now be deducted from your paycheck, your ***monthly take-home pay*** *(salário líquido) would be* ***slightly lower*** *than today.*

G1. Would you accept having your *carteira de trabalho* signed under these conditions?

(a) Definitely yes

(b) Probably yes

(c) Not sure

(d) Probably no

(e) Definitely no

G2. [If G1 = (c), (d), or (e)] What would be the **main** reason you might not accept?

(a) My take-home pay would decrease (INSS deductions)

(b) I would lose flexibility in my schedule

(c) I could lose government benefits (Bolsa Família, BPC, etc.)

(d) I don't plan to stay at this job for long

(e) I prefer to work on my own terms

(f) I'm afraid the employer might fire me after regularizing

(g) I don't trust that the benefits will actually be paid

(h) Other: _____

G3. Now consider a **modified** version of the same situation. Everything is the same as before, except: your employer also **increases your gross salary** so that your **monthly take-home pay** (*salário líquido*), after INSS deductions, remains **exactly the same** as today. Under these new conditions, would you accept having your *carteira de trabalho* signed?

(a) Definitely yes

(b) Probably yes

(c) Not sure

(d) Probably no

(e) Definitely no

G4. [If G3 = (c), (d), or (e)] Even with your take-home pay fully protected, what would be the **main** reason you might not accept the formal contract?

(a) I would lose flexibility or autonomy in my schedule

(b) I could lose government benefits (Bolsa Família, BPC, etc.)

(c) I don't plan to stay at this job for long

(d) I'm afraid the employer might fire me after regularizing

(e) I don't trust that the benefits will actually be paid / honored

(f) I find the current arrangement (without a formal contract) more suitable for my situation

(g) Other: _____

G5. Has a labor inspector (*fiscal do trabalho*) ever visited any workplace where you were employed?

(a) Yes

(b) No

(c) Don't know / don't remember

G6. [If G5 = (a)] Thinking about that inspection: at the time, were you working **without** a signed *carteira de trabalho*? If so, what happened afterward?

(a) I had a signed *carteira de trabalho* at the time

(b) I was working without it, and the employer offered to sign it — I accepted

(c) I was working without it, and the employer offered to sign it — I declined

(d) I was working without it, and the employer did not offer to sign it

(e) Don't know / don't remember

9 Module H: Attention Check and Survey Quality

H1. Earlier in this survey, we asked about your employment situation. Just to confirm: do you currently have a signed *carteira de trabalho*?

(a) Yes

(b) No

(c) Not sure

H2. How carefully did you read and think about the job comparisons in this survey?

(a) Very carefully

(b) Somewhat carefully

(c) Not very carefully

(d) I didn't pay much attention

H3. Were there any questions in this survey that were confusing or hard to understand?

(a) No, everything was clear

(b) Yes (please describe): _____